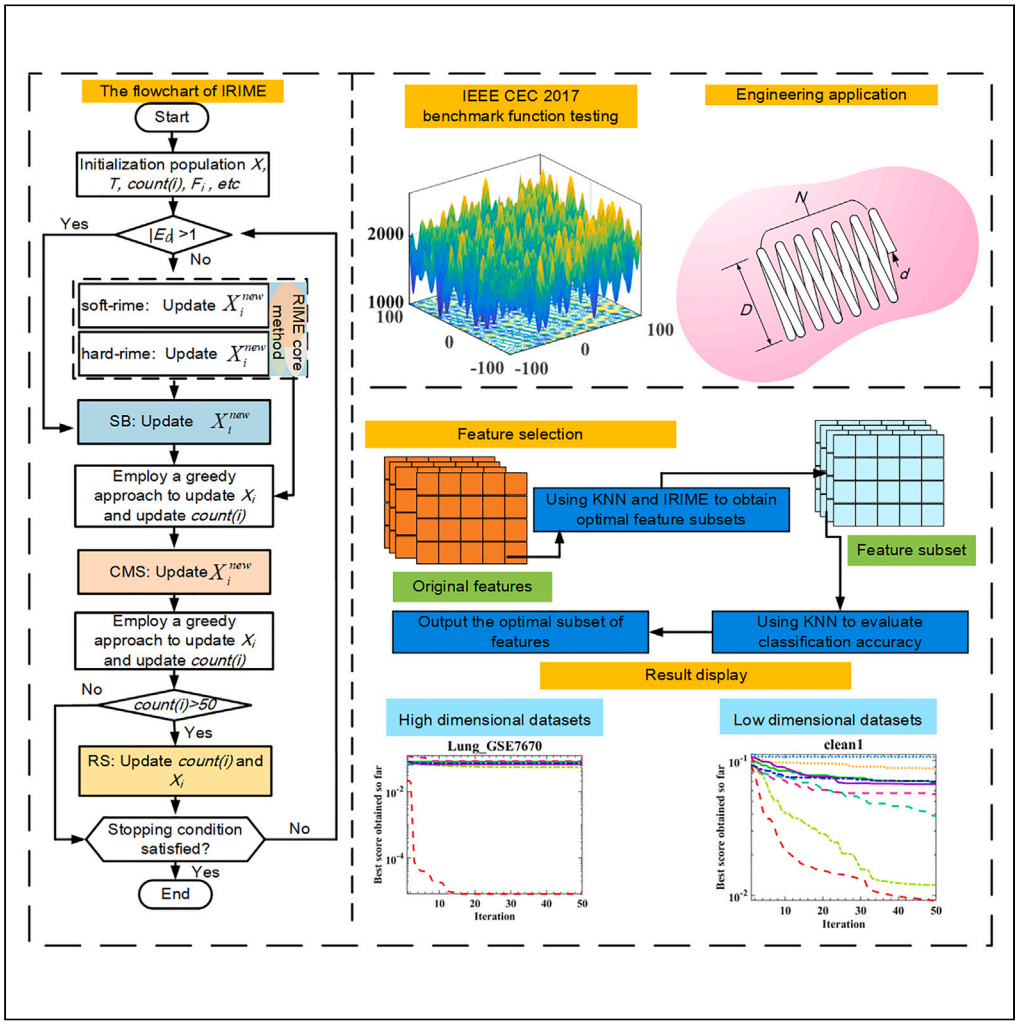
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# IRIME: Mitigating exploitation-exploration imbalance in RIME optimization for feature selection




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## Highlights

This study introduces the SB and CMS-RS into RIME, named IRIME

IRIME addresses the drawbacks of balance between exploration and exploitation

IRIME performs well in benchmark functions and real-world engineering problems

IRIME has outstanding performance in feature selection

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# IRIME: Mitigating exploitation-exploration imbalance in RIME optimization for feature selection

Jinpeng Huang,[1] Yi Chen,[1,*] Ali Asghar Heidari,[2] Lei Liu,[3] Huiling Chen,[1,5,*] and Guoxi Liang[4,*]

## SUMMARY

**Rime optimization algorithm (RIME) encounters issues such as an imbalance between exploitation and exploration, susceptibility to local optima, and low convergence accuracy when handling problems. This paper introduces a variant of RIME called IRIME to address these drawbacks. IRIME integrates the soft besiege (SB) and composite mutation strategy (CMS) and restart strategy (RS). To comprehensively validate IRIME's performance, IEEE CEC 2017 benchmark tests were conducted, comparing it against many advanced algorithms. The results indicate that the performance of IRIME is the best. In addition, applying IRIME in four engineering problems reflects the performance of IRIME in solving practical problems. Finally, the paper proposes a binary version, bIRIME, that can be applied to feature selection problems. bIRIMR performs well on 12 low-dimensional datasets and 24 high-dimensional datasets. It outperforms other advanced algorithms in terms of the number of feature subsets and classification accuracy. In conclusion, bIRIME has great potential in feature selection.**

## INTRODUCTION

In engineering design optimization problems, balancing resource allocation and constraint conditions is often challenging.[1] When facing real-world problems, the optimization process frequently involves multiple variables and diverse constraints, significantly increasing the difficulty of optimization.[2] Engineering optimization also often considers factors such as performance and cost, leading to complex problems with multiple variables and objectives. Among optimization problems, feature selection is a problem widely investigated by scholars, especially in today's era of rapid information growth, where data abundance leads to issues like data redundancy, high computational costs, and weakened model generalization abilities. Feature selection plays a crucial role in reducing computational expenses, simplifying models, and enhancing their generalization capabilities. Commonly used feature selection methods include filter, wrapper, and embedded methods.[3–6] Filter methods primarily determine the importance of features to the target variable based on statistical properties between features or the relationship between feature variables and target variables. While filter methods can be independently analyzed without involving machine learning, they overlook other connections between features. Embedded methods, although capable of uncovering complex relationships between features, require consideration of intricate parameters and structures and are influenced by the machine learning model. Meanwhile, wrapper methods, favored by many researchers because of their straightforward nature and ease of implementation, face a challenge when dealing with an n-dimensional feature dataset, resulting in possible combinations of features.[7,8] Faced with such complex computations, researchers have started using metaheuristic algorithms as a feasible solution for wrapper methods.

Heuristics are problem-solving strategies that use basic principles or shortcuts to quickly uncover approximate answers, generally valuing speed above accuracy.[9–13] Metaheuristics, on the other hand, work at a higher abstraction level, directing the search of solution spaces.[14] They enable the search of optimum or nearly optimal solutions across several issue domains by continually refining and adapting heuristic techniques, so overcoming the constraints of individual problem settings.[15] Metaheuristic algorithms represent advanced optimization techniques that simulate certain biological or physical phenomena found in nature. These algorithms can generally be categorized into physics-based, swarm intelligence-based, and evolution-inspired. Physics-based metaheuristic algorithms, such as the sine cosine algorithm (SCA),[16] RUNge Kutta optimizer (RUN),[17] weighted mean of vectors (INFO),[18] simulated annealing (SA),[19] gravitational search algorithm (GSA),[20] and rime optimization algorithm (RIME),[21] draw inspiration from different natural entities. SCA is inspired by trigonometric functions like sine and cosine, simulating their properties for the search process. SA is inspired by material cooling from high temperatures to a stable state, involving the probabilistic selection of optimal solutions. Newton's law of universal gravitation inspires GSA. The algorithm is modeled based on this

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**Table 1. IEEE CEC 2017 benchmark functions (Search Range: $[-100, 100]^D$)**

| Class | No. | Functions | Optimum |
|---|---|---|---|
| Unimodal | F1 | Shifted and Rotated Bent Cigar Function | 100 |
| | F2 | Shifted and Rotated Bent Sum of Different Power Function | 200 |
| | F3 | Shifted Rotated Zakharov Function | 300 |
| Multimodal | F4 | Shifted and Rotated Rosenbrock's Function | 400 |
| | F5 | Shifted and Rotated Rastrigin's Function | 500 |
| | F6 | Shifted and Rotated Expanded Scaffer's F6 Function | 600 |
| | F7 | Shifted and Rotated Lunacek Bi_Rastrigin's Function | 700 |
| | F8 | Shifted and Rotated Non-Continuous Rastrigin's Function | 800 |
| | F9 | Shifted and Rotated Levy Function | 900 |
| | F10 | Shifted and Rotated Schwefel's Function | 1000 |
| Hybrid | F11 | Hybrid Function 1 ($N$ = 3) | 1100 |
| | F12 | Hybrid Function 2 ($N$ = 3) | 1200 |
| | F13 | Hybrid Function 3 ($N$ = 3) | 1300 |
| | F14 | Hybrid Function 4 ($N$ = 4) | 1400 |
| | F15 | Hybrid Function 5 ($N$ = 4) | 1500 |
| | F16 | Hybrid Function 6 ($N$ = 4) | 1600 |
| | F17 | Hybrid Function 6 ($N$ = 5) | 1700 |
| | F18 | Hybrid Function 6 ($N$ = 5) | 1800 |
| | F19 | Hybrid Function 6 ($N$ = 5) | 1900 |
| | F20 | Hybrid Function 6 ($N$ = 6) | 2000 |
| Composition | F21 | Composition Function 1 ($N$ = 3) | 2100 |
| | F22 | Composition Function 2 ($N$ = 3) | 2200 |
| | F23 | Composition Function 3 ($N$ = 4) | 2300 |
| | F24 | Composition Function 4 ($N$ = 4) | 2400 |
| | F25 | Composition Function 5 ($N$ = 5) | 2500 |
| | F26 | Composition Function 6 ($N$ = 5) | 2600 |
| | F27 | Composition Function 7 ($N$ = 6) | 2700 |
| | F28 | Composition Function 8 ($N$ = 6) | 2800 |
| | F29 | Composition Function 9 ($N$ = 3) | 2900 |
| | F30 | Composition Function 10 ($N$ = 3) | 3000 |

concept, with individual fitness values treated as mass in the gravitational formula. An adaptive gravitational constant is introduced into GSA. RIME simulates the growth of rime-ice in nature, modeling both soft-rime and hard-rime and eventually incorporating a greedy selection strategy. Swarm intelligence-based metaheuristic algorithms have seen rapid development, featuring algorithms like particle swarm optimizer (PSO),[22] water cycle algorithm (WCA),[23] gray wolf optimizer (GWO),[24] hunger games search (HGS),[25] slime mold algorithm (SMA),[26,27] Harris hawks optimizer (HHO),[28] moth-flame optimization (MFO),[29] liver cancer algorithm (LCA),[30] parrot optimizer (PO),[31] colony predation algorithm (CPA),[32] among others. SMA draws inspiration from the foraging process of slime mold, including aspects like capturing, encircling, and approaching food. HHO mathematically models the soft besiege (SB) and hard besiege processes of Harris hawks. MFO involves the mathematical modeling of moths' attraction to flames. In MFO, the number of flames is adjusted based on the iteration count, and flames are selected according to their fitness values. The author models moths' spiraling flight behavior when they are close to flames. Evolution-inspired metaheuristic algorithms primarily include differential evolution (DE),[33] genetic algorithm (GA),[34] and biogeography-based optimization (BBO).[35] DE operates through mutation, crossover, and selection operations, guiding individuals toward potentially better solutions. GA treats each individual as a chromosome, facilitating genetic operations among chromosomes to achieve search outcomes. BBO models migration and mutation in biogeography, relying on migration probability for population updates. These algorithms possess robust optimization capabilities and are expected to demonstrate superior performance in various applications such as fault identification,[36] vehicle communication,[37] text privacy,[38] hemodialysis prediction,[39] target tracking,[40] economic emission,[41,42] and intrusion detection.[43]

As scholars delve into the study of metaheuristic algorithms, numerous outstanding variants have emerged and been successfully applied in various domains.[44–47] Ozsoydan et al.[48] modified the mutation mechanism of elite wolves, proposing a new variant of GWO that effectively tackled multiple combinatorial problems and the 0–1 knapsack problem. Dhargupta et al.[49] utilized spearman's rank correlation coefficient to

**Table 2. The parameters of the algorithm involved**

| Algorithm | Parameters |
|---|---|
| RIME | $w = 5, E = \sqrt{t/T}$ |
| SCA | $c_1 = rand, c_2 = rand, l = 2$ |
| WOA | $b = 1, r_1 = rand, r_2 = rand$ |
| DE | $beta \in [0.2, 0.8], pcr = 0.2$ |
| SSA | $c_1 = rand, c_2 = rand, l = 2$ |
| PSO | $w_{max} = 0.9, w_{min} = 0.2, c_1 = c_2 = 2$ |
| MFO | $b = 1, a = -1 + FEs(-1/MaxFEs), t = (a-1)rand + 1$ |
| GWO | $r_1 = rand, r_2 = rand, a = 2 - FEs(2/MaxFEs)$ |
| BA | $Q_{min} = 0, Q_{max} = 2$ |
| CS | $beta = 1.5, pa = 0.25$ |
| FA | $\alpha = 0.5, beta_{min} = 0.2$ |
| CPA | $a = \exp(9 - 18FEs/MaxFEs), S0 = a(1 - FEs/MaxFEs)$ |
| HHO | $c = 2(1 - (FEs/MaxFEs)), Escaping_Energy = c(2rand - 1)$ |
| EBOwithCMAR | $memorysize = 6$ |
| LSHADE_cnEpSi | $pb = 0.4, ps = 0.5$ |
| ALCPSO | $w = 0.4, c_1 = 2, c_2 = 2, T = 2$ |
| CLPSO | $c = 1.49445$ |
| LSHADE | $p_best_rate = 0.11, \max_popsize = popsize$ |
| SADE | $numst = 4$ |
| JADE | $c = 0.1, crm = 0.5, fm = 0.5$ |
| RCBA | $Q_{min} = 0, Q_{max} = 2$ |
| EPSO | $nsize = 3, LP = 50$ |
| CBA | $Q_{min} = 0, Q_{max} = 2$ |
| LWOA | $b = 1, beta = 1.5$ |

determine whether wolf packs engage in opposition learning, enhancing the convergence speed and capability of populations within GWO. Deng et al.[50] divided the SMA into two populations dynamically, adjusting population sizes to balance the algorithm's exploitation and exploration capabilities and successfully applying it to real-world engineering problems. Samantaray et al.[51] combined SMA with PSO, successfully applying this hybrid approach to predict flood flow rates. Tan et al.[52] combined the whale optimization algorithm (WOA) with the equilibrium optimizer, validating its performance on benchmark test sets. Wang et al.[53] enhanced population diversity in WOA by reducing intra-population similarity, ultimately applying it to multi-threshold image segmentation tasks. Kumar et al.[54] augmented the global search capability of HHO through opposition learning, successfully addressing multi-objective hydrothermal power generation scheduling problems. Tian et al.[55] proposed a novel initialization method, incorporating elite opposition learning to improve HHO populations, ultimately applying it to engineering problems. Tiwari et al.[56] addressed issues such as poor population diversity and inadequate exploration capabilities in DE. They improved DE by incorporating ideas from PSO to enhance its global search ability. Additionally, they changed the crossover rate of DE, proposing a new crossover rate, and introduced a new selection method to further promote DE's convergence capabilities, successfully applied to engineering design optimization problems. Pham et al.[57] proposed a strategy of opposition learning and roulette selection to improve the global optimization capability of SCA. The enhanced SCA maintains a balanced exploration and exploitation similar to the original SCA but with increased stability, ideal for challenging real-world optimization problems. Huang et al.[58] combined various strategies, including Nelder-Mead simplex, opposition learning, and spiral strategy, to enhance beluga whale optimization (BWO). Combining different strategies at different algorithm stages improved BWO's performance, successfully applied it to engineering design problems, and tested it on the CEC benchmark dataset. Gomes et al.[59] proposed and compared a hybrid algorithm with GA. They applied metaheuristic algorithms to channel parameter estimation and successfully demonstrated GA's advantages over the hybrid algorithm in channel parameter estimation. Gundogdu et al.[60] successfully applied an improved GWO to photovoltaic systems. They improved GWO to escape local optima, enhancing performance in photovoltaic system applications. Yu et al.[61] improved the teaching-learning-based optimization algorithm (TLBO) using reinforcement learning to enhance TLBO's update phase, successfully applied to wind farm data problems. Moustafa et al.[62] applied mantis search algorithm (MSA) to economic dispatch in combined heat and power systems, drawing inspiration from collective intelligence of mantises. Al-Areeq et al.[63] utilized a hybrid two-population intelligence algorithm for flood hazard assessment. Tu et al.[64] combined GWO and HHO into HGWO, improving collective search ability, convergence speed, and accuracy compared to GWO, applied to real-world engineering problems. Combining optimization algorithms is a crucial approach to improving metaheuristic algorithms. Silva et al.[65]

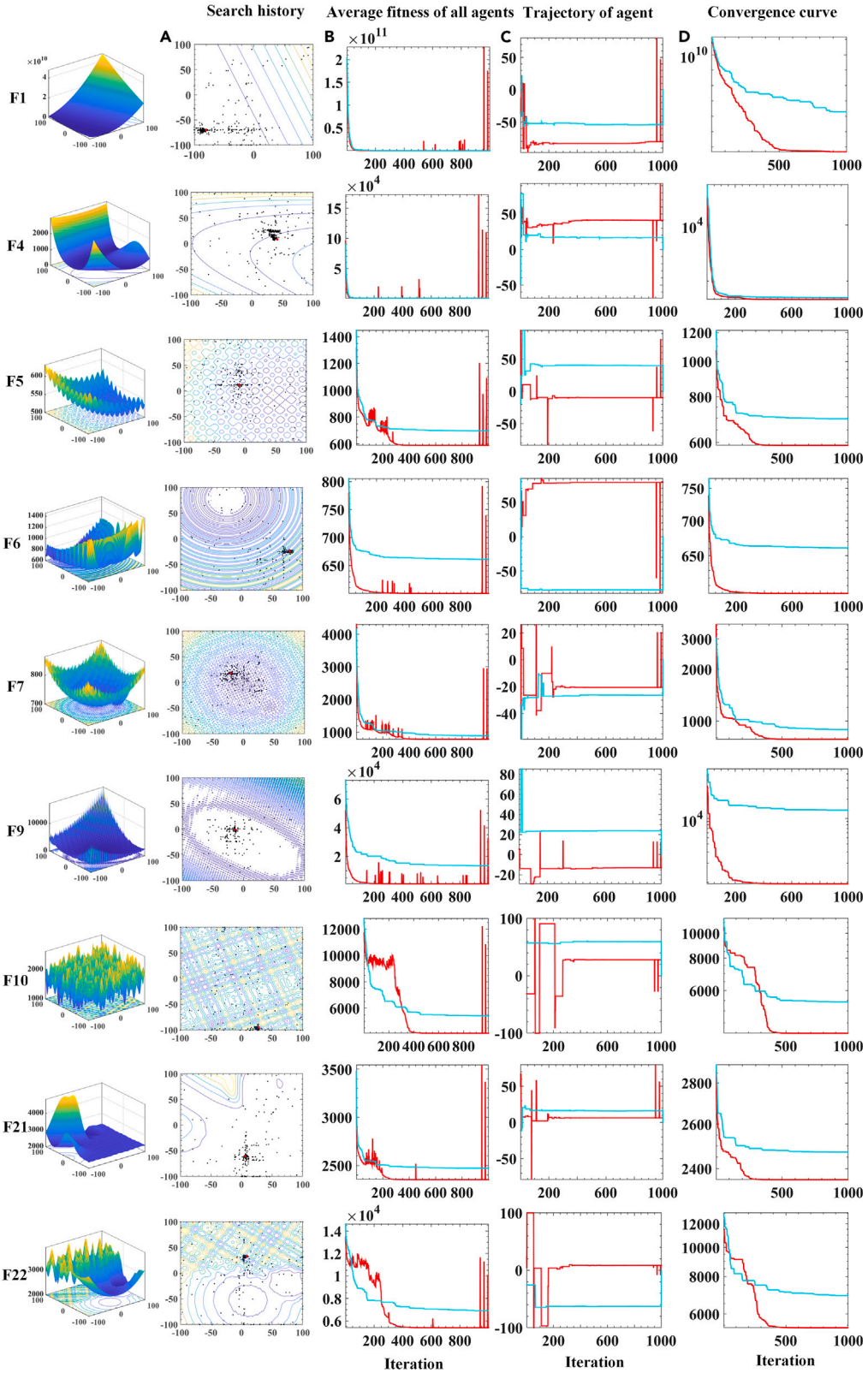



**Figure 1. The history trajectory analysis for IRIME**

(A) The search trajectory of IRIME, (B) Average fitness of IRIME, (C) One-dimensional trajectory of IRIME, (D) convergence curves for IRIME (red) and RIME (blue).

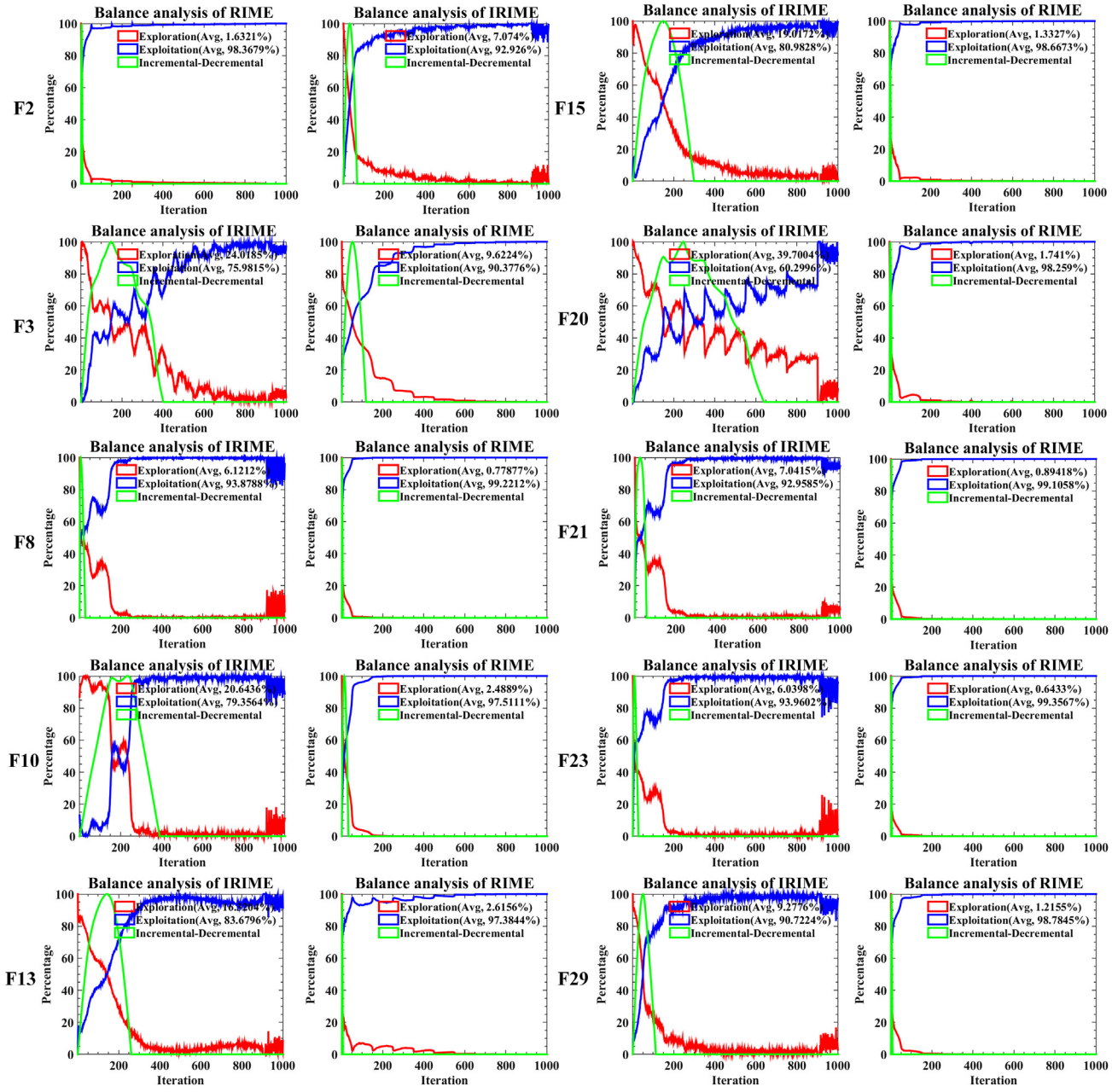



**Figure 2. Balance analysis for IRIME and RIME**

combined ant colony optimization (ACO) and GA, enhancing ACO's convergence capability and mitigating the algorithm's tendency to get stuck in local optima, applied to sustainable solution problems. Moreover, metaheuristic algorithms have found extensive applications in the domain of feature selection.

Peng et al.[66] adopted hierarchical strategies to enhance HHO, conducting feature selection on both low and high-dimensional datasets. Yu et al.[67] improved WOA using various strategies, including sine initialization and employed a kernel extreme learning machine as a classifier for feature selection. AbdelAty et al.[68] utilized chaos theory to boost the convergence capability of the hunter-prey optimization algorithm, successfully applying it to feature selection. Al-Khatib et al.[69] enhanced lemurs optimization by integrating local search strategies and opposition learning, evaluating feature selection performance on UCI datasets. Zaimoglu et al.[70] employed different chaos learning methods to improve the herd optimization algorithm and conducted feature selection tests across multiple classifiers. Chhabra et al.[71] improved bald eagle search by incorporating three distinct enhancement strategies at different stages of algorithm execution, successfully applying them to feature selection. Pan et al.[72] improved the initialization strategy of GWO and enhanced GWO using differential and

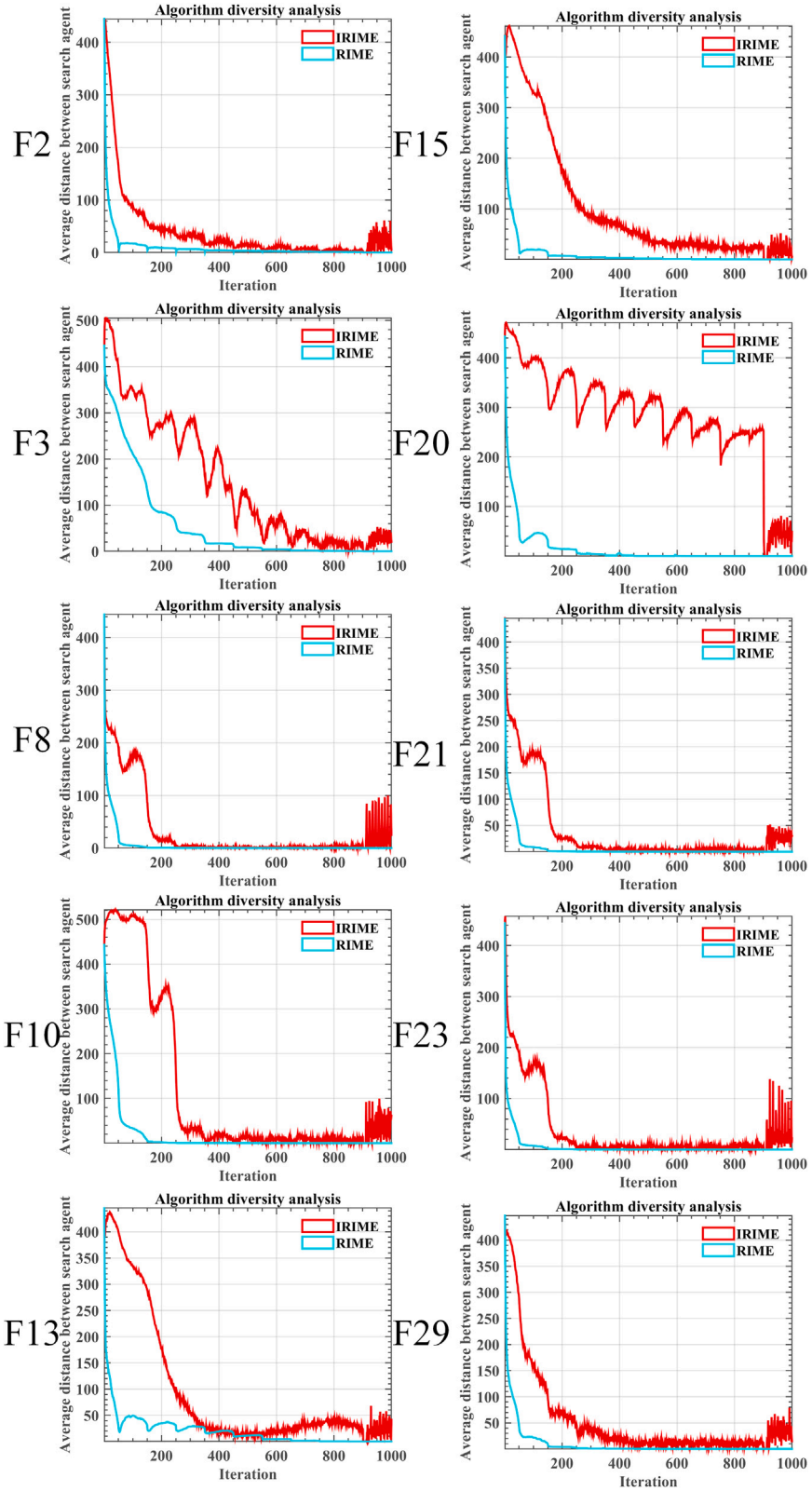



Figure 3. Algorithm diversity analysis for IRIME and RIME

**Table 3. Comparison results between IRIME and IRIMEs**

| Overall Rank | | | |
|---|---|---|---|
| Algorithm | Rank | +/ = /- | Avg |
| IRIME50 | **1** | ~ | **1.7** |
| IRME30 | 3 | 10/19/1 | 2.4 |
| IRIME100 | 2 | 7/18/5 | 1.866667 |

competition-guided strategies for feature selection in high-dimensional data. Askr et al.[73] used various strategies to enhance the exploration and exploitation capabilities of the golden jackal optimization (GJO) algorithm. They proposed a binary form of GJO and tested it for feature selection on multiple high-dimensional datasets. Wang et al.[74] made improvements to the transfer function, introducing a new function specifically targeting the deficiencies of the GWO in handling feature selection problems. They applied different enhancement methods to elite and ordinary wolves within the population to enhance the balance between exploration and exploitation in GWO. Ye et al.[75] enhanced the optimization capability of the hybrid breeding optimization using the elite opposition mechanism and Levy flight strategy. They combined different classifiers for intrusion detection feature selection problems. Yang et al.[76] focused on feature space, dividing it into regions and proposing a new initialization strategy. They successfully applied the golden eagle optimizer to solve feature selection problems in medium to small-dimensional spaces. Chakraborty et al.[77] improved the WOA using a horizontal crossing strategy and collaborative hunting. They introduced a binary version of WOA and combined it with the K-nearest neighbor for feature selection on UCI datasets. Abdelrazek et al.[78] incorporated different chaotic mappings into the dwarf mongoose optimization algorithm (DMO), enabling DMO to better adapt to wrapper-based feature selection methods. The improved algorithm was validated on various UCI datasets, demonstrating competitive performance compared to other metaheuristic algorithms. Mostafa et al.[79] used spider wasp optimization to enhance DE, improving DE's problem-solving capabilities and incorporating methods to enhance solution quality, specifically applied to feature selection. As per the no free lunch (NFL)[80] theorem, no single algorithm can address all feature selection tasks, especially in complex optimization environments, where RIME tends to get stuck in local optima and encounter slow convergence issues. Hence, this paper develops a variant of RIME to enhance its performance in complex optimization environments and with intricate datasets. RIME, a new algorithm proposed by Su in 2023,[21] has seen limited use in feature selection studies.

Research and application of RIME is underway. Yu et al.[81] combined the triangular game search strategy and random follower search strategy to improve RIME, enhancing its global search capability and inter-population information exchange capabilities. The enhanced RIME was then applied in the diagnostic process of pulmonary hypertension. Yang et al.[82] applied the improved RIME in photovoltaic systems to maintain temperature stability. Zhong et al.[83] improved RIME by utilizing Latin hypercube sampling and distance-based selection mechanisms and enhanced the hard-rime process, ultimately applying the improved RIME to engineering design problems. Zhu et al.[84] improved RIME using the Gaussian diffusion and interactive mechanism strategy, which effectively solved multi-threshold image segmentation problems. Li et al.[85] also applied the improved RIME in multi-threshold image segmentation.

In this paper, to further enhance the capability of RIME in feature selection applications, we introduced SB, composite mutation strategy, and restart strategy (CMS-RS) into RIME, naming it IRIME. SB expands the search space of RIME, increases population diversity in the early stage, and effectively prevents the problem of local optima caused by greedy strategies. Additionally, CMS encourages more in-depth exploitation at the current position of RIME, to some extent, enhancing RIME's exploitation abilities. RS keeps an eye on whether RIME falls into local optima and restarts when it does. The combination of these mechanisms involves adaptive parameters and does not run in a singular form like classical PSO and DE, but instead has multiple optimization methods. In sum, the combination of these approaches balances the exploration and exploitation abilities of RIME. To validate IRIME's performance, this study conducted tests on the IEEE CEC 2017 benchmark tests and compared them with other advanced algorithms, demonstrating significant advantages for IRIME. In addition, IRIME's performance in engineering design problems also reflects its ability to solve practical problems. Finally, it applied to feature selection in low-dimensional and high-dimensional datasets. In summary, this paper's primary contributions encompass.

- Proposed a variant of RIME named IRIME.
- This paper effectively enhances the population diversity of RIME by using SB, expands the search space, and enhances the exploratory ability.

**Table 4. Comparison results between IRIME and RIMEs**

| Overall Rank | | | |
|---|---|---|---|
| Algorithm | Rank | +/ = /- | Avg |
| IRIME | **1** | ~ | **1.433333** |
| SBRIME | 3 | 23/4/3 | 3.133333 |
| CMSRSRIME | 2 | 9/18/3 | 2.066667 |
| RIME | 4 | 28/2/0 | 3.366667 |

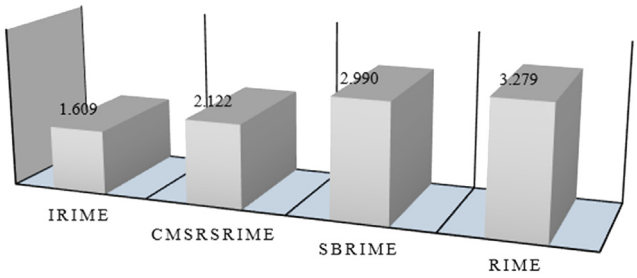



**Figure 4. The Friedman ranking of IRIME and RIMEs**

- This paper integrates CMS-RS to improve the exploitation capacity of RIME and explores new solutions when stuck in a local optimum.
- IRIME has demonstrated excellent performance in IEEE CEC 2017 benchmark functions and demonstrated the ability to solve practical problems in engineering design.
- The paper proposes a binary version of IRIME applied to feature selection problems, which respectively achieves good results on high- and low-dimensional datasets.

## RESULTS AND DISCUSSION

### Experimental design and analysis of results

A series of systematic experiments were conducted in this study to validate the efficacy of the RIME variant. The IEEE CEC 2017 benchmark functions were utilized,[86] comprising functions categorized into four types: simple unimodal (F1-F3), simple multimodal (F4-F10), hybrid (F11-F20), and composite functions (F21-F30),[87] as shown in Table 1. The IEEE CEC 2017 benchmark functions were a set of standard functions used during the 2017 IEEE congress on evolutionary computation for evaluating the performance of evolutionary algorithms and other optimization algorithms.[88,89] These functions are designed to test different optimization problem settings and have undergone extensive research and validation to ensure that they pose a certain level of complexity and diversity, effectively evaluating the performance of optimization algorithms. The experiments involved historical trajectory analysis, balance and diversity analyses of IRIME, stability analysis, and ablation studies. A comparison was made against 13 conventional algorithms and 11 advanced algorithms. In addition, apply IRIME to 4 practical engineering problems to verify its ability to solve engineering problems. Finally, it was applied to 12 low-dimensional and 24 high-dimensional datasets to validate its performance in feature selection. To ensure statistical significance in the experimental results, non-parametric statistical tests such as the Wilcoxon signed-rank test[90] were employed, with a significance level set at 0.05. Additionally, average (AVG) and standard deviation (STD) analyses were used, and ranking was conducted using the Friedman test.[91] In testing the IEEE CEC 2017 benchmark function, all experiments referred to previous research to minimize bias as much as possible.

The experiments in this study were conducted using MATLAB R2020a on a system running the Windows 11 operating system, powered by an Intel(R) Core(TM) i5-12400 12th generation processor clocked at 2.50 GHz. The relevant parameters for the algorithms tested alongside IRIME are listed in Table 2.

#### *Qualitative analysis of IRIME*

*Analysis of historical search trajectories.* In Figure 1, F1 represents a unimodal function. Observing Figures 1B and 1D, it is apparent that IRIME converges to smaller fitness values. Moreover, the sudden increase in the average fitness of all agents in later iterations is due to the CMS-RS initiating a restart upon identifying local optima, thereby exploring new solutions and enhancing the possibility of discovering potential solutions. In Figure 1A, the red dots indicate the positions where the best solution has been found so far, while the black dots represent the trajectory points during the search. Initially, individuals are randomly distributed in the solution space, but with IRIME iterations, they gradually approach the peak of the unimodal function. In Figure 1C, the one-dimensional trajectory also shows that IRIME has a broader search space in both the initial and final stages than RIME. In the middle stage of IRIME, there is a bias toward exploitation: initially influenced by SB and later influenced by RS, with CMS contributing more to exploitation in the middle stage. For functions F4-F10, representing simple multimodal functions, Figure 1A shows that IRIME's individuals are distributed across each peak at the onset. Gradually, IRIME discovers better peaks and exploits them. In Figures 1B and 1D, IRIME consistently achieves better fitness values than RIME. When IRIME gets stuck in local optima, the CMS-RS opens up new spaces. Figure 1C also indicates that IRIME initially explores multiple peaks and gradually converges to a better peak, which is evident in the selected region, which shows better fitness values than RIME. Similarly, in composite functions F21 and F22, IRIME initially exhibits a larger search space than RIME and converges to better fitness values later, discovering regions where RIME fails to reach.

*Balance and diversity analysis.* This section utilizes the IEEE CEC 2017 benchmark functions to evaluate the balance and diversity of IRIME and RIME. As depicted in Figures 2 and 3, the blue line represents the algorithm in an exploitation phase, the red line illustrates

**Table 5. Comparison of IRIME with RIMEs**

| | F1 | | F2 | | F3 | |
|---|---|---|---|---|---|---|
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **1.000000E+02** | 4.341879E-07 | 1.208397E+04 | 1.692978E+04 | 3.000675E+02 | 6.544860E-02 |
| CMSRSRIME | **1.000000E+02** | 6.787999E-07 | **9.608267E+03** | 2.901084E+04 | **3.000137E+02** | 1.440701E-02 |
| SBRIME | 6.912754E+03 | 3.342332E+03 | 1.903663E+04 | 3.287184E+04 | 3.019264E+02 | 7.738878E-01 |
| RIME | 6.740600E+03 | 4.683236E+03 | 1.725470E+04 | 4.617363E+04 | 3.016142E+02 | 7.393312E-01 |
| | F4 | | F5 | | F6 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **4.049188E+02** | 1.660907E+01 | 5.612429E+02 | 1.547367E+01 | 6.000036E+02 | 1.901825E-03 |
| CMSRSRIME | 4.119034E+02 | 2.428008E+01 | **5.592430E+02** | 1.695952E+01 | **6.000025E+02** | 1.635151E-03 |
| SBRIME | 4.822844E+02 | 2.432146E+01 | 5.940733E+02 | 2.538615E+01 | 6.008424E+02 | 7.132176E-01 |
| RIME | 4.934413E+02 | 3.982441E+01 | 5.792156E+02 | 2.035945E+01 | 6.003355E+02 | 3.274886E-01 |
| | F7 | | F8 | | F9 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **7.854226E+02** | 1.459656E+01 | **8.592915E+02** | 1.276500E+01 | **9.000000E+02** | 1.360593E-11 |
| CMSRSRIME | 7.875552E+02 | 1.590540E+01 | 8.632096E+02 | 1.265242E+01 | 9.031533E+02 | 9.154477E+00 |
| SBRIME | 8.092856E+02 | 2.294600E+01 | 8.825459E+02 | 2.181813E+01 | 1.832479E+03 | 8.498307E+02 |
| RIME | 8.196870E+02 | 2.584547E+01 | 8.795674E+02 | 2.032560E+01 | 1.300574E+03 | 5.075069E+02 |
| | F10 | | F11 | | F12 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **3.370405E+03** | 6.680517E+02 | **1.142597E+03** | 2.458383E+01 | 3.778128E+03 | 1.308558E+03 |
| CMSRSRIME | 3.527053E+03 | 4.907452E+02 | 1.148122E+03 | 2.141947E+01 | **3.759469E+03** | 1.719976E+03 |
| SBRIME | 3.618576E+03 | 5.236817E+02 | 1.276824E+03 | 7.583414E+01 | 6.747955E+06 | 3.731713E+06 |
| RIME | 3.402741E+03 | 6.191174E+02 | 1.231434E+03 | 6.460259E+01 | 4.432634E+06 | 2.177431E+06 |
| | F13 | | F14 | | F15 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **1.334285E+03** | 8.006018E+00 | **1.444883E+03** | 1.161457E+01 | **1.559461E+03** | 2.867685E+01 |
| CMSRSRIME | 1.338540E+03 | 9.647202E+00 | 1.449102E+03 | 1.254535E+01 | 1.569876E+03 | 3.745820E+01 |
| SBRIME | 4.021342E+03 | 3.087106E+03 | 1.564542E+03 | 8.300780E+01 | 4.034866E+03 | 3.706339E+03 |
| RIME | 4.693434E+03 | 2.467541E+03 | 1.504226E+03 | 3.706804E+01 | 6.473199E+03 | 4.852897E+03 |
| | F16 | | F17 | | F18 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **2.153008E+03** | 2.439803E+02 | **1.937497E+03** | 8.960966E+01 | 1.363427E+04 | 8.005798E+03 |
| CMSRSRIME | 2.256912E+03 | 2.333114E+02 | 1.999397E+03 | 1.131100E+02 | **1.223189E+04** | 1.042454E+04 |
| SBRIME | 2.290189E+03 | 2.808756E+02 | 2.041873E+03 | 8.707292E+01 | 7.422992E+04 | 5.051162E+04 |
| RIME | 2.348066E+03 | 2.163382E+02 | 2.026551E+03 | 1.694096E+02 | 1.018748E+05 | 7.327155E+04 |
| | F19 | | F20 | | F21 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **2.014868E+03** | 2.572843E+02 | 2.189910E+03 | 8.610377E+01 | 2.134962E+03 | 3.507469E+01 |
| CMSRSRIME | 2.925334E+03 | 3.347182E+03 | **2.181714E+03** | 9.010563E+01 | **2.125485E+03** | 3.190956E+01 |
| SBRIME | 4.612865E+03 | 4.464702E+03 | 2.361028E+03 | 1.289487E+02 | 2.207251E+03 | 3.255049E+01 |
| RIME | 7.662323E+03 | 6.550908E+03 | 2.290457E+03 | 1.005639E+02 | 2.214770E+03 | 3.275008E+01 |

(*Continued on next page*)

**Table 5. *Continued***

| | F22 | | F23 | | F24 | |
|---|---|---|---|---|---|---|
| | AVG | STD | AVG | STD | AVG | STD |
| | F22 | | F23 | | F24 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **2.259745E+03** | 1.569471E+01 | **2.500000E+03** | 8.301258E-06 | **2.600000E+03** | 1.888236E-13 |
| CMSRSRIME | 2.263672E+03 | 1.606741E+01 | 2.845609E+03 | 1.510916E+01 | 3.394899E+03 | 1.332673E+01 |
| SBRIME | 2.311125E+03 | 3.239709E+01 | 2.500972E+03 | 8.935810E-01 | 2.600053E+03 | 5.355676E-02 |
| RIME | 2.282943E+03 | 2.054662E+01 | 2.884192E+03 | 3.074977E+01 | 3.360063E+03 | 1.826656E+02 |
| | F25 | | F26 | | F27 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | 2.707120E+03 | 3.899517E+01 | **2.800000E+03** | 7.208242E-10 | 2.920793E+03 | 1.138842E+02 |
| CMSRSRIME | 2.918437E+03 | 2.484449E+01 | 5.090513E+03 | 1.921404E+02 | 3.437174E+03 | 3.633907E+01 |
| SBRIME | **2.700552E+03** | 3.402318E-01 | 2.800529E+03 | 4.689096E-01 | **2.902733E+03** | 9.562687E-01 |
| RIME | 2.965889E+03 | 4.976697E+01 | 4.887692E+03 | 9.898190E+02 | 3.547105E+03 | 7.414412E+01 |
| | F28 | | F29 | | F30 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | 3.031860E+03 | 7.474622E+01 | 3.255024E+03 | 1.067390E+02 | **5.978167E+03** | 3.490858E+03 |
| CMSRSRIME | 4.714342E+03 | 8.194617E+02 | 3.418242E+03 | 1.152990E+02 | 1.069414E+04 | 4.913513E+03 |
| SBRIME | **3.000791E+03** | 6.112466E-01 | **3.109261E+03** | 4.379583E+00 | 6.810517E+04 | 8.243885E+04 |
| RIME | 3.350192E+03 | 3.439794E+02 | 3.545178E+03 | 1.500152E+02 | 3.456162E+04 | 2.103248E+04 |

the algorithm in an exploration phase, and the green line indicates an increasing trend when the exploration outweighs the exploitation or a decreasing trend otherwise. As shown in the graphs for unimodal functions F2 and F3, the variant of RIME proposed in this paper, IRIME, tends to be explored more extensively in the early stages. This appropriate increase in the exploration phase expedites the algorithm's convergence and mitigates the susceptibility to local optima. In contrast, RIME spends less time exploring F2 and F3, particularly with only 1.6321% in F2. This results in slow convergence of the algorithm in unimodal functions. However, the integration of SB effectively enhances IRIME's exploratory capability, accelerating convergence. As observed in the diversity graph, IRIME's individuals are initially distributed across a broader space due to SB's influence, leading to higher diversity than RIME. During the mid-phase, extensive exploitation occurs, and a sudden rise in diversity toward the end is attributed to CMS-RS's role, which detects the algorithm's local entrapment and enhances IRIME's precision. In the case of simple multimodal functions F8 and F10, RIME exhibits minimal exploration, especially in F10. This starkly contrasts IRIME, where the exploration capability exceeds 20%, while RIME's exploration capability is only around 2%. Consequently, RIME is highly prone to local optima, only developing around specific peaks and failing to explore potentially more fruitful regions. The diversity curve further demonstrates that IRIME possesses greater initial population diversity and engages in substantial exploitation in the mid-phase, and after exploitation stagnation, IRIME attempts to break out of local optima to find better solutions. For hybrid functions F13, F15, and F20, RIME only explores about 1%, whereas IRIME explores more extensively. Additionally, the population's diversity increases. CMS-RS also plays a role in discovering better solutions in the later phase. A similar pattern emerges for functions F21, F23, and F29, where RS and CMS-RS balance the exploration and exploitation capabilities of the original RIME, increasing population diversity and enhancing convergence accuracy. This also empowers IRIME to escape local optima. In conclusion, the combination of SB and CMS-RS equips IRIME with superior balance and diversity, allowing it to escape local optima more effectively.

### *Parameter sensitivity experiment*

The selection of parameters critically influences algorithm's performance; therefore, conducting a parameter sensitivity analysis is essential.[92] This analysis evaluates how different parameter values affect the performance of the algorithm, thereby optimizing algorithm efficiency and ensuring robustness under various conditions.[93,94] In this paper, most parameters used in IRIME are supported by theoretical or empirical justifications from original papers. The only point of contention is the threshold at which the restart strategy in CMS-RS begins to execute. As mentioned earlier, the threshold is set at 50. To verify the appropriateness of this threshold, this paper conducted experiments at threshold values of 30, 50, and 100, represented as IRIME30, IRIME50, and IRIME100, respectively.

The experimental results are shown in Table 3. The symbols "+/ = /-" represent whether IRIME performs significantly better, equal to, or worse than other algorithms in this experiment on the Wilcoxon signed-rank test. From the table, it can be seen that the choice of threshold

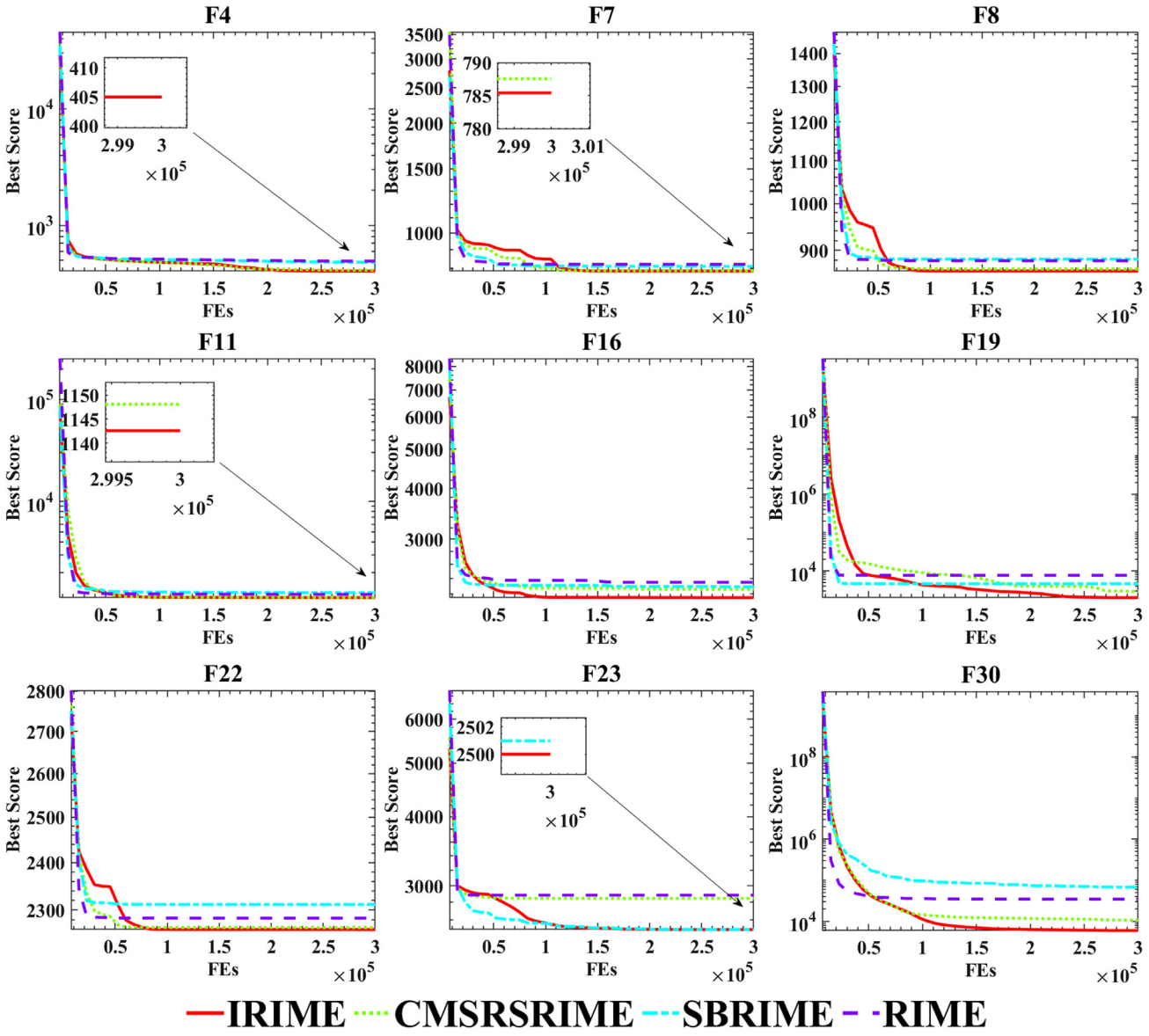



**Figure 5. Convergence curves of IRIME and RIMEs at IEEE CEC 2017**

affects the overall performance of IRIME. However, the differences are not pronounced for most functions. As per the results, the performance difference between IRIME30 and IRIME50 is insignificant in 19 functions, while between IRIME100 and IRIME50, the difference is insignificant in 18 functions. On the whole, the overall performance of the IRIME algorithm is better when the threshold is set at 50. As shown in the table, the average ranking of IRIME50 is smaller, with a final score of 1.7.

### *The influence of SB and CMS-RS*

In this section, we specifically examine the precise impact of SB and CMS-RS on RIME using the IEEE CEC 2017 benchmark functions. In this experiment, the population size was set to 30, problem dimensionality was set to 30, and the maximum iteration count was 300,000. To eliminate potential incidental influences, each algorithm ran independently 30 times. SBRIME represents RIME integrated with SB, while CMSRSRIME denotes RIME integrated with CMS-RS. The symbols "+/ = /-" represent whether IRIME performs significantly better, equal to, or worse than other algorithms in this experiment on the Wilcoxon signed-rank test.

Table 4 shows that the average ranking of IRIME is 1.43333, securing the top position. This indicates that combining these two methods improves RIME in various aspects. Compared to the original RIME, 28 out of 30 benchmark functions perform better, while two functions converge to the same outcome. This suggests that the incorporation of SB and CMS-RS has not adversely affected RIME. We conducted

Table 6. *p*−value of Wilcoxon signed−rank test between IRIME and other RIMEs

| | F1 | F2 | F3 | F4 | F5 | F6 |
|---|---|---|---|---|---|---|
| CMSRSRIME | 2.9574621307E-03 | 1.0746884002E-01 | 3.4052567233E-05 | 4.7794743855E-01 | 4.7794743855E-01 | 6.0350064738E-03 |
| SBRIME | 1.7343976283E-06 | 4.1653380739E-01 | 1.7343976283E-06 | 1.9209211049E-06 | 2.1630223984E-05 | 1.7343976283E-06 |
| RIME | 1.7343976283E-06 | 3.4934556237E-01 | 1.7343976283E-06 | 1.7343976283E-06 | 3.3788544377E-03 | 1.7343976283E-06 |
| | F7 | F8 | F9 | F10 | F11 | F12 |
| CMSRSRIME | 7.3432529144E-01 | 2.5364409755E-01 | 1.6367234818E-01 | 4.0483472216E-01 | 1.1092566513E-01 | 4.6528258188E-01 |
| SBRIME | 3.5888445045E-04 | 1.2505680433E-04 | 1.7343976283E-06 | 9.3675596532E-02 | 1.7343976283E-06 | 1.7343976283E-06 |
| RIME | 4.0715116266E-05 | 7.1570338462E-04 | 1.7343976283E-06 | 7.0356369987E-01 | 2.3534209951E-06 | 1.7343976283E-06 |
| | F13 | F14 | F15 | F16 | F17 | F18 |
| CMSRSRIME | 1.2543823903E-01 | 4.7794743855E-01 | 2.8947707171E-01 | 6.8713630797E-02 | 8.7296677536E-03 | 3.3885615525E-01 |
| SBRIME | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 | 5.4462503972E-02 | 2.2248266458E-04 | 1.7343976283E-06 |
| RIME | 1.7343976283E-06 | 2.1266360107E-06 | 1.7343976283E-06 | 2.1052603409E-03 | 2.8485956185E-02 | 1.7343976283E-06 |
| | F19 | F20 | F21 | F22 | F23 | F24 |
| CMSRSRIME | 3.0861485053E-01 | 7.0356369987E-01 | 1.5885549929E-01 | 1.6502656562E-01 | 1.7343976283E-06 | 1.7343976283E-06 |
| SBRIME | 2.3534209951E-06 | 4.8602606067E-05 | 2.3534209951E-06 | 4.7292023374E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| RIME | 1.7343976283E-06 | 4.1955098606E-04 | 1.7343976283E-06 | 8.1877534396E-05 | 1.7343976283E-06 | 1.7343976283E-06 |
| | F25 | F26 | F27 | F28 | F29 | F30 |
| CMSRSRIME | 1.9209211049E-06 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7333066442E-06 | 1.9729484516E-05 | 1.6046383717E-04 |
| SBRIME | 3.1123151154E-05 | 1.7343976283E-06 | 3.1123151154E-05 | 5.7096495243E-02 | 4.7292023374E-06 | 3.1816794110E-06 |
| RIME | 1.7343976283E-06 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 | 1.9209211049E-06 | 1.9209211049E-06 |

a Friedman ranking, as depicted in Figure 4. From the Friedman ranking, it is evident that IRIME ranks first, and the algorithm's performance improves with each additional mechanism integrated.

Table 5 presents specific comparative data of the algorithms, with bold text highlighting the best results obtained among all algorithms. It also includes convergence graphs, as depicted in Figure 5. The graph and the table show that solely incorporating SB or CMS-RS does not lead the algorithm to perform optimally. Solely adding SB enhances population diversity, boosting RIME's exploratory capability. This improvement is evident in RIME's performance on composite functions like F22 and F23. However, it does not manifest advantages in unimodal functions like F3 or multimodal functions like F8. Solely incorporating CMS-RS strengthens the algorithm's performance on unimodal functions such as F1, multimodal functions like F4 and F5, and hybrid functions like F11 and F12. However, CMS-RS does not demonstrate significant effects on composite functions, mainly because it effectively improves RIME's convergence capability, meeting the requirements for local exploitation and, to some extent, providing the ability to escape local optima. Yet, it offers less in terms of population diversity and weaker exploratory capabilities. Only through the comprehensive integration of SB and CMS-RS can IRIME achieve the top-ranking position. Additionally, Table 6 provides the algorithm's *p*-values. From a statistical perspective, IRIME holds a dominant position across most functions.

### *Stability testing of IRIME*

Stability experiments were conducted in this section to validate the stability of IRIME. The parameter settings remained mostly similar to the previous experiments, except for variations in dimensions (30, 50, and 100). The specific experimental data is shown in Table 7. The table shows that IRIME outperforms RIME in 30, 29, and 29 benchmark functions across different dimensions, demonstrating significantly better stability than RIME. Particularly on functions like F6 and F26, as the dimensions increase, RIME's convergence to the optimal value also grows, but IRIME maintains convergence at similarly low optimal values. A Friedman ranking was computed, as shown in Figure 6, demonstrating IRIME's significant advantage as the dimensionality increases. In summary, with changing problem dimensions, IRIME sustains its competitiveness, showcasing remarkable stability and maintaining strong performance.

### *Comparison with conventional algorithms*

In this section, IRIME was primarily compared against 13 conventional algorithms: RIME,[21] SCA,[16] WOA,[95] DE,[33] SSA,[96] PSO,[97] MFO,[29] GWO,[98] BA,[99] CS,[91] FA,[100] CPA[32] and HHO.[28] Table 8 illustrates that IRIME achieves an average ranking of 1.733333, indicating its exceptional performance across all 30 benchmark test functions. Detailed experimental data is provided in Table 9. Upon careful comparison with SCA, WOA, DE, PSO, MFO, GWO, FA, and RIME, IRIME does not exhibit noticeably poorer performance than these algorithms. Furthermore, compared to other algorithms such as CPA, HHO, etc., there are many cases where IRIME is significantly superior. For a visual representation of IRIME's performance, convergence curve plots were generated, as depicted in Figure 7. These curves demonstrate IRIME's distinctive characteristics compared to RIME, especially evident in simple unimodal functions F1, multimodal functions F4, F5, F7, F8 and F10, hybrid

**Table 7. Stability testing of IRIME and RIME**

| | 30 | | 50 | | 100 | |
|---|---|---|---|---|---|---|
| Metric | IRIME | RIME | IRIME | RIME | IRIME | RIME |
| AVG | **1.000000E+02** | 1.028739E+04 | **8.150018E+03** | 3.498054E+04 | **3.587010E+04** | 5.599304E+05 |
| STD | 6.624418E-07 | 1.822468E+04 | 1.360977E+04 | 1.030577E+04 | 2.748387E+04 | 1.181859E+05 |
| AVG | **6.779600E+03** | 3.152160E+04 | **3.707965E+13** | 8.353509E+13 | **9.181274E+56** | 1.820439E+80 |
| STD | 8.247370E+03 | 6.815905E+04 | 1.077243E+14 | 2.980378E+14 | 3.252605E+57 | 9.948538E+80 |
| AVG | **3.000804E+02** | 3.017968E+02 | 9.127787E+02 | **5.914656E+02** | 9.806014E+04 | **9.242816E+04** |
| STD | 8.807603E-02 | 7.545155E-01 | 3.786835E+02 | 1.216696E+02 | 1.448744E+04 | 1.933928E+04 |
| AVG | **4.161644E+02** | 4.904783E+02 | **4.920837E+02** | 5.228727E+02 | **6.357680E+02** | 6.980471E+02 |
| STD | 2.870048E+01 | 3.720042E+01 | 1.411291E+01 | 4.746566E+01 | 4.853969E+01 | 4.424859E+01 |
| AVG | **5.619552E+02** | 5.836654E+02 | **6.267921E+02** | 6.818772E+02 | **8.695937E+02** | 9.965034E+02 |
| STD | 1.790301E+01 | 1.722221E+01 | 2.854613E+01 | 3.959170E+01 | 6.564103E+01 | 8.890611E+01 |
| AVG | **6.000026E+02** | 6.002950E+02 | **6.000213E+02** | 6.027180E+02 | **6.002694E+02** | 6.182405E+02 |
| STD | 1.369855E-03 | 2.141152E-01 | 8.625810E-03 | 1.841028E+00 | 1.523005E-01 | 4.611132E+00 |
| AVG | **7.840374E+02** | 8.168247E+02 | **8.740425E+02** | 9.318452E+02 | **1.181254E+03** | 1.294812E+03 |
| STD | 1.300727E+01 | 2.777899E+01 | 1.874723E+01 | 4.792427E+01 | 6.421225E+01 | 9.746528E+01 |
| AVG | **8.621855E+02** | 8.839896E+02 | **9.318084E+02** | 9.770503E+02 | **1.184238E+03** | 1.324732E+03 |
| STD | 1.320405E+01 | 1.835873E+01 | 2.928632E+01 | 3.136398E+01 | 6.472031E+01 | 8.014492E+01 |
| AVG | **9.000181E+02** | 1.124300E+03 | **9.045256E+02** | 3.715535E+03 | **1.520706E+03** | 1.770827E+04 |
| STD | 8.398819E-02 | 2.651242E+02 | 8.449065E+00 | 1.840252E+03 | 7.826171E+02 | 7.679602E+03 |
| AVG | 3.403375E+03 | **3.318342E+03** | **6.472167E+03** | 6.607002E+03 | **1.483534E+04** | 1.553603E+04 |
| STD | 5.195037E+02 | 4.690800E+02 | 7.651949E+02 | 7.082011E+02 | 1.171622E+03 | 1.278887E+03 |
| AVG | **1.144627E+03** | 1.235619E+03 | **1.213375E+03** | 1.566197E+03 | **1.639799E+03** | 2.527533E+03 |
| STD | 1.801979E+01 | 7.120005E+01 | 2.596203E+01 | 1.123408E+02 | 1.378813E+02 | 2.185059E+02 |
| AVG | **3.259220E+03** | 3.185041E+06 | **6.215314E+03** | 8.247759E+06 | **2.146365E+04** | 9.556442E+07 |
| STD | 9.613443E+02 | 2.533263E+06 | 2.619565E+03 | 3.820874E+06 | 9.818197E+03 | 3.444275E+07 |
| AVG | **1.339929E+03** | 3.903406E+03 | **1.038753E+04** | 5.893687E+04 | **2.624556E+04** | 2.400604E+05 |
| STD | 1.078343E+01 | 3.101797E+03 | 7.408806E+03 | 4.089735E+04 | 1.241422E+04 | 8.196934E+04 |
| AVG | **1.443579E+03** | 1.497542E+03 | **1.466908E+03** | 1.651542E+03 | **1.532119E+03** | 1.599748E+04 |
| STD | 9.549785E+00 | 2.906395E+01 | 1.731724E+01 | 5.900106E+01 | 2.628161E+01 | 7.791170E+03 |
| AVG | **1.560320E+03** | 7.576501E+03 | **2.017887E+03** | 4.234476E+03 | **8.630684E+03** | 5.690617E+04 |
| STD | 3.352494E+01 | 6.640569E+03 | 7.957409E+02 | 2.180854E+03 | 9.731233E+03 | 2.010337E+04 |
| AVG | **2.189042E+03** | 2.341539E+03 | **2.641367E+03** | 2.967189E+03 | **5.169371E+03** | 5.974952E+03 |
| STD | 2.485710E+02 | 3.008552E+02 | 3.246147E+02 | 4.208589E+02 | 6.400325E+02 | 5.933374E+02 |
| AVG | **1.932272E+03** | 2.019690E+03 | **2.303726E+03** | 2.577466E+03 | **4.363901E+03** | 4.793590E+03 |
| STD | 1.219315E+02 | 1.098334E+02 | 2.243114E+02 | 2.717036E+02 | 4.906362E+02 | 5.352419E+02 |
| AVG | **1.321118E+04** | 1.282609E+05 | **1.076984E+05** | 6.718929E+05 | **8.873835E+05** | 2.215444E+06 |
| STD | 1.040334E+04 | 9.873021E+04 | 4.224163E+04 | 4.070633E+05 | 4.590897E+05 | 7.540629E+05 |
| AVG | **1.996303E+03** | 8.668593E+03 | **9.349882E+03** | 1.402825E+04 | **6.842005E+03** | 4.190986E+04 |
| STD | 2.888304E+02 | 7.195163E+03 | 1.038581E+04 | 1.488471E+04 | 3.688717E+03 | 2.169956E+04 |
| AVG | **2.171758E+03** | 2.261173E+03 | **2.579609E+03** | 2.792553E+03 | **4.184473E+03** | 4.741982E+03 |
| STD | 7.861606E+01 | 9.435565E+01 | 2.374063E+02 | 2.490740E+02 | 4.799121E+02 | 4.920016E+02 |
| AVG | **2.129702E+03** | 2.194063E+03 | **2.212182E+03** | 2.270509E+03 | **2.250000E+03** | 2.250002E+03 |
| STD | 3.233579E+01 | 2.859524E+01 | 3.470840E+01 | 3.920999E+01 | 1.705788E-06 | 1.661872E-04 |
| AVG | **2.261555E+03** | 2.278551E+03 | **2.339057E+03** | 2.389399E+03 | **2.350000E+03** | **2.350000E+03** |
| STD | 1.980115E+01 | 1.982225E+01 | 2.854209E+01 | 4.896348E+01 | 8.275093E-09 | 9.695178E-06 |
| AVG | **2.500000E+03** | 2.877338E+03 | **2.522419E+03** | 3.249900E+03 | **2.500380E+03** | 3.959631E+03 |

(*Continued on next page*)

**Table 7. *Continued***

| | 30 | | 50 | | 100 | |
|---|---|---|---|---|---|---|
| Metric | IRIME | RIME | IRIME | RIME | IRIME | RIME |
| STD | 9.799078E-06 | 2.785766E+01 | 1.227458E+02 | 5.579856E+01 | 3.389571E-01 | 8.897769E+01 |
| AVG | **2.600000E+03** | 3.299834E+03 | **2.600000E+03** | 3.829701E+03 | **2.600019E+03** | 5.362005E+03 |
| STD | 2.533334E-13 | 2.579968E+02 | 4.619013E-06 | 6.618639E+01 | 2.013730E-02 | 9.411192E+01 |
| AVG | **2.707120E+03** | 2.973604E+03 | **2.881642E+03** | 3.051111E+03 | **2.700225E+03** | 3.340697E+03 |
| STD | 3.899625E+01 | 5.501906E+01 | 1.531719E+02 | 2.489686E+01 | 2.330694E-01 | 6.969830E+01 |
| AVG | **2.800000E+03** | 5.014069E+03 | **2.800000E+03** | 7.360208E+03 | **2.800134E+03** | 1.570043E+04 |
| STD | 4.776900E-13 | 9.979314E+02 | 2.963824E-05 | 1.592764E+03 | 1.101589E-01 | 1.136894E+03 |
| AVG | **2.918219E+03** | 3.594628E+03 | **2.971923E+03** | 3.994026E+03 | **3.097434E+03** | 5.449449E+03 |
| STD | 9.978696E+01 | 8.565568E+01 | 2.188521E+02 | 1.697183E+02 | 5.004242E+02 | 3.176788E+02 |
| AVG | **3.023799E+03** | 3.412600E+03 | **3.275639E+03** | 3.342462E+03 | **3.000172E+03** | 3.407231E+03 |
| STD | 7.300070E+01 | 4.842390E+02 | 1.299708E+02 | 3.360505E+01 | 1.083659E-01 | 4.757740E+01 |
| AVG | **3.240401E+03** | 3.560112E+03 | **3.569520E+03** | 4.500026E+03 | **5.375933E+03** | 6.954458E+03 |
| STD | 6.760153E+01 | 1.833220E+02 | 2.907660E+02 | 3.105699E+02 | 4.990865E+02 | 5.511103E+02 |
| AVG | **6.144891E+03** | 4.017941E+04 | **2.279465E+04** | 7.241399E+05 | **9.420026E+03** | 6.578093E+06 |
| STD | 3.632263E+03 | 2.698645E+04 | 1.915455E+03 | 4.691778E+05 | 2.867600E+03 | 2.615942E+06 |

functions F11, F12, F13, F14, F17, and F18, and composite functions F21 and F22. This notable performance is primarily attributed to balancing RIME's exploration and exploitation abilities by SB and CMS-RS, enabling IRIME's ability to escape local optima. Moreover, the Friedman ranking chart in Figure 8 positions IRIME at the top. To indicate the statistical significance of IRIME's superiority over other algorithms, a table presenting *p*-values of the Wilcoxon signed-rank test is included in Table10.

### *Comparison with advanced algorithms*

To further validate IRIME's performance, this study compared it with some advanced algorithms using the IEEE CEC 2017 benchmark test suite. These algorithms include EBOwithCMAR,[101] LSHADE_cnEpSi,[86] ALCPSO,[102] CLPSO,[103] LSHADE,[104] SADE,[105] JADE,[106] RCBA,[107] EPSO,[108] CBA,[109] and LWOA,[110] The specific experimental data is detailed in Table 11. The results in the table highlight that IRIME, alongside some advanced algorithms, achieves performance near the theoretical optimum in functions such as F1, F3, F6, and F9. Compared to exceptional variants of DE like JADE and SADE, IRIME demonstrates some drawbacks in multimodal functions (F4, F5, F7, and F10) and hybrid functions (F12, F16, F17, and F18). This can be attributed to the limitations of SB and CMS-RS in improving convergence accuracy. Nevertheless, these limitations do not significantly impact IRIME's overall performance. IRIME can also find very good results on multimodal functions F7 and F8, indicating that IRIME is not uniformly poor on multimodal functions. In addition, the hybrid functions F11, F13, F14, and F15 can also reflect excellent results, demonstrating that the disadvantage of IRIME on hybrid functions is not significant. Particularly in composite functions (F23, F24, F25, F26, F27, F28 and F29), IRIME exhibits advantages that aren't present in these advanced algorithms, such as EBOwithCMAR, LSHADE_cnEpSi, ALCPSO, and CLPSO. When compared to other successful improvements in swarm intelligence algorithms like RCBA, CBA, and LWOA, IRIME outperforms them in convergence capability, especially in simple unimodal functions such as F1 and F2. Despite potential shortcomings in convergence accuracy, IRIME's strong exploration abilities and the balance between exploration and exploitation elevate its average ranking to the top among these algorithms, as depicted in Table 12.

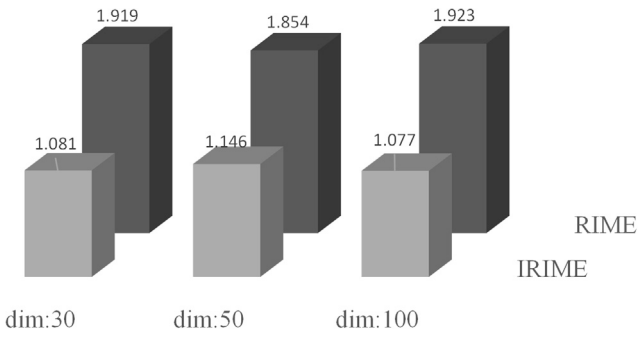



**Figure 6. Friedman ranking of IRIME in different dimensions**

**Table 8. Comparison results between IRIME and conventional algorithms**

Overall Rank

| Algorithm | Rank | +/ = /- | Avg |
|---|---|---|---|
| IRIME | **1** | ~ | **1.733333** |
| SCA | 13 | 30/0/0 | 12.2 |
| WOA | 11 | 27/3/0 | 10.3 |
| DE | 3 | 25/2/3 | 5 |
| SSA | 6 | 29/0/1 | 5.833333 |
| PSO | 9 | 30/0/0 | 8.366667 |
| MFO | 12 | 30/0/0 | 10.93333 |
| GWO | 8 | 28/2/0 | 7.566667 |
| BA | 10 | 28/1/1 | 9.366667 |
| CS | 5 | 27/2/1 | 5.766667 |
| FA | 14 | 30/0/0 | 12.3 |
| CPA | 2 | 18/4/8 | 3.333333 |
| HHO | 7 | 22/2/6 | 6.9 |
| RIME | 3 | 29/1/0 | 5 |

To provide a clear visual representation of IRIME's performance, this study developed convergence curve graphs, in Figure 9. From the graph, it is evident that IRIME does not exhibit outstanding convergence speed. The red curve initially positions relatively higher among numerous algorithms. However, unlike other algorithms that directly converge near a local optimum and struggle to escape, IRIME demonstrates more excellent exploitation capabilities. Despite its slower initial convergence, IRIME and CMS-RS effectively explore and break away from local optima. This ability enables IRIME to converge to better positions in functions such as F9, F13, and F27. Likewise, to assess whether IRIME statistically outperforms these advanced algorithms, a Friedman ranking graph was generated in Figure 10. It is evident from the graph that IRIME secures the top rank among these algorithms, with a value of 3.71. JADE is still a strong algorithm, and IRIME has a slight advantage over JADE. Variations of other DEs, such as SADE and SHADE, also rank closely behind IRIME and JADE, demonstrating excellent performance. It is worth mentioning that EBOwithCMAR also achieves results similar to DE variants and ranks fourth, even more competitive than SHADE. Specific *p*-values are available in Table 13. From the *p*-value of the Wilcoxon signed-rank test, it can be seen that IRIME is significantly superior to other algorithms in most cases.

### *The experiments for engineering design*

To further validate the performance of IRIME in practical applications, this paper applies IRIME to five real-world engineering problems, including the tension compression string problem (TCSP), cantilever beam problem (CBP), I-beam problem (IBP), and Belleville spring problem (BSP). It compares it with some algorithms that perform excellently in engineering design. The maximum number of iterations in the engineering problems is set to 2000, and the population size is 50. The purpose of this setup is to follow the original RIME[21] paper.

*TCSP.* The TCSP[111] is an optimization problem involving three variables: the number of effective coils ($N$), average coil diameter ($D$), and wire diameter ($d$). As shown in Figure 11, the TCSP problem can be formulated as follows:

$$Consider \quad \vec{x} = [x_1 x_2 x_3] = [dDN]$$

$$Objective\ function \quad f(\vec{x})_{\min} = x_1^2 x_2 x_3 + 2x_1^2 x_2$$

$$Subject\ to \quad h_1(\vec{x}) = 1 - \frac{x_2^3 x_3}{71785 x_1^4} \leq 0,$$

$$h_2(\vec{x}) = \frac{4x_2^2 - x_1 x_2}{12566(x_2 x_1^3 - x_1^4)} + \frac{1}{5180 x_1^2} - 1 \leq 0,$$

$$h_3(\vec{x}) = 1 - \frac{140.45 x_1}{x_2^3 x_3} \leq 0,$$

$$h_4(\vec{x}) = \frac{x_1 + x_2}{1.5} - 1 \leq 0,$$

**Table 9. Comparison of IRIME with conventional algorithms**

| | F1 | | F2 | | F3 | |
|---|---|---|---|---|---|---|
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **1.000000E+02** | 8.228886E-07 | **1.092587E+04** | 2.557057E+04 | 3.000768E+02 | 8.230567E-02 |
| SCA | 1.841382E+10 | 3.722769E+09 | 7.136067E+32 | 1.990604E+33 | 3.706668E+04 | 4.666736E+03 |
| WOA | 6.004626E+06 | 6.340405E+06 | 2.066092E+29 | 1.113738E+30 | 1.338299E+04 | 5.360271E+03 |
| DE | 1.333382E+02 | 1.663241E+02 | 1.983923E+25 | 3.478891E+25 | 1.996025E+04 | 3.770712E+03 |
| SSA | 2.792499E+03 | 3.017484E+03 | 4.056705E+06 | 2.112276E+07 | 3.000000E+02 | 8.818389E-09 |
| PSO | 1.389773E+08 | 1.936371E+07 | 2.561524E+12 | 2.515226E+12 | 6.259001E+02 | 3.466948E+01 |
| MFO | 1.178499E+10 | 5.277154E+09 | 4.606288E+38 | 2.268979E+39 | 1.139089E+05 | 5.365509E+04 |
| GWO | 3.656408E+09 | 2.680374E+09 | 8.477831E+28 | 2.813243E+29 | 3.158195E+04 | 9.978276E+03 |
| BA | 3.979009E+05 | 1.875329E+05 | 2.001667E+02 | 9.128709E-01 | 3.000746E+02 | 4.876542E-02 |
| CS | 1.000000E+10 | 0.000000E+00 | 1.000000E+10 | 0.000000E+00 | 1.185032E+04 | 2.895928E+03 |
| FA | 1.422799E+10 | 1.490877E+09 | 2.637550E+34 | 2.817855E+34 | 5.846299E+04 | 8.407209E+03 |
| CPA | 2.114254E+03 | 2.746738E+03 | 1.107555E+05 | 5.629322E+05 | **3.000000E+02** | 1.708056E-07 |
| HHO | 1.397816E+07 | 2.883833E+06 | 2.473100E+12 | 7.349195E+12 | 2.801396E+03 | 1.243863E+03 |
| RIME | 7.124393E+03 | 4.505485E+03 | 2.058783E+04 | 5.563495E+04 | 3.016689E+02 | 7.524137E-01 |
| | F4 | | F5 | | F6 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **4.047558E+02** | 1.615793E+01 | **5.592333E+02** | 1.350497E+01 | 6.000030E+02 | 2.016607E-03 |
| SCA | 1.374591E+03 | 2.771825E+02 | 7.588440E+02 | 1.443074E+01 | 6.417769E+02 | 5.229569E+00 |
| WOA | 5.840244E+02 | 5.335661E+01 | 6.972605E+02 | 4.197744E+01 | 6.644217E+02 | 1.299889E+01 |
| DE | 4.901688E+02 | 3.989320E+01 | 6.123099E+02 | 1.004958E+01 | **6.000000E+02** | 0.000000E+00 |
| SSA | 4.880627E+02 | 3.685931E+01 | 6.083748E+02 | 3.205830E+01 | 6.293176E+02 | 1.206613E+01 |
| PSO | 4.559716E+02 | 3.206651E+01 | 6.922583E+02 | 2.502204E+01 | 6.343293E+02 | 1.113896E+01 |
| MFO | 1.249248E+03 | 7.468984E+02 | 7.099112E+02 | 3.925173E+01 | 6.444544E+02 | 9.505611E+00 |
| GWO | 6.646073E+02 | 1.181156E+02 | 5.870130E+02 | 2.823283E+01 | 6.064975E+02 | 3.413389E+00 |
| BA | 4.492754E+02 | 5.335580E+01 | 7.378640E+02 | 4.005934E+01 | 6.715805E+02 | 1.149586E+01 |
| CS | 4.193988E+02 | 2.958199E+01 | 6.269874E+02 | 2.190149E+01 | 6.269248E+02 | 8.245281E+00 |
| FA | 1.482195E+03 | 1.539432E+02 | 7.535040E+02 | 1.239110E+01 | 6.437613E+02 | 3.892201E+00 |
| CPA | 4.741417E+02 | 4.920473E+01 | 6.277521E+02 | 2.556818E+01 | **6.000000E+02** | 1.993686E-07 |
| HHO | 5.452018E+02 | 3.950364E+01 | 6.738793E+02 | 1.994907E+01 | 6.526138E+02 | 3.710877E+00 |
| RIME | 4.869630E+02 | 2.962177E+01 | 5.771337E+02 | 1.783255E+01 | 6.003665E+02 | 2.821859E-01 |
| | F7 | | F8 | | F9 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **7.883173E+02** | 1.821927E+01 | **8.592822E+02** | 1.301643E+01 | 9.000454E+02 | 1.386272E-01 |
| SCA | 1.203477E+03 | 5.847467E+01 | 1.077141E+03 | 2.148756E+01 | 7.055291E+03 | 1.483905E+03 |
| WOA | 1.305250E+03 | 1.357140E+02 | 1.079100E+03 | 5.445599E+01 | 9.026235E+03 | 3.453609E+03 |
| DE | 8.431410E+02 | 9.738404E+00 | 9.123994E+02 | 6.583571E+00 | **9.000000E+02** | 2.985563E-14 |
| SSA | 8.727399E+02 | 4.585326E+01 | 9.195604E+02 | 3.175260E+01 | 3.902221E+03 | 1.700013E+03 |
| PSO | 9.195953E+02 | 1.906608E+01 | 1.041707E+03 | 3.675778E+01 | 6.310334E+03 | 2.485837E+03 |
| MFO | 1.223923E+03 | 2.587419E+02 | 1.007102E+03 | 4.956693E+01 | 8.133813E+03 | 2.578392E+03 |
| GWO | 8.687501E+02 | 5.299589E+01 | 8.966145E+02 | 2.657562E+01 | 2.549040E+03 | 8.862296E+02 |
| BA | 1.672298E+03 | 2.052284E+02 | 1.121454E+03 | 5.642799E+01 | 1.596200E+04 | 4.562495E+03 |
| CS | 8.931793E+02 | 2.914000E+01 | 9.358801E+02 | 2.485981E+01 | 5.137085E+03 | 1.624264E+03 |
| FA | 1.378286E+03 | 4.499559E+01 | 1.053642E+03 | 1.212715E+01 | 5.904095E+03 | 5.642448E+02 |
| CPA | 8.489678E+02 | 3.506446E+01 | 8.985005E+02 | 2.920458E+01 | 3.316187E+03 | 7.771718E+02 |

(*Continued on next page*)

**Table 9. *Continued***

| | F7 | | F8 | | F9 | |
|---|---|---|---|---|---|---|
| | AVG | STD | AVG | STD | AVG | STD |
| HHO | 1.302266E+03 | 9.362298E+01 | 1.076979E+03 | 4.260311E+01 | 7.620443E+03 | 1.098960E+03 |
| RIME | 8.108969E+02 | 2.663949E+01 | 8.771250E+02 | 1.542515E+01 | 1.282633E+03 | 7.780124E+02 |
| | F10 | | F11 | | F12 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **3.113983E+03** | 4.683815E+02 | **1.130403E+03** | 1.468068E+01 | **3.319172E+03** | 7.705038E+02 |
| SCA | 7.981345E+03 | 3.150563E+02 | 3.022295E+03 | 6.445754E+02 | 1.574493E+09 | 3.979278E+08 |
| WOA | 5.934370E+03 | 7.789161E+02 | 1.481984E+03 | 8.519552E+01 | 1.382265E+08 | 7.101250E+07 |
| DE | 5.672989E+03 | 2.624536E+02 | 1.152053E+03 | 8.966502E+00 | 2.025299E+04 | 4.813425E+04 |
| SSA | 4.452536E+03 | 7.094846E+02 | 1.344872E+03 | 7.211084E+01 | 9.182351E+06 | 5.040467E+06 |
| PSO | 5.643005E+03 | 5.073707E+02 | 1.347653E+03 | 5.680875E+01 | 8.995054E+07 | 3.766610E+07 |
| MFO | 5.147688E+03 | 8.203648E+02 | 8.873480E+03 | 7.933486E+03 | 1.033950E+09 | 1.367383E+09 |
| GWO | 3.885633E+03 | 9.366685E+02 | 2.439886E+03 | 1.375237E+03 | 2.211911E+08 | 5.231662E+08 |
| BA | 5.665602E+03 | 6.360143E+02 | 1.463662E+03 | 1.486178E+02 | 7.653362E+06 | 7.277742E+06 |
| CS | 4.458374E+03 | 2.296132E+02 | 1.185109E+03 | 1.742362E+01 | 9.666740E+09 | 1.825338E+09 |
| FA | 7.802109E+03 | 2.450349E+02 | 4.351898E+03 | 6.829350E+02 | 2.681007E+09 | 3.497603E+08 |
| CPA | 3.335836E+03 | 4.680002E+02 | 1.153764E+03 | 2.469058E+01 | 6.094811E+04 | 5.596317E+04 |
| HHO | 4.635003E+03 | 5.314861E+02 | 1.373707E+03 | 9.325560E+01 | 5.779730E+07 | 3.642617E+07 |
| RIME | 3.409716E+03 | 5.427260E+02 | 1.236769E+03 | 6.668725E+01 | 3.654135E+06 | 2.400589E+06 |
| | F13 | | F14 | | F15 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **1.336538E+03** | 8.473889E+00 | **1.443824E+03** | 8.556915E+00 | 1.553682E+03 | 2.436474E+01 |
| SCA | 1.135809E+08 | 3.597453E+07 | 1.857652E+05 | 8.431296E+04 | 4.841776E+06 | 2.356802E+06 |
| WOA | 1.000657E+05 | 7.298211E+04 | 2.780113E+05 | 1.802183E+05 | 5.291920E+04 | 8.617273E+04 |
| DE | 1.412306E+03 | 3.210871E+02 | 1.464071E+03 | 7.837059E+00 | **1.550860E+03** | 1.334677E+01 |
| SSA | 9.785285E+04 | 1.030243E+05 | 2.465348E+04 | 2.001924E+04 | 3.396406E+04 | 1.937665E+04 |
| PSO | 2.672529E+06 | 6.920589E+05 | 2.900206E+04 | 1.903076E+04 | 3.161740E+05 | 1.329552E+05 |
| MFO | 8.890284E+07 | 2.019387E+08 | 4.818146E+05 | 1.166317E+06 | 4.578076E+04 | 4.589563E+04 |
| GWO | 1.493141E+07 | 2.605109E+07 | 1.207818E+05 | 1.325602E+05 | 2.080886E+06 | 1.127817E+07 |
| BA | 2.056385E+05 | 1.475900E+05 | 1.764583E+04 | 8.026951E+03 | 9.857857E+04 | 9.742760E+04 |
| CS | 1.403952E+03 | 2.910464E+01 | 1.467773E+03 | 8.666795E+00 | 1.566742E+03 | 1.286895E+01 |
| FA | 4.300832E+08 | 1.161389E+08 | 3.545473E+05 | 1.403319E+05 | 5.339371E+07 | 1.919045E+07 |
| CPA | 3.945263E+03 | 2.400788E+03 | 1.467093E+03 | 3.012936E+01 | 7.282572E+03 | 4.871670E+03 |
| HHO | 1.949828E+05 | 8.419094E+04 | 4.341608E+04 | 3.603063E+04 | 3.486027E+04 | 1.511152E+04 |
| RIME | 4.237048E+03 | 2.386911E+03 | 1.521076E+03 | 4.668063E+01 | 7.385270E+03 | 4.797001E+03 |
| | F16 | | F17 | | F18 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | 2.157519E+03 | 2.319810E+02 | **1.942563E+03** | 8.414615E+01 | **1.227806E+04** | 8.274997E+03 |
| SCA | 3.297912E+03 | 2.433252E+02 | 2.544122E+03 | 1.996151E+02 | 1.843845E+06 | 9.558486E+05 |
| WOA | 3.325774E+03 | 4.574786E+02 | 2.621453E+03 | 3.052444E+02 | 4.481267E+06 | 3.637390E+06 |
| DE | **2.025687E+03** | 1.610609E+02 | 1.965870E+03 | 4.022178E+01 | 3.533861E+05 | 1.845308E+05 |
| SSA | 2.392214E+03 | 2.408053E+02 | 2.084263E+03 | 1.671643E+02 | 6.859217E+04 | 5.922140E+04 |
| PSO | 2.668135E+03 | 2.316809E+02 | 2.379098E+03 | 2.458484E+02 | 8.997152E+04 | 4.816071E+04 |
| MFO | 3.191869E+03 | 4.313203E+02 | 2.382983E+03 | 2.636714E+02 | 2.996153E+05 | 6.695035E+05 |

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Table 9. *Continued*

| | F16 | | F17 | | F18 | |
|---|---|---|---|---|---|---|
| | AVG | STD | AVG | STD | AVG | STD |
| GWO | 2.252737E+03 | 2.427924E+02 | 1.984203E+03 | 1.120046E+02 | 7.415711E+05 | 1.403432E+06 |
| BA | 3.490398E+03 | 5.264603E+02 | 2.771914E+03 | 3.374694E+02 | 6.745901E+04 | 2.947320E+04 |
| CS | 2.458258E+03 | 1.646827E+02 | 2.093107E+03 | 8.895653E+01 | 2.839660E+04 | 8.739249E+03 |
| FA | 3.171391E+03 | 1.900067E+02 | 2.556076E+03 | 1.239653E+02 | 2.123477E+06 | 7.703717E+05 |
| CPA | 2.638873E+03 | 3.589458E+02 | 2.035829E+03 | 1.520298E+02 | 8.337727E+04 | 5.771694E+04 |
| HHO | 2.853986E+03 | 4.348123E+02 | 2.518518E+03 | 2.613917E+02 | 1.479073E+06 | 1.536378E+06 |
| RIME | 2.361515E+03 | 2.358123E+02 | 1.997354E+03 | 1.314076E+02 | 1.082342E+05 | 6.657172E+04 |
| | F19 | | F20 | | F21 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **2.332181E+03** | 1.864812E+03 | **2.192876E+03** | 9.492175E+01 | **2.136135E+03** | 3.352597E+01 |
| SCA | 2.168794E+07 | 1.687208E+07 | 2.650003E+03 | 1.360133E+02 | 3.142386E+03 | 2.110077E+02 |
| WOA | 7.039075E+05 | 7.090756E+05 | 2.762562E+03 | 1.563571E+02 | 2.275860E+03 | 3.276134E+01 |
| DE | 4.947694E+03 | 2.473822E+03 | 2.216929E+03 | 4.611997E+01 | 2.187477E+03 | 1.809598E+01 |
| SSA | 1.667896E+05 | 1.140472E+05 | 2.432897E+03 | 1.772135E+02 | 2.203093E+03 | 3.248284E+01 |
| PSO | 3.430261E+05 | 1.514502E+05 | 2.653687E+03 | 1.678977E+02 | 2.177850E+03 | 3.668529E+01 |
| MFO | 3.757899E+05 | 1.837212E+06 | 2.678688E+03 | 2.917192E+02 | 2.874566E+03 | 8.105786E+02 |
| GWO | 5.753903E+04 | 6.494282E+04 | 2.350499E+03 | 1.051505E+02 | 2.337764E+03 | 7.529951E+01 |
| BA | 2.695612E+05 | 1.461260E+05 | 2.978931E+03 | 2.096352E+02 | 2.161001E+03 | 4.171884E+01 |
| CS | 1.931255E+03 | 5.385502E+00 | 2.480485E+03 | 8.794099E+01 | 2.141894E+03 | 3.215894E+01 |
| FA | 3.416584E+07 | 1.587037E+07 | 2.599760E+03 | 9.585333E+01 | 3.244746E+03 | 1.713522E+02 |
| CPA | 4.131226E+03 | 2.932877E+03 | 2.409347E+03 | 1.245356E+02 | 2.192308E+03 | 2.812837E+01 |
| HHO | 1.482658E+05 | 9.248775E+04 | 2.802366E+03 | 2.084324E+02 | 2.264632E+03 | 2.386389E+01 |
| RIME | 8.102371E+03 | 6.772024E+03 | 2.285830E+03 | 1.169991E+02 | 2.197692E+03 | 3.982284E+01 |
| | F22 | | F23 | | F24 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **2.265552E+03** | 1.375038E+01 | **2.500000E+03** | 2.412039E-05 | **2.600000E+03** | 2.234190E-13 |
| SCA | 2.469766E+03 | 1.862005E+01 | 3.280074E+03 | 5.293000E+01 | 3.860830E+03 | 7.261677E+01 |
| WOA | 2.434663E+03 | 4.591771E+01 | 3.152556E+03 | 1.486188E+02 | 2.826444E+03 | 4.640745E+02 |
| DE | 2.312139E+03 | 9.803931E+00 | 2.873856E+03 | 1.165131E+01 | 3.396319E+03 | 7.143519E+00 |
| SSA | 2.302442E+03 | 2.696481E+01 | 2.899683E+03 | 4.591268E+01 | 2.600461E+03 | 1.754068E+00 |
| PSO | 2.424424E+03 | 2.981972E+01 | 4.664458E+03 | 5.424066E+02 | 2.667356E+03 | 4.579105E+00 |
| MFO | 2.397898E+03 | 4.542225E+01 | 2.955630E+03 | 3.010570E+01 | 3.492602E+03 | 4.045296E+01 |
| GWO | 2.294783E+03 | 2.927255E+01 | 2.885035E+03 | 4.101812E+01 | 3.028401E+03 | 3.720900E+02 |
| BA | 2.490033E+03 | 5.712484E+01 | 3.541796E+03 | 2.055549E+02 | 2.846532E+03 | 4.871311E+02 |
| CS | 2.344756E+03 | 2.801735E+01 | 2.913872E+03 | 2.132308E+01 | 2.869770E+03 | 2.713068E+02 |
| FA | 2.445136E+03 | 1.294870E+01 | 3.103677E+03 | 1.621235E+01 | 3.691747E+03 | 1.601300E+01 |
| CPA | 2.313971E+03 | 2.535603E+01 | **2.500000E+03** | 0.000000E+00 | **2.600000E+03** | 0.000000E+00 |
| HHO | 2.428248E+03 | 2.689161E+01 | **2.500000E+03** | 0.000000E+00 | **2.600000E+03** | 0.000000E+00 |
| RIME | 2.288698E+03 | 1.933506E+01 | 2.875461E+03 | 1.894064E+01 | 3.218714E+03 | 3.332643E+02 |
| | F25 | | F26 | | F27 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **2.700000E+03** | 3.014049E-06 | **2.800000E+03** | 7.966475E-13 | 2.917703E+03 | 9.696226E+01 |
| SCA | 3.620928E+03 | 1.263230E+02 | 7.789088E+03 | 1.025837E+03 | 4.061770E+03 | 1.175104E+02 |

(*Continued on next page*)

**Table 9. *Continued***

| | F25 | | F26 | | F27 | |
|---|---|---|---|---|---|---|
| | AVG | STD | AVG | STD | AVG | STD |
| WOA | 2.717575E+03 | 9.626031E+01 | 3.542405E+03 | 1.941872E+03 | 3.968988E+03 | 2.352272E+02 |
| DE | 2.912818E+03 | 4.515466E+00 | 5.310729E+03 | 3.259103E+02 | 3.440492E+03 | 2.054747E+01 |
| SSA | 2.956126E+03 | 4.508026E+01 | 2.803337E+03 | 1.825686E+01 | 3.603933E+03 | 8.474287E+01 |
| PSO | 2.953735E+03 | 3.770860E+01 | 3.409442E+03 | 3.172684E+01 | 5.031577E+03 | 7.931823E+02 |
| MFO | 3.699591E+03 | 7.905559E+02 | 6.711784E+03 | 5.120422E+02 | 3.590907E+03 | 6.526928E+01 |
| GWO | 3.212052E+03 | 1.861504E+02 | 4.866584E+03 | 9.148374E+02 | 3.681051E+03 | 9.477657E+01 |
| BA | 3.013904E+03 | 8.224810E+01 | 5.309482E+03 | 3.576900E+03 | 3.906917E+03 | 1.275351E+02 |
| CS | 2.907161E+03 | 8.298785E+00 | 3.789000E+03 | 1.166225E+03 | 3.509616E+03 | 7.246362E+01 |
| FA | 4.108952E+03 | 1.520293E+02 | 7.293080E+03 | 1.459334E+02 | 3.902377E+03 | 8.965520E+01 |
| CPA | **2.700000E+03** | 0.000000E+00 | **2.800000E+03** | 0.000000E+00 | **2.900000E+03** | 0.000000E+00 |
| HHO | **2.700000E+03** | 0.000000E+00 | **2.800000E+03** | 0.000000E+00 | **2.900000E+03** | 0.000000E+00 |
| RIME | 2.956893E+03 | 4.093764E+01 | 5.213364E+03 | 8.405995E+02 | 3.566066E+03 | 7.106036E+01 |
| | F28 | | F29 | | F30 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | 3.034850E+03 | 8.227287E+01 | 3.279330E+03 | 8.123477E+01 | 5.583571E+03 | 2.871580E+03 |
| SCA | 5.650824E+03 | 5.471978E+02 | 4.271864E+03 | 2.801958E+02 | 7.855927E+06 | 1.863115E+07 |
| WOA | 3.308537E+03 | 6.514672E+02 | 4.361104E+03 | 4.091531E+02 | 1.826287E+06 | 1.767812E+06 |
| DE | 3.941911E+03 | 6.995067E+02 | 3.466897E+03 | 8.981603E+01 | 6.628119E+04 | 2.352216E+04 |
| SSA | 3.305292E+03 | 3.583736E+02 | 3.771022E+03 | 1.643812E+02 | 1.179055E+06 | 1.017255E+06 |
| PSO | 3.311009E+03 | 1.132244E+02 | 4.050026E+03 | 2.446224E+02 | 2.561684E+06 | 1.127760E+06 |
| MFO | 4.866595E+03 | 7.300049E+02 | 4.138832E+03 | 2.317888E+02 | 1.941961E+06 | 2.782699E+06 |
| GWO | 3.705572E+03 | 2.929996E+02 | 3.524327E+03 | 1.794326E+02 | 1.259130E+06 | 4.317675E+06 |
| BA | 3.415627E+03 | 5.750275E+02 | 4.654428E+03 | 4.556327E+02 | 1.054830E+06 | 6.501027E+05 |
| CS | 3.228935E+03 | 4.607668E+01 | 3.677880E+03 | 1.126906E+02 | 7.589453E+03 | 1.446341E+03 |
| FA | 4.068826E+03 | 1.044341E+02 | 4.559695E+03 | 1.650738E+02 | 1.171501E+08 | 3.086506E+07 |
| CPA | **3.000000E+03** | 0.000000E+00 | **3.100000E+03** | 0.000000E+00 | **3.200000E+03** | 0.000000E+00 |
| HHO | **3.000000E+03** | 0.000000E+00 | **3.100000E+03** | 0.000000E+00 | **3.200000E+03** | 0.000000E+00 |
| RIME | 3.549195E+03 | 6.610575E+02 | 3.609103E+03 | 1.811955E+02 | 3.838771E+04 | 2.085730E+04 |

$$\begin{aligned} &\textit{Variaable ranges} \quad 0.05 \le x_1 \le 2.00, \\ &\qquad\qquad\qquad\quad 0.25 \le x_2 \le 1.30, \\ &\qquad\qquad\qquad\quad 2.00 \le x_3 \le 15.0 \end{aligned}$$

As shown in Table 14, compared with the other five excellent algorithms, IRIME can achieve better results, with the final optimal value being **0.012665**.

*CBP.* In CBP,[117] the goal is to optimize the performance of the beam while minimizing its weight. There are five variables representing the height of the cross-section, as shown in Figure 12. The mathematical expression of this problem is as follows:

$$\begin{aligned} &\textit{Consider} && \vec{z} = [z_1 z_2 z_3 z_4 z_5] \\ &\textit{Objective function} && f(\vec{z})_{\min} = 0.6224(z_1 + z_2 + z_3 + z_4 + z_5) \\ &\textit{Subject to} && g(\vec{z}) = \frac{61}{z_1^2} + \frac{27}{z_2^2} + \frac{19}{z_3^2} + \frac{7}{z_4^2} + \frac{1}{z_5^2} - 1 \le 0 \\ &\textit{Variable range} && 0.01 \le z_1, z_2, z_3, z_4, z_5 \le 100 \end{aligned}$$

Table 15 shows that IRIME can ultimately achieve **1.339957**, which is unattainable by the other four algorithms compared in CBP.

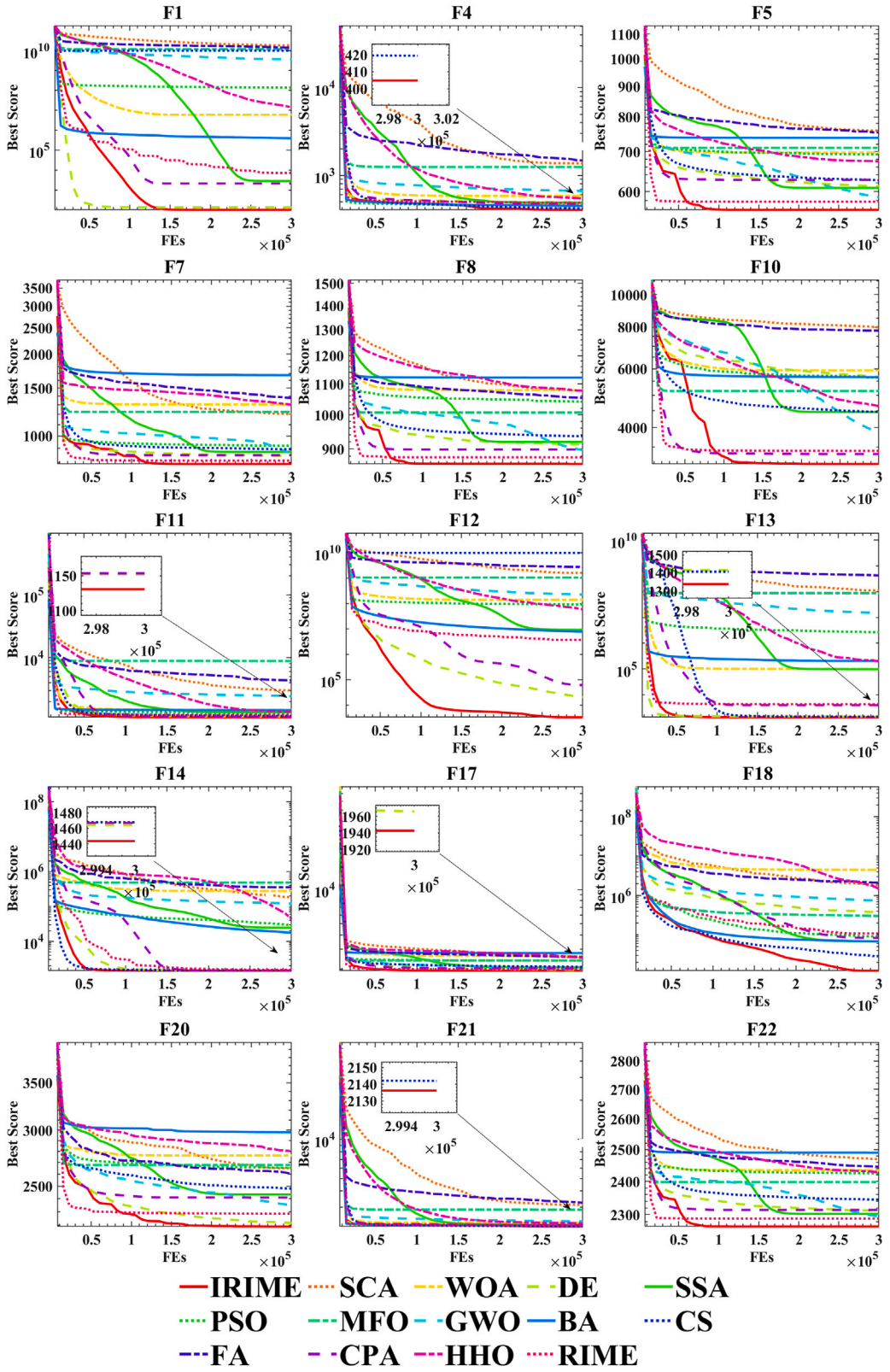



Figure 7. Convergence curves of IRIME and conventional algorithms at IEEE CEC 2017

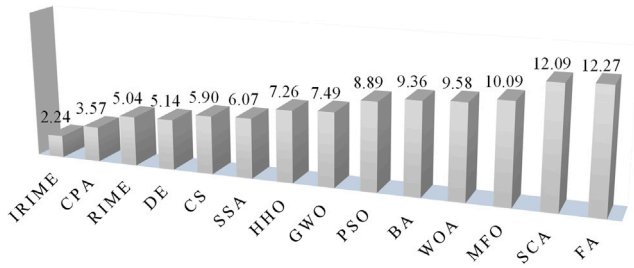



**Figure 8. The Friedman ranking of IRIME and conventional algorithms**

*IBP.* IBP [120] is about reducing vertical displacement in the design process of I-beam. This problem involves four variables as shown in Figure 13. The specific expression of IBP is as follows:

$$\begin{aligned} &\textit{Consider} && \vec{x} = [x_1 x_2 x_3 x_4] \\ &\textit{Objective function} && f(\vec{x})_{\min} = \frac{5000}{\frac{x_3(x_2 - 2x_4)}{12} + \frac{x_1 x_4^3}{6} + 2x_1 x_4 \left(\frac{x_2 - x_4}{2}\right)^2} \\ &\textit{Subject to} && g(\vec{x}) = 2x_1 x_3 + x_3(x_2 - 2x_4) \\ &\textit{Variablerange} && 10 \le x_1 \le 50 \\ &&& 10 \le x_2 \le 80 \\ &&& 0.9 \le x_3 \le 5 \\ &&& 0.9 \le x_4 \le 5 \end{aligned}$$

From Table 16, it can be seen that under the same running environment as RIME, IRIME will have better results. Compared with the other algorithms, IRIME can also achieve very good results, and ultimately IRIME can get **0.013074**.

*BSP.* BSP[122] involves four variables to make the Belleville spring's mass as small as possible while satisfying constraints. This problem involves four variables, as shown in Figure 14. Its mathematical expression is as follows:

$$\textit{Consider} \quad \vec{x} = [x_1 x_2 x_3 x_4] = [D_e D_i t h]$$

$$\begin{aligned} \textit{Objectivefunction} \quad & f(\vec{x})_{\min} = 0.07075 t\pi\left(D_e^2 - D_i^2\right) \\ & K = \frac{D_e}{D_i}, P_{\max} = 5400, \alpha = \left(\frac{K-1}{K}\right)^2 \frac{6}{\pi \ln K}, a = \frac{t}{h}, \\ & \beta = \left(\frac{K-1}{\ln K} - 1\right) \frac{6}{\pi \ln K}, \gamma = \left(\frac{K-1}{2}\right) \frac{6}{\pi \ln K}, E = 30e6Psi \\ & \mu = 0.3, \delta_{\max} = 0.2, S = 200KPsi, D_{\max} = 12.01, H = 2, \delta_l = hf(a) \end{aligned}$$

$$\begin{aligned} \textit{Subjectto} \quad & h_1(\vec{x}) = \frac{4E\delta_{\max}}{\alpha D_e^2 (1-\mu^2)} \left[\gamma t + \beta\left(h - \frac{\delta_{\max}}{2}\right)\right] - S \le 0, \\ & h_2(\vec{x}) = P_{\max} - \frac{4E\delta_{\max}}{\alpha D_e^2 (1-\mu^2)} \left[t(h - \delta_{\max})\left(h - \frac{\delta_{\max}}{2}\right) + t^3\right] \le 0, \\ & h_3(\vec{x}) = \delta_{\max} - \delta_l \le 0, \\ & h_4(\vec{x}) = t + h - H \le 0, \\ & h_5(\vec{x}) = D_e - D_{\max} \le 0, \\ & h_6(\vec{x}) = D_i - D_e \le 0, \\ & h_7(\vec{x}) = \frac{h}{D_e - D_i} - 0.3 \le 0, \end{aligned}$$

Table 10. *p*–value of Wilcoxon signed–rank test between IRIME and conventional algorithms

| | F1 | F2 | F3 | F4 | F5 | F6 |
|---|---|---|---|---|---|---|
| SCA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| WOA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| DE | 5.7516532694E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| SSA | 1.7343976283E-06 | 1.3594767037E-04 | 1.7343976283E-06 | 2.8785992194E-06 | 3.5152372790E-06 | 1.7343976283E-06 |
| PSO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| MFO | 1.7343976283E-06 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| GWO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 2.1630223984E-05 | 1.7343976283E-06 |
| BA | 1.7343976283E-06 | 9.3385961710E-06 | 5.9993592817E-01 | 1.4772761749E-04 | 1.7343976283E-06 | 1.7343976283E-06 |
| CS | 4.3204630578E-08 | 1.7126865599E-06 | 1.7343976283E-06 | 5.4462503972E-02 | 1.9209211049E-06 | 1.7343976283E-06 |
| FA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| CPA | 1.7343976283E-06 | 5.5774268620E-01 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| HHO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| RIME | 1.7343976283E-06 | 7.3432529144E-01 | 1.7343976283E-06 | 2.1266360107E-06 | 6.1564062070E-04 | 1.7343976283E-06 |
| | F7 | F8 | F9 | F10 | F11 | F12 |
| SCA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| WOA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| DE | 1.7343976283E-06 | 1.7343976283E-06 | 1.6976242501E-06 | 1.7343976283E-06 | 2.1630223984E-05 | 1.7343976283E-06 |
| SSA | 1.7343976283E-06 | 3.1816794110E-06 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| PSO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| MFO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| GWO | 2.6033283895E-06 | 3.5152372790E-06 | 1.7343976283E-06 | 8.9187274245E-05 | 1.7343976283E-06 | 1.7343976283E-06 |
| BA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| CS | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 |
| FA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| CPA | 1.9209211049E-06 | 4.7292023374E-06 | 1.7343976283E-06 | 2.0588822306E-01 | 4.5335631776E-04 | 1.7343976283E-06 |
| HHO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| RIME | 1.1973382000E-03 | 4.8602606067E-05 | 1.7343976283E-06 | 1.3974564120E-02 | 3.1816794110E-06 | 1.7343976283E-06 |
| | F13 | F14 | F15 | F16 | F17 | F18 |
| SCA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| WOA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| DE | 3.5008956820E-02 | 1.7343976283E-06 | 8.6121251974E-01 | 3.6826128416E-02 | 4.2766688017E-02 | 1.7343976283E-06 |
| SSA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 4.5335631776E-04 | 8.9443006475E-04 | 1.7343976283E-06 |
| PSO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 3.8821823861E-06 | 2.1266360107E-06 | 1.7343976283E-06 |
| MFO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 2.1266360107E-06 | 1.7343976283E-06 | 1.9209211049E-06 |
| GWO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 6.8713630797E-02 | 1.8462187723E-01 | 1.7343976283E-06 |
| BA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| CS | 1.7343976283E-06 | 2.8785992194E-06 | 1.7518393580E-02 | 4.4493372835E-05 | 1.6394463017E-05 | 2.3534209951E-06 |
| FA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| CPA | 1.7343976283E-06 | 3.5888445045E-04 | 1.9209211049E-06 | 2.8434237746E-05 | 6.4242118722E-03 | 4.2856858692E-06 |
| HHO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 6.3391355731E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| RIME | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 3.3788544377E-03 | 4.4918903765E-02 | 1.7343976283E-06 |
| | F19 | F20 | F21 | F22 | F23 | F24 |
| SCA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| WOA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 3.1250000000E-02 |
| DE | 2.3704477026E-05 | 1.8462187723E-01 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 | 1.7343976283E-06 |

(*Continued on next page*)

Table 10. *Continued*

| | F19 | F20 | F21 | F22 | F23 | F24 |
|---|---|---|---|---|---|---|
| SSA | 1.7343976283E-06 | 8.4660816904E-06 | 2.1266360107E-06 | 6.3391355731E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| PSO | 1.7343976283E-06 | 1.9209211049E-06 | 4.9915540124E-03 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| MFO | 8.4660816904E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| GWO | 1.7343976283E-06 | 2.5967125848E-05 | 1.7343976283E-06 | 6.8922902968E-05 | 1.7343976283E-06 | 1.3183388898E-04 |
| BA | 1.7343976283E-06 | 1.7343976283E-06 | 4.9915540124E-03 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| CS | 4.4918903765E-02 | 1.9209211049E-06 | 8.5895825870E-02 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| FA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| CPA | 5.3069919381E-05 | 2.3704477026E-05 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.0000000000E+00 |
| HHO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.0000000000E+00 |
| RIME | 3.4052567233E-05 | 2.9574621307E-03 | 6.9837831475E-06 | 1.1499217544E-04 | 1.7343976283E-06 | 1.7343976283E-06 |
| | F25 | F26 | F27 | F28 | F29 | F30 |
| SCA | 1.7343976283E-06 | 2.5630832507E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| WOA | 5.3710937500E-02 | 1.2500000000E-01 | 1.7343976283E-06 | 4.1652148500E-01 | 1.7343976283E-06 | 2.3534209951E-06 |
| DE | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 2.1266360107E-06 | 1.7343976283E-06 |
| SSA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| PSO | 1.7343976283E-06 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| MFO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7300371293E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| GWO | 2.5630832507E-06 | 8.2980993064E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 5.2164934470E-06 | 1.7343976283E-06 |
| BA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 2.3534209951E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| CS | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 3.8821823861E-06 | 1.7343976283E-06 | 9.6265892907E-04 |
| FA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| CPA | 1.9531250000E-03 | 1.0000000000E+00 | 1.7343976283E-06 | 1.7191759139E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| HHO | 1.9531250000E-03 | 1.0000000000E+00 | 1.7343976283E-06 | 1.7191759139E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| RIME | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 2.1266360107E-06 |

$$\textit{Variaableranges} \quad 1 \le R, R_0, Q \le 16, \quad 1e-6 \le \mu \le 16e-6,$$

As shown in Table 17, in the BSP, IRIME can achieve excellent results with **1.979675**.

## The experiments for feature selection

In this section, a variant form of IRIME, called bIRIME, was introduced. bIRIME was tested for feature selection using a K-nearest neighbor classifier. The results indicate that bIRIME outperforms IRIME in both low-dimensional and high-dimensional datasets. This superiority is primarily observed in having fewer feature subsets while maintaining higher accuracy levels. To reduce bias in the experiment, this paper follows the same experiment steps and validation methods as those researchers.[125,126]

### *Simulation experiments*

In this study, we conducted experiments using both UCI datasets[127] and the SBCB machine learning library microarray datasets.[128] Tables 18, 19, and 20 demonstrate our selection of 12 high-dimensional datasets and 12 low-dimensional datasets from the UCI dataset collection. Among the low-dimensional datasets, the categories range from 2-class to 7-class classifications. The features vary from 11 to 326, and the sample sizes range from 73 to 6598. The high-dimensional datasets primarily consist of medical gene expression data such as Colon, Leukemia, and Lung_Cancer. These datasets typically exhibit numerous features with relatively fewer samples. Due to the high feature count, noise and filtration often lead to insufficient classification accuracy, making feature selection critical. To further demonstrate bIRIME's performance in high-dimensional datasets, we utilized SBCB microarray data and selected 12 high-dimensional data. These datasets possess a substantial number of features, ranging from 22,277 to 54,675. To comprehensively display bIRIME's performance, it was compared against bMFO,[129] bGWO,[130] bSMA,[131] bALO,[132] BBA,[133] BSSA,[134] bWOA,[135] and bHHO.[136] The dimension size depends on the dataset's dimensionality, and the population size is set to 20. The parameters involved in these algorithms are detailed in Table 21.

Figure 15 vividly illustrates the feature selection process. Data preprocessing is conducted, and necessary preparations are made for cross-validation. Subsequently, bIRIME is employed to update the population and select pertinent features. Finally, the chosen features are utilized

**Table 11. Comparison of IRIME with advanced algorithms**

| | F1 | | F2 | | F3 | |
|---|---|---|---|---|---|---|
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **1.000000E+02** | 6.154959E-07 | 1.010703E+04 | 1.303957E+04 | **3.000649E+02** | 5.911368E-02 |
| EBOwithCMAR | **1.000000E+02** | 7.463907E-15 | 2.456040E+21 | 1.345229E+22 | 2.400802E+04 | 3.386547E+04 |
| LSHADE_cnEpSi | **1.000000E+02** | 1.355622E-09 | 2.409411E+21 | 9.065340E+21 | 3.000060E+02 | 1.496863E-02 |
| ALCPSO | 1.090225E+03 | 1.626553E+03 | 1.080386E+19 | 4.265054E+19 | 2.652894E+04 | 4.688938E+03 |
| CLPSO | 1.064836E+02 | 1.516141E+01 | 6.521139E+13 | 1.787106E+14 | 8.761694E+03 | 2.317848E+03 |
| LSHADE | **1.000000E+02** | 3.369109E-14 | 3.545959E+12 | 1.851603E+13 | 5.041967E+03 | 1.485133E+04 |
| SADE | **1.000000E+02** | 4.856153E-06 | **2.000000E+02** | 0.000000E+00 | 3.224915E+02 | 1.121460E+02 |
| JADE | **1.000000E+02** | 2.447205E-14 | 4.353151E+11 | 2.250183E+12 | 2.152864E+03 | 4.413241E+03 |
| RCBA | 1.865639E+04 | 6.844783E+03 | 2.203667E+02 | 2.627767E+01 | 3.005611E+02 | 1.705823E-01 |
| EPSO | 6.291322E+02 | 8.551875E+02 | 1.672571E+13 | 6.510333E+13 | 5.967043E+03 | 1.583147E+03 |
| CBA | 1.418221E+05 | 7.288431E+05 | 1.079177E+04 | 1.420128E+04 | 3.142839E+02 | 5.455682E+00 |
| LWOA | 5.842765E+05 | 1.266102E+05 | 5.940121E+05 | 8.651440E+05 | 3.226856E+02 | 7.158850E+00 |
| | F4 | | F5 | | F6 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | 4.073550E+02 | 2.050446E+01 | 5.660927E+02 | 1.848911E+01 | 6.000037E+02 | 1.694265E-03 |
| EBOwithCMAR | **4.002521E+02** | 8.386622E-01 | **5.236547E+02** | 6.702270E+00 | 6.001585E+02 | 2.583089E-01 |
| LSHADE_cnEpSi | 4.055120E+02 | 1.717506E+01 | 5.380104E+02 | 1.063740E+01 | 6.017437E+02 | 1.478015E+00 |
| ALCPSO | 5.308335E+02 | 5.904290E+01 | 6.161281E+02 | 2.700655E+01 | 6.053601E+02 | 5.947913E+00 |
| CLPSO | 4.711778E+02 | 2.235273E+01 | 5.533829E+02 | 9.235232E+00 | **6.000000E+02** | 7.313105E-14 |
| LSHADE | 4.045201E+02 | 1.653684E+01 | 5.353546E+02 | 9.099230E+00 | 6.002333E+02 | 2.393252E-01 |
| SADE | 4.329790E+02 | 3.720693E+01 | 5.492836E+02 | 1.048001E+01 | 6.000063E+02 | 3.290097E-02 |
| JADE | 4.042788E+02 | 1.628497E+01 | 5.336508E+02 | 8.884047E+00 | **6.000000E+02** | 0.000000E+00 |
| RCBA | 4.743177E+02 | 3.834853E+01 | 7.518372E+02 | 5.131458E+01 | 6.655776E+02 | 1.003048E+01 |
| EPSO | 4.548814E+02 | 5.265618E+01 | 6.600886E+02 | 2.641726E+01 | 6.000003E+02 | 5.045327E-04 |
| CBA | 5.134138E+02 | 3.745535E+01 | 7.559000E+02 | 5.956804E+01 | 6.661605E+02 | 1.013116E+01 |
| LWOA | 5.100688E+02 | 4.313991E+01 | 7.240533E+02 | 4.400702E+01 | 6.517513E+02 | 9.359289E+00 |
| | F7 | | F8 | | F9 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | 7.870448E+02 | 1.584953E+01 | 8.597845E+02 | 1.429043E+01 | **9.000000E+02** | 1.461630E-11 |
| EBOwithCMAR | **7.617788E+02** | 1.086112E+01 | **8.241612E+02** | 6.468610E+00 | 1.010946E+03 | 1.865535E+02 |
| LSHADE_cnEpSi | 7.834569E+02 | 1.377100E+01 | 8.383194E+02 | 9.682021E+00 | 1.216868E+03 | 3.280355E+02 |
| ALCPSO | 8.599645E+02 | 3.415210E+01 | 8.992732E+02 | 3.062454E+01 | 1.812602E+03 | 6.849134E+02 |
| CLPSO | 7.839056E+02 | 6.953906E+00 | 8.472955E+02 | 7.448280E+00 | 9.157469E+02 | 7.964412E+00 |
| LSHADE | 7.815275E+02 | 1.571445E+01 | 8.335645E+02 | 9.503744E+00 | 9.902918E+02 | 7.764578E+01 |
| SADE | 7.749352E+02 | 1.226535E+01 | 8.473600E+02 | 9.523944E+00 | 9.273077E+02 | 3.178418E+01 |
| JADE | 7.650131E+02 | 8.822871E+00 | 8.336426E+02 | 7.420753E+00 | 9.052861E+02 | 1.104424E+01 |
| RCBA | 2.010604E+03 | 2.965324E+02 | 1.136632E+03 | 6.332926E+01 | 9.032123E+03 | 2.911425E+03 |
| EPSO | 9.549782E+02 | 1.501968E+01 | 9.418434E+02 | 4.081173E+01 | 9.061145E+02 | 7.897042E+00 |
| CBA | 2.101250E+03 | 3.264006E+02 | 1.133645E+03 | 4.797118E+01 | 9.231007E+03 | 2.672835E+03 |
| LWOA | 1.120801E+03 | 8.984370E+01 | 1.063691E+03 | 5.696527E+01 | 8.045517E+03 | 2.647910E+03 |

(*Continued on next page*)

**Table 11.** *Continued*

| | F10 | | F11 | | F12 | |
|---|---|---|---|---|---|---|
| | AVG | STD | AVG | STD | AVG | STD |
| | F10 | | F11 | | F12 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | 3.605941E+03 | 4.881660E+02 | **1.139898E+03** | 1.331120E+01 | 3.089471E+03 | 8.489044E+02 |
| EBOwithCMAR | **2.525258E+03** | 3.088550E+02 | 1.286875E+03 | 9.828048E+01 | 2.880533E+03 | 4.158699E+02 |
| LSHADE_cnEpSi | 2.624054E+03 | 2.002851E+02 | 1.318372E+03 | 8.695477E+01 | 2.866196E+03 | 4.327095E+02 |
| ALCPSO | 3.954499E+03 | 6.489345E+02 | 1.229450E+03 | 7.662722E+01 | 1.033384E+04 | 1.464053E+04 |
| CLPSO | 2.880675E+03 | 2.651593E+02 | 1.142415E+03 | 1.084536E+01 | 9.469498E+05 | 1.654078E+06 |
| LSHADE | 2.752554E+03 | 4.173304E+02 | 1.270254E+03 | 6.932573E+01 | 2.949696E+03 | 4.558499E+02 |
| SADE | 2.879776E+03 | 3.917646E+02 | 1.200905E+03 | 4.515427E+01 | 4.726730E+03 | 2.647403E+03 |
| JADE | 2.623302E+03 | 2.383964E+02 | 1.212654E+03 | 7.928624E+01 | **2.766356E+03** | 3.864192E+02 |
| RCBA | 5.680930E+03 | 9.100438E+02 | 1.398023E+03 | 1.096345E+02 | 3.280932E+06 | 1.606840E+06 |
| EPSO | 6.247404E+03 | 5.886293E+02 | 1.167354E+03 | 3.521652E+01 | 7.818073E+03 | 1.446235E+04 |
| CBA | 5.949017E+03 | 7.647499E+02 | 1.484663E+03 | 1.029935E+02 | 4.628344E+07 | 2.003444E+07 |
| LWOA | 4.946551E+03 | 7.285654E+02 | 1.335759E+03 | 6.241386E+01 | 2.401700E+07 | 1.462717E+07 |
| | F13 | | F14 | | F15 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **1.337193E+03** | 1.082334E+01 | **1.446440E+03** | 1.111946E+01 | **1.573745E+03** | 6.698064E+01 |
| EBOwithCMAR | 2.309903E+03 | 6.775576E+02 | 1.700054E+03 | 1.581119E+02 | 1.608253E+03 | 5.932891E+01 |
| LSHADE_cnEpSi | 3.344365E+03 | 7.163644E+02 | 1.759389E+03 | 1.762675E+02 | 1.636242E+03 | 6.554775E+01 |
| ALCPSO | 2.184079E+03 | 1.462129E+03 | 1.544889E+03 | 7.325523E+01 | 1.647693E+03 | 1.043166E+02 |
| CLPSO | 1.395122E+03 | 6.029177E+01 | 2.709409E+03 | 1.346364E+03 | 1.591891E+03 | 4.846005E+01 |
| LSHADE | 1.638179E+03 | 4.617010E+02 | 1.632836E+03 | 1.126944E+02 | 1.603103E+03 | 6.605244E+01 |
| SADE | 1.407279E+03 | 6.616253E+01 | 1.557814E+03 | 7.421909E+01 | 1.676438E+03 | 2.594247E+02 |
| JADE | 1.348047E+03 | 3.333131E+01 | 1.582810E+03 | 1.344301E+02 | 1.609297E+03 | 7.385639E+01 |
| RCBA | 7.265341E+04 | 6.402215E+04 | 4.696516E+03 | 2.785539E+03 | 5.711193E+04 | 4.796568E+04 |
| EPSO | 3.093509E+03 | 1.468986E+03 | 2.316170E+03 | 7.585056E+02 | 4.753825E+03 | 2.553358E+03 |
| CBA | 9.975372E+04 | 8.820378E+04 | 3.683850E+04 | 2.700027E+04 | 1.212623E+05 | 9.118596E+04 |
| LWOA | 1.022851E+05 | 8.880648E+04 | 6.616154E+03 | 4.192332E+03 | 5.954205E+04 | 5.615821E+04 |
| | F16 | | F17 | | F18 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | 2.188709E+03 | 2.197289E+02 | 1.961515E+03 | 8.368609E+01 | 1.701529E+04 | 1.205528E+04 |
| EBOwithCMAR | 1.993310E+03 | 1.574016E+02 | 1.868368E+03 | 5.873619E+01 | 1.060575E+04 | 3.319645E+04 |
| LSHADE_cnEpSi | 2.006091E+03 | 1.640479E+02 | 1.900019E+03 | 1.040879E+02 | 2.028030E+03 | 7.770539E+01 |
| ALCPSO | 2.375960E+03 | 2.883184E+02 | 2.135919E+03 | 1.673665E+02 | 3.233300E+05 | 3.245044E+05 |
| CLPSO | 2.064265E+03 | 1.272278E+02 | 1.909224E+03 | 3.344437E+01 | 1.218734E+05 | 8.742301E+04 |
| LSHADE | 2.053002E+03 | 2.111169E+02 | 1.876846E+03 | 6.557391E+01 | **1.997827E+03** | 7.261495E+01 |
| SADE | 2.026547E+03 | 1.726456E+02 | **1.808450E+03** | 3.439651E+01 | 1.079344E+04 | 6.893049E+03 |
| JADE | **1.948349E+03** | 1.473068E+02 | 1.853893E+03 | 8.751333E+01 | 1.589941E+04 | 5.056289E+04 |
| RCBA | 3.142629E+03 | 4.326378E+02 | 2.802676E+03 | 3.220768E+02 | 7.429505E+04 | 5.466366E+04 |
| EPSO | 2.041652E+03 | 2.649307E+02 | 1.931363E+03 | 7.198191E+01 | 1.677313E+05 | 7.516923E+04 |
| CBA | 3.304683E+03 | 5.172750E+02 | 3.035659E+03 | 3.673759E+02 | 1.371818E+05 | 1.114421E+05 |
| LWOA | 2.845689E+03 | 3.162336E+02 | 2.368900E+03 | 1.988964E+02 | 3.029696E+05 | 2.802489E+05 |

(*Continued on next page*)

Table 11. *Continued*

| | F19 | | F20 | | F21 | |
|---|---|---|---|---|---|---|
| | AVG | STD | AVG | STD | AVG | STD |
| | F19 | | F20 | | F21 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | 1.959884E+03 | 6.476027E+01 | 2.155432E+03 | 5.634312E+01 | 2.135293E+03 | 3.274789E+01 |
| EBOwithCMAR | 2.027494E+03 | 5.613013E+01 | 2.191757E+03 | 7.615655E+01 | **2.100907E+03** | 1.563620E+00 |
| LSHADE_cnEpSi | 2.099848E+03 | 7.203404E+01 | 2.181353E+03 | 7.953380E+01 | 2.126027E+03 | 3.902697E+01 |
| ALCPSO | 7.603640E+03 | 6.793607E+03 | 2.447291E+03 | 1.420227E+02 | 2.202135E+03 | 3.840544E+01 |
| CLPSO | **1.955767E+03** | 2.567386E+01 | 2.196185E+03 | 6.826542E+01 | 2.175023E+03 | 1.811541E+01 |
| LSHADE | 2.040357E+03 | 7.342819E+01 | 2.205063E+03 | 8.215525E+01 | 2.112883E+03 | 2.853235E+01 |
| SADE | 2.255521E+03 | 4.320845E+02 | **2.107392E+03** | 5.241777E+01 | 2.173371E+03 | 2.708825E+01 |
| JADE | 2.136178E+03 | 4.362812E+02 | 2.150406E+03 | 5.833396E+01 | 2.120825E+03 | 3.842830E+01 |
| RCBA | 7.723117E+03 | 4.296416E+03 | 2.964025E+03 | 2.131662E+02 | 2.188202E+03 | 2.425089E+01 |
| EPSO | 4.506391E+03 | 3.811221E+03 | 2.311569E+03 | 1.462999E+02 | 2.181575E+03 | 2.455679E+01 |
| CBA | 4.467737E+05 | 3.324954E+05 | 2.944890E+03 | 2.356614E+02 | 2.227517E+03 | 3.350044E+01 |
| LWOA | 9.466100E+04 | 6.146595E+04 | 2.609928E+03 | 2.214594E+02 | 2.215128E+03 | 3.341065E+01 |
| | F22 | | F23 | | F24 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | 2.259506E+03 | 1.682965E+01 | **2.500000E+03** | 1.948270E-05 | **2.600000E+03** | 2.234190E-13 |
| EBOwithCMAR | **2.227091E+03** | 7.221436E+00 | 2.832506E+03 | 1.693732E+01 | 2.601079E+03 | 5.909505E+00 |
| LSHADE_cnEpSi | 2.246370E+03 | 1.094491E+01 | 2.906475E+03 | 5.638362E+01 | 2.826014E+03 | 3.289360E+02 |
| ALCPSO | 2.302378E+03 | 2.860447E+01 | 3.068606E+03 | 2.059907E+02 | 3.123298E+03 | 4.347163E+02 |
| CLPSO | 2.253704E+03 | 7.132959E+00 | 2.841313E+03 | 8.614610E+00 | 2.636610E+03 | 1.413640E+02 |
| LSHADE | 2.239367E+03 | 1.035199E+01 | 2.837869E+03 | 1.690434E+01 | 3.322111E+03 | 2.046145E+02 |
| SADE | 2.252534E+03 | 1.170676E+01 | 2.839123E+03 | 2.000302E+01 | 2.600000E+03 | 0.000000E+00 |
| JADE | 2.236709E+03 | 6.770296E+00 | 2.826564E+03 | 1.041841E+01 | 2.889271E+03 | 3.212827E+02 |
| RCBA | 2.497327E+03 | 6.821186E+01 | 3.666152E+03 | 3.324807E+02 | 2.963443E+03 | 6.156478E+02 |
| EPSO | 2.326335E+03 | 4.395933E+01 | 2.842157E+03 | 1.645687E+01 | 2.600237E+03 | 1.299120E+00 |
| CBA | 2.508071E+03 | 7.161797E+01 | 3.530369E+03 | 2.816484E+02 | 2.823170E+03 | 5.107849E+02 |
| LWOA | 2.437273E+03 | 5.166642E+01 | 3.091381E+03 | 9.601964E+01 | 2.740857E+03 | 3.580514E+02 |
| | F25 | | F26 | | F27 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **2.707107E+03** | 3.892667E+01 | **2.800000E+03** | 8.775726E-13 | **2.918588E+03** | 1.018113E+02 |
| EBOwithCMAR | 2.964937E+03 | 4.444758E+01 | 2.867373E+03 | 1.035210E+02 | 3.630655E+03 | 9.660895E+01 |
| LSHADE_cnEpSi | 2.988170E+03 | 5.291435E+01 | 4.396978E+03 | 1.077869E+03 | 3.733814E+03 | 1.371558E+02 |
| ALCPSO | 2.979627E+03 | 4.998191E+01 | 5.062450E+03 | 1.458940E+03 | 3.817345E+03 | 2.051964E+02 |
| CLPSO | 2.906255E+03 | 8.572836E+00 | 3.874735E+03 | 8.128199E+02 | 3.509834E+03 | 2.967095E+01 |
| LSHADE | 2.927482E+03 | 3.570898E+01 | 4.835357E+03 | 3.439051E+02 | 3.534700E+03 | 6.625671E+01 |
| SADE | 3.022715E+03 | 4.584665E+01 | 2.800000E+03 | 3.377779E-13 | 3.474498E+03 | 3.861081E+01 |
| JADE | 2.934809E+03 | 3.928321E+01 | 4.005817E+03 | 9.835249E+02 | 3.535885E+03 | 7.435108E+01 |
| RCBA | 2.998403E+03 | 1.137192E+02 | 6.186807E+03 | 3.541301E+03 | 3.886796E+03 | 1.714723E+02 |
| EPSO | 2.971111E+03 | 5.719272E+01 | 3.387140E+03 | 9.647026E+02 | 3.544143E+03 | 9.782839E+01 |
| CBA | 3.019975E+03 | 1.188225E+02 | 5.897695E+03 | 3.939380E+03 | 3.908455E+03 | 1.994220E+02 |
| LWOA | 2.754015E+03 | 1.012658E+02 | 4.201834E+03 | 2.352678E+03 | 3.835631E+03 | 1.280903E+02 |

(*Continued on next page*)

**Table 11. *Continued***

| | F28 | | F29 | | F30 | |
|---|---|---|---|---|---|---|
| | AVG | STD | AVG | STD | AVG | STD |
| | F28 | | F29 | | F30 | |
| | AVG | STD | AVG | STD | AVG | STD |
| IRIME | **3.005807E+03** | 3.180635E+01 | **3.264609E+03** | 7.599340E+01 | 5.323413E+03 | 2.519083E+03 |
| EBOwithCMAR | 3.220699E+03 | 5.279574E+01 | 3.314116E+03 | 8.246615E+01 | 6.601682E+03 | 4.179781E+03 |
| LSHADE_cnEpSi | 3.278647E+03 | 5.287670E+01 | 3.400352E+03 | 1.112338E+02 | 4.480632E+03 | 4.524749E+02 |
| ALCPSO | 3.419074E+03 | 4.964794E+02 | 3.767528E+03 | 2.488379E+02 | 9.323689E+04 | 1.782215E+05 |
| CLPSO | 3.280910E+03 | 1.712080E+01 | 3.358882E+03 | 6.498981E+01 | 1.531679E+04 | 6.821514E+03 |
| LSHADE | 3.273343E+03 | 6.553523E+01 | 3.396029E+03 | 1.152126E+02 | **4.770491E+03** | 2.514803E+03 |
| SADE | 3.273393E+03 | 3.954162E+01 | 3.269617E+03 | 4.273034E+01 | 8.067340E+03 | 2.739648E+03 |
| JADE | 3.273310E+03 | 3.944772E+01 | 3.368281E+03 | 9.272310E+01 | 5.674743E+03 | 7.892053E+03 |
| RCBA | 3.405828E+03 | 4.996942E+02 | 4.703163E+03 | 3.687816E+02 | 3.265553E+05 | 2.072308E+05 |
| EPSO | 3.282789E+03 | 3.920240E+01 | 3.322935E+03 | 8.698164E+01 | 5.535669E+03 | 2.114442E+03 |
| CBA | 3.609872E+03 | 8.361012E+02 | 4.929752E+03 | 4.003558E+02 | 1.991735E+06 | 1.063884E+06 |
| LWOA | 3.362448E+03 | 6.242544E+02 | 3.872058E+03 | 3.305692E+02 | 1.342063E+06 | 6.054879E+05 |

for classification, and the average of the best results from the 10-fold cross-validation is taken as the evaluation metric (average fitness value, average error rate, and average number of selected features).

The average of fitness is specifically shown in Table 22, the average number of selected feature subsets is in Table 23, and the average error rate is in Table 24. It's evident from the data in the tables that bIRIME outperforms all other algorithms across all 36 datasets. When observed closely, the average number of selected feature subsets is the smallest on each dataset, accompanied by the lowest average classification error rate—achieving first place in all rankings. On the low-dimensional datasets, bIRIME exhibits outstanding performance. Based on the fitness values, bIRIME yields exceptional results, especially on medical datasets like wdbc and Dermatology. bIRIME can select an average of 4.2 and 6.7 features, with the fewest selected features and remarkably low error rates—outperforming all other algorithms. Some datasets even achieve 100% accuracy, such as Dermatology, IonosphereEW, JPNdata, and penglungEW. Additionally, the STD of bIRIME is extremely low, indicating its stability. To visualize the performance of bIRIME, convergence curves are plotted in Figure 16. These curves indicate that bIRIME converges to better results across the 12 low-dimensional datasets. Furthermore, several high-dimensional microarray medical datasets, such as Leukemia, Brain_Tumor1, and Brain_Tumor2 were considered for an in-depth analysis of high-dimensional data. Even with thousands of features, bIRIME can reduce these datasets to two-digit figures, while other algorithms stay around 4,000 to 5,000 features. The best-performing bGWO still has significantly more features than bIRIME, and bIRIME maintains low classification error rates. For instance, in

**Table 12. Comparison results between IRIME and advanced algorithms**

| Overall Rank | | | |
|---|---|---|---|
| Algorithm | Rank | +/ = /- | Avg |
| IRIME | **1** | ~ | **3.533333** |
| EBOwithCMAR | 3 | 15/4/11 | 4.066667 |
| LSHADE_cnEpSi | 6 | 15/5/10 | 5.6 |
| ALCPSO | 9 | 29/1/0 | 8.8 |
| CLPSO | 7 | 20/4/6 | 5.666667 |
| LSHADE | 5 | 16/3/11 | 4.766667 |
| SADE | 4 | 15/6/9 | 4.5 |
| JADE | 2 | 14/3/13 | 3.6 |
| RCBA | 11 | 29/0/1 | 9.933333 |
| EPSO | 8 | 26/2/2 | 7 |
| CBA | 12 | 29/1/0 | 10.9 |
| LWOA | 10 | 30/0/0 | 9.5 |

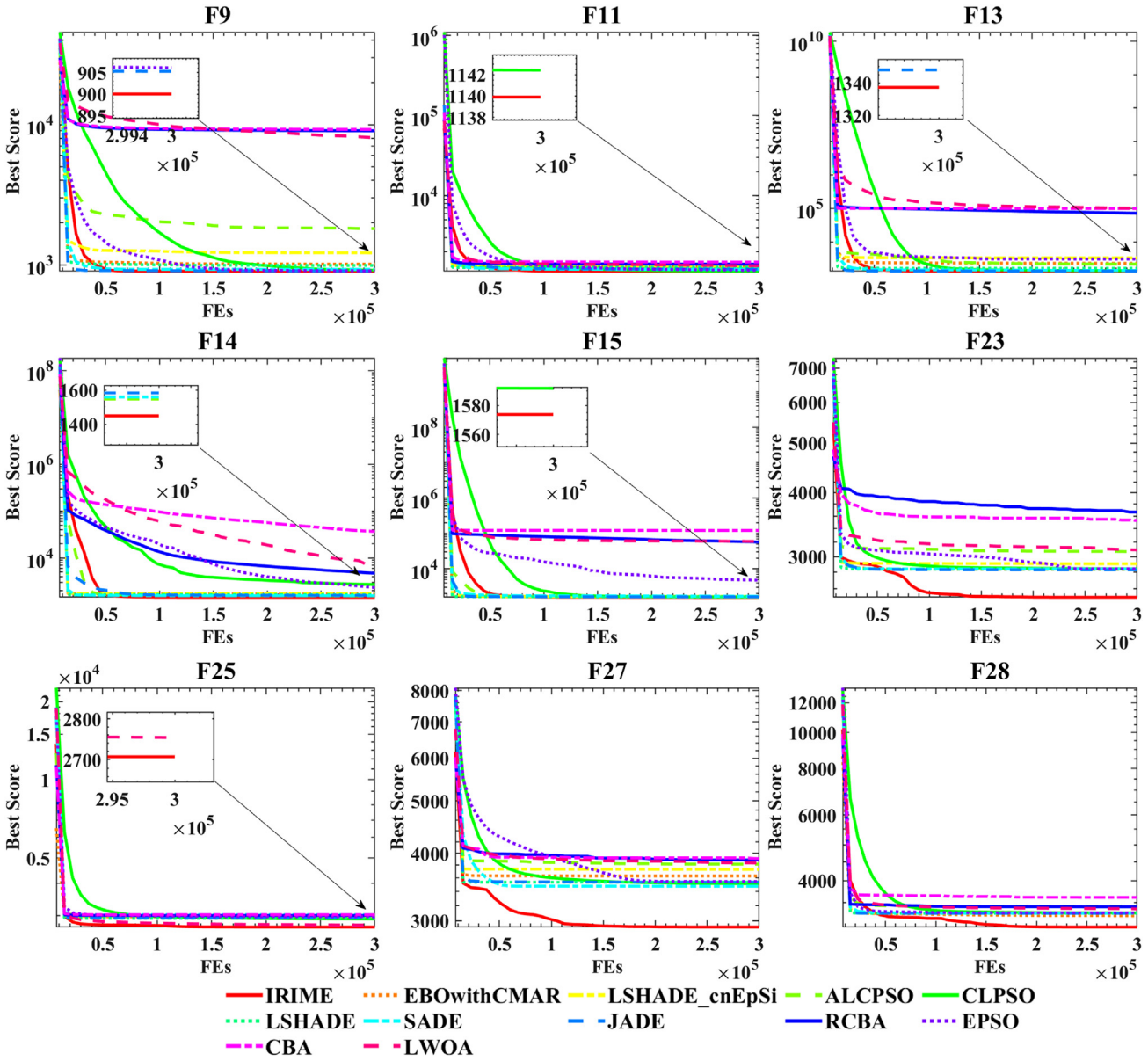



**Figure 9. Convergence curves of IRIME and advanced algorithms at IEEE CEC 2017**

Tumors_9 and Tumors_11, which are multi-classification problems, bIRIME reduces the dataset features to around 611.8 and 1310, respectively, while maintaining error rates at approximately 0.05 and 0.02. Corresponding convergence curve graphs are presented in Figure 17, highlighting bIRIME's superior fitness values compared to other algorithms, demonstrating its competitive edge. To further demonstrate the algorithm's performance on high-dimensional medical microarray gene expression datasets, 12 datasets from SBCB were selected, ranging from dimensions 22277 to 54675. bIRIME drastically reduces the dataset features, decreasing from over ten thousand features, while other algorithms struggle, to just a few dozen features. Particularly in low-sample, binary classification problems such as Liver_GSE14520_U133_2 and Lung_GSE7670, Lung_GSE63459, bIRIME achieves an outstanding error rate of 0. Moreover, the convergence curve graph in Figure 18 vividly illustrates bIRIME's significantly superior performance compared to other algorithms. IRIME's relatively balanced exploration and exploitation capabilities enhance its performance, making it more suitable for complex feature selection problems and particularly prominent in high-dimensional feature selection. Additionally, combining the binary version of IRIME in wrapper-based feature selection, the wrapper-based feature selection method can select a feature set that maximizes model performance by comprehensively considering the interrelationships between features and optimizing the final predictive model. This ultimately leads to improved model performance.[137,138] In conclusion, bIRIME demonstrates excellent results across 12 low-dimensional datasets and 24 high-dimensional medical

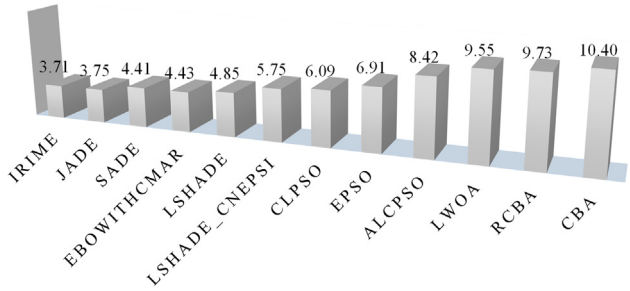



**Figure 10. The Friedman ranking of IRIME and advanced algorithms**

gene expression datasets, ranking first among the compared algorithms. It significantly reduces the number of features while maintaining algorithm accuracy. Overall, bIRIME exhibits excellent performance in feature selection.

### *Time cost*

Figure 19 shows the time each algorithm takes with low-dimensional data, from which it can be seen that bIRIME takes slightly more time, primarily due to the additional computation of fitness values, increasing the algorithm's time, but from the figure, it can be observed that the time increase is not very noticeable. Figure 20 displays the time each algorithm takes with high-dimensional data, from which it is discernible that bIRIME has a higher time cost, with a significant difference. It is worth mentioning that Figures 17 and 18 show that bIRIME can converge at the 2/5 mark of the convergence curve in almost all high-dimensional datasets, such as Colon, Leukemia2, Lung_Cancer, Tumors_9, Tumors_11, and Leukemia1. This indicates that when bIRIME achieves peak performance, the time difference with other algorithms is not as extensive as indicated in Figure 20. To sum it up, while bIRIME does come with a time cost, it indeed manages to enhance performance.

## Conclusions and future directions

RIME is an emerging metaheuristic algorithm that suffers from issues like imbalanced exploration and exploitation, making it prone to local optima. To address these limitations, this paper introduces a variant of RIME, incorporating SB to facilitate inter-population information exchange, enhancing population diversity and subsequently bolstering exploration capabilities. The addition of composite mutation strategy and restart strategy further amplifies RIME's exploitation ability and equips it with the capability to escape local optima. To evaluate IRIME's performance, the study utilizes the IEEE CEC 2017 benchmark functions. In the initial analysis of historical trajectories, the search process of IRIME can be preliminarily obtained. The average fitness value shows that the function of CMS-RS allows RIME to reach a better fitness value quickly, and in the late stage of the population, IRIME can jump out of the local optimal situation. The convergence curve can initially show that IRIME is superior to RIME, and it can also be seen from the one-dimensional trajectory analysis that IRIME will have a broader search in the early and late stages compared to RIME. From the balance analysis, it can be seen that IRIME has an advantage over RIME in terms of balance. IRIME has solved the problem of the weak exploration ability of RIME. From the diversity analysis, it can be seen that the population diversity of IRIME is more abundant, allowing IRIME to explore a broader space and providing the ability to jump out locally. This is also the role that SB plays in it. The stability analysis shows that IRIME expands to more dimensions, and its stability is better than that of RIME. When compared with conventional algorithms, IRIME has significant advantages. In addition, when compared with advanced algorithms, which include EBOwithCMAR, LSHADE_cnEpSi, LSHADE, SADE, and JADE, which performed outstandingly in IEEE CEC 2017, IRIME also has an advantage. For functions like F1, IRIME can converge around the optimal theoretical value, and IRIME also has an advantage in composite functions. However, compared to excellent algorithms such as JADE and EBOwithCMAR, IRIME shows some disadvantages in multimodal and hybrid functions, which is a direction for future improvement. Nevertheless, the overall performance of IRIME is still the first to be compared to these advanced algorithms. To verify the performance of IRIME in practical applications, IRIME was applied to four engineering problems: TCSP, IBP, BSP, and CBP. This demonstrated that IRIME also has good application potential and performance in practical engineering directions. Additionally, the paper introduces bIRIME, a binary version validated across 12 UCI low-dimensional datasets and 24 high-dimensional medical datasets. The validation underscores bIRIME's substantial potential in addressing feature selection problems, notably in high-dimensional datasets, where it significantly reduces feature numbers and improves KNN classification accuracy.

Moving forward, the focus remains on refining the algorithm IRIME and exploring its application in various domains, including engineering optimization, multi-objective optimization, and image segmentation. There are many engineering design problems with complex conditions that need to be optimized, including multi-objective engineering design problems. Moreover, the multi-objective problem is also a problem studied by many researchers, which includes some workshop scheduling and other economic problems. In addition, medical image segmentation is also a focal point of research. While solving the existing problems of IRIME, such as poor performance on multimodal functions, the potential application of IRIME to these problems will be explored. In addition, we urgently need to verify the potential of IRIME in real-time applications.

**Table 13. *p*–value of Wilcoxon signed–rank test between IRIME and advanced algorithms**

| | F1 | F2 | F3 | F4 | F5 | F6 |
|---|---|---|---|---|---|---|
| EBOwithCMAR | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.4772761749E-04 | 1.7343976283E-06 | 6.8922902968E-05 |
| LSHADE_cnEpSi | 1.7343976283E-06 | 1.9209211049E-06 | 1.0246327833E-05 | 5.7096495243E-02 | 8.4660816904E-06 | 1.7343976283E-06 |
| ALCPSO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 3.5152372790E-06 | 1.7343976283E-06 |
| CLPSO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 2.4147041360E-03 | 1.7343976283E-06 |
| LSHADE | 1.7343976283E-06 | 2.3534209951E-06 | 2.7652741970E-03 | 1.9645813713E-03 | 1.0246327833E-05 | 2.3534209951E-06 |
| SADE | 3.1123151154E-05 | 5.6061165275E-06 | 1.0201069525E-01 | 3.8811142877E-04 | 8.1877534396E-05 | 5.3069919381E-05 |
| JADE | 1.7343976283E-06 | 5.3069919381E-05 | 5.7096495243E-02 | 2.5967125848E-05 | 2.1266360107E-06 | 1.7343976283E-06 |
| RCBA | 1.7343976283E-06 | 3.1816794110E-06 | 1.7343976283E-06 | 3.8821823861E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| EPSO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.8909720230E-04 | 1.7343976283E-06 | 1.9209211049E-06 |
| CBA | 1.7343976283E-06 | 9.5899017214E-01 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| LWOA | 1.7343976283E-06 | 9.3156585911E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| | F7 | F8 | F9 | F10 | F11 | F12 |
| EBOwithCMAR | 1.2380795615E-05 | 1.7343976283E-06 | 1.7343976283E-06 | 2.1266360107E-06 | 1.7343976283E-06 | 3.1849060258E-01 |
| LSHADE_cnEpSi | 3.7093530785E-01 | 1.6394463017E-05 | 1.7343976283E-06 | 3.5152372790E-06 | 1.7343976283E-06 | 5.7164578367E-01 |
| ALCPSO | 1.7343976283E-06 | 3.4052567233E-05 | 1.7343976283E-06 | 5.4462503972E-02 | 2.6033283895E-06 | 4.2856858692E-06 |
| CLPSO | 3.7093530785E-01 | 3.3172583108E-04 | 1.7343976283E-06 | 7.6908593028E-06 | 3.8203416302E-01 | 1.7343976283E-06 |
| LSHADE | 1.9861020995E-01 | 4.7292023374E-06 | 1.7343976283E-06 | 3.8821823861E-06 | 1.7343976283E-06 | 7.8126371015E-01 |
| SADE | 2.7652741970E-03 | 4.5335631776E-04 | 1.7343976283E-06 | 1.6394463017E-05 | 2.6033283895E-06 | 6.1564062070E-04 |
| JADE | 1.9729484516E-05 | 1.7343976283E-06 | 1.3601107968E-05 | 2.3534209951E-06 | 4.8602606067E-05 | 2.7029156618E-02 |
| RCBA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| EPSO | 1.7343976283E-06 | 3.1816794110E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 8.9443006475E-04 | 5.7924461898E-05 |
| CBA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| LWOA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 3.8821823861E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| | F13 | F14 | F15 | F16 | F17 | F18 |
| EBOwithCMAR | 1.7343976283E-06 | 1.7343976283E-06 | 3.1617649367E-03 | 8.9443006475E-04 | 2.6134309227E-04 | 3.5888445045E-04 |
| LSHADE_cnEpSi | 1.7343976283E-06 | 1.7343976283E-06 | 3.3172583108E-04 | 2.2551238908E-03 | 1.1748106348E-02 | 1.7343976283E-06 |
| ALCPSO | 1.7343976283E-06 | 1.7343976283E-06 | 4.4493372835E-05 | 6.4242118722E-03 | 4.0715116266E-05 | 1.7343976283E-06 |
| CLPSO | 1.7343976283E-06 | 1.7343976283E-06 | 2.0671113567E-02 | 1.6565526979E-02 | 6.8358564253E-03 | 1.9209211049E-06 |
| LSHADE | 1.7343976283E-06 | 1.7343976283E-06 | 3.1617649367E-03 | 2.8485956185E-02 | 4.5335631776E-04 | 1.7343976283E-06 |
| SADE | 3.4052567233E-05 | 1.7343976283E-06 | 2.4519030932E-01 | 1.8325799472E-03 | 1.7343976283E-06 | 1.1748106348E-02 |
| JADE | 1.6502656562E-01 | 6.8922902968E-05 | 2.7029156618E-02 | 7.5136622549E-05 | 4.8969007053E-04 | 6.1564062070E-04 |
| RCBA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 | 2.1266360107E-06 |
| EPSO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 2.1826721596E-02 | 1.7790738330E-01 | 1.7343976283E-06 |
| CBA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| LWOA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 4.2856858692E-06 | 3.1816794110E-06 | 1.7343976283E-06 |
| | F19 | F20 | F21 | F22 | F23 | F24 |
| EBOwithCMAR | 2.2248266458E-04 | 6.2682812500E-02 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 | 1.0000000000E+00 |
| LSHADE_cnEpSi | 1.3601107968E-05 | 1.3059163494E-01 | 2.8021440811E-01 | 2.9574621307E-03 | 1.7343976283E-06 | 1.8712393203E-06 |
| ALCPSO | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 3.1816794110E-06 | 1.7343976283E-06 | 5.4553418355E-06 |
| CLPSO | 3.8203416302E-01 | 3.1617649367E-03 | 6.3197567644E-05 | 2.4519030932E-01 | 1.7343976283E-06 | 1.7343976283E-06 |
| LSHADE | 2.6134309227E-04 | 6.5641143410E-02 | 1.5927015021E-03 | 4.8602606067E-05 | 1.7343976283E-06 | 1.7333066442E-06 |
| SADE | 7.6908593028E-06 | 6.4242118722E-03 | 1.2505680433E-04 | 8.5895825870E-02 | 1.7343976283E-06 | 1.0000000000E+00 |
| JADE | 1.0569503498E-04 | 7.8126371015E-01 | 3.8723026479E-02 | 1.9729484516E-05 | 1.7343976283E-06 | 1.2062023719E-04 |
| RCBA | 1.7343976283E-06 | 1.7343976283E-06 | 3.5152372790E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| EPSO | 1.0246327833E-05 | 8.1877534396E-05 | 9.3156585911E-06 | 1.0246327833E-05 | 1.7343976283E-06 | 5.3372636003E-07 |

(*Continued on next page*)

**Table 13. *Continued***

| | F19 | F20 | F21 | F22 | F23 | F24 |
|---|---|---|---|---|---|---|
| CBA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| LWOA | 1.7343976283E-06 | 1.7343976283E-06 | 1.9209211049E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| | F25 | F26 | F27 | F28 | F29 | F30 |
| EBOwithCMAR | 1.9209211049E-06 | 2.4414062500E-04 | 1.7343976283E-06 | 1.7333066442E-06 | 3.0009891313E-02 | 1.5885549929E-01 |
| LSHADE_cnEpSi | 1.7343976283E-06 | 3.6947913156E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 6.3391355731E-06 | 4.9498046028E-02 |
| ALCPSO | 1.7343976283E-06 | 1.6741785257E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.9209211049E-06 |
| CLPSO | 1.9209211049E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 2.8434237746E-05 | 1.7343976283E-06 |
| LSHADE | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 2.8434237746E-05 | 2.1826721596E-02 |
| SADE | 1.7343976283E-06 | 1.2500000000E-01 | 1.7343976283E-06 | 1.7343976283E-06 | 6.8835929823E-01 | 1.8909720230E-04 |
| JADE | 1.7343976283E-06 | 3.9632291146E-05 | 1.7343976283E-06 | 1.7300371293E-06 | 3.3172583108E-04 | 6.8358564253E-03 |
| RCBA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| EPSO | 1.7343976283E-06 | 2.1320754352E-04 | 1.7343976283E-06 | 1.7343976283E-06 | 2.0671113567E-02 | 4.2843002855E-01 |
| CBA | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |
| LWOA | 2.1630223984E-05 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 | 1.7343976283E-06 |

## Limitations of the study

Of course, the current IRME still has some limitations. First, on the IEEE CEC 2017 test functions, despite reaching first place compared to many algorithms, both convergence accuracy and the capability of multimodal functions need to be improved. Second, for engineering design problems, there exist some unstable situations where it does not always converge to the best value. Third, in feature selection issues, the binary conversion method only tested one, and other conversion functions need to be tested. Moreover, only one type of classifier was used on the classifier, and no tests were conducted on other classifiers. Fourth, in feature selection, its running time would be longer than that of other algorithms because the added mechanism would increase the complexity of RIME. In summary, when applying IRIME to practical problems in situations with different classifiers, IRIME may encounter worse situations than KNN, and there might be an occurrence of overfitting. In addition, the selection of significance level is also an important factor affecting the experiment in this article.

## STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- RESOURCE AVAILABILITY
  - Lead contact

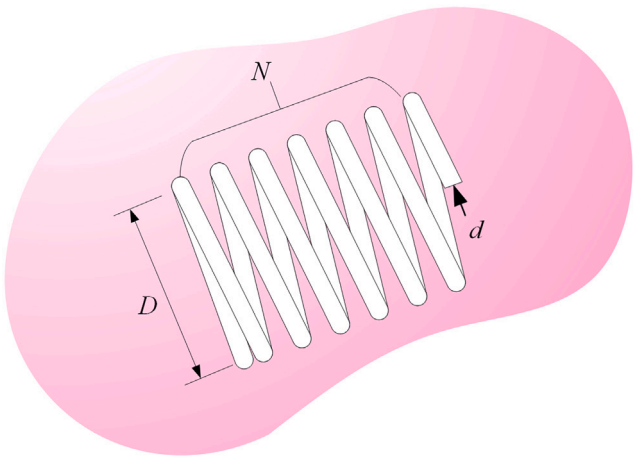



**Figure 11. Schematic diagram of TCSP**

**Table 14. Comparison of IRIME optimization results with literature for the TCSP**

| | Optimal values for variables | | | |
|---|---|---|---|---|
| Methods | *d* | *D* | *N* | Optimum result |
| IRIME | **0.051694** | **0.356844** | **11.28156** | **0.012665** |
| CPSO[112] | 0.015728 | 0.357644 | 11.244543 | 0.0126747 |
| ES[113] | 0.051989 | 0.363965 | 10.890522 | 0.012681 |
| GA[114] | 0.05148 | 0.351661 | 11.632201 | 0.0127048 |
| IHS[115] | 0.051154 | 0.349871 | 12.076432 | 0.0126706 |
| EWOA[116] | 0.051961 | 0.363306 | 10.91296 | 0.012667 |


- Materials availability
- Data and code availability

- METHOD DETAILS
  - RIME
  - The proposed IRIME
  - Framework of proposed IRIME
  - Computational complexity analysis
  - K-nearest neighbor classifier
  - Binary IRIME


## SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.isci.2024.110561.


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## AUTHOR CONTRIBUTIONS

H.J.P.: Writing – Original Draft, Writing – Review and Editing, Software, Visualization, and Investigation. C.Y.: Writing – Original Draft, Writing – Review and Editing, Software, Visualization, Investigation, Conceptualization, Methodology, and Formal Analysis. A.A.H.: Writing – Review and Editing, Software, Visualization, and Investigation. L.L.: Writing – Original Draft, Writing – Review and Editing, Software, Visualization, and Investigation. C.H.L.: Conceptualization, Methodology, Formal Analysis, Investigation, Writing – Review and Editing, Funding

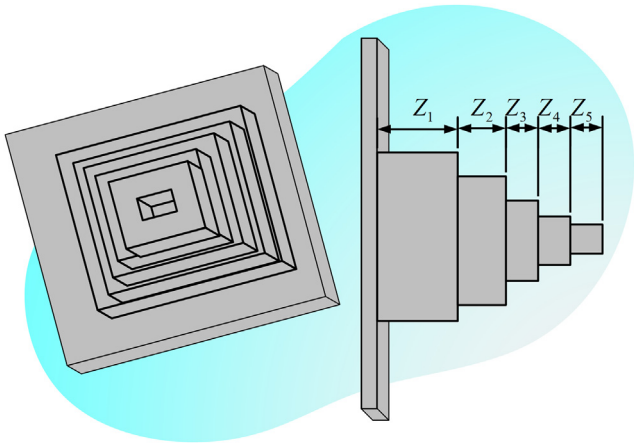



**Figure 12. Schematic diagram of CBP**

**Table 15. Comparison of IRIME optimization results with literature for the CBP**

| | Optimal values for variables | | | | | |
|---|---|---|---|---|---|---|
| Methods | $z_1$ | $z_2$ | $z_3$ | $z_4$ | $z_5$ | Optimum result |
| IRIME | **6.01674** | **5.310676** | **4.49418** | **3.499961** | **2.152106** | **1.339957** |
| CMHHO[117] | 6.012090792 | 5.308024726 | 4.492675534 | 3.50272647 | 2.158177954 | 1.33996575 |
| CS[91] | 6.0089 | 5.3049 | 4.5023 | 3.5077 | 2.1504 | 1.33999 |
| GCA_I[118] | 6.01000 | 5.3000 | 4.49000 | 3.49000 | 2.15000 | 1.34 |
| MFO[119] | 5.984871773 | 5.316726924 | 4.497332586 | 3.513616468 | 2.161620293 | 1.339988086 |

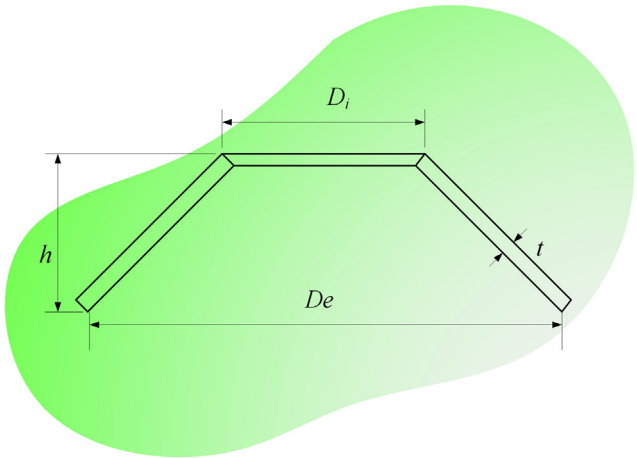



**Figure 13. Schematic diagram of IBP**

Acquisition, Supervision, and Project administration. L.G.X.: Conceptualization, Methodology, Formal Analysis, Investigation, Writing – Review and Editing, Funding Acquisition, Supervision, and Project administration.

## DECLARATION OF INTERESTS

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work the authors didn't use generative AI and AI-assisted technologies in the writing process.

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**Table 16. Comparison of IRIME optimization results with literature for the IBP**

| | Optimal values for variables | | | | |
|---|---|---|---|---|---|
| Methods | $x_1$ | $x_2$ | $x_3$ | $x_4$ | Optimum result |
| IRIME | **50** | **80** | **0.9** | **2.321792** | **0.013074** |
| RIME[21] | 50 | 80 | 0.9 | 2.321676 | 0.01308 |
| SOS[121] | 50 | 80 | 0.9 | 2.3218 | 0.131 |
| ARSM[120] | 37.05 | 80 | 1.71 | 2.31 | 0.0157 |
| CS[119] | 50 | 80 | 0.9 | 2.3217 | 0.131 |

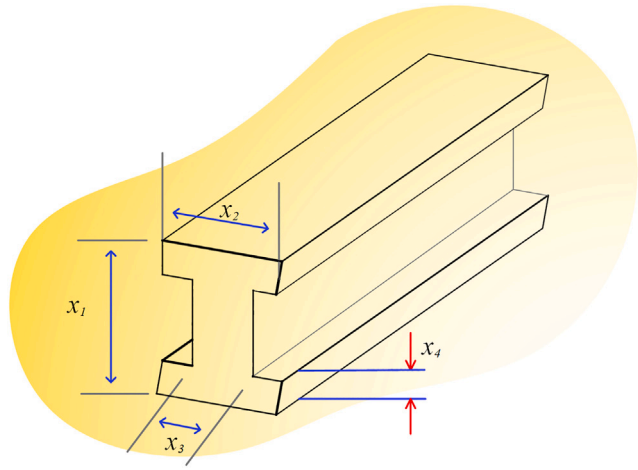



**Figure 14. Schematic diagram of BSP**

**Table 17. Comparison of IRIME optimization results with literature for the BSP**

| | Optimal values for variables | | | | |
|---|---|---|---|---|---|
| Methods | $x_1$ | $x_2$ | $x_3$ | $x_4$ | Optimum result |
| IRIME | 12.01 | 10.03047 | 0.204143 | 0.2 | 1.979675 |
| AGOA[123] | 12.0019 | 10.02024 | 0204158 | 0.20013 | 1.980342 |
| Gene AS I[124] | 11.627 | 9.354 | 0.205 | 0.201 | 2.018070 |
| Gene AS II[124] | 11.499 | 9.268 | 0.210 | 0.204 | 2.162560 |

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**Table 18. Characteristics of UCI low dimensional datasets**

| Datasets | Samples | Features | Classes | Datasets | Samples | Features | Classes |
|---|---|---|---|---|---|---|---|
| WineEW | 178 | 13 | 3 | segment | 2310 | 19 | 7 |
| BreastEW | 569 | 30 | 2 | penglungEW | 73 | 326 | 7 |
| clean1 | 476 | 166 | 2 | JPNdata | 152 | 11 | 2 |
| clean2 | 6598 | 166 | 2 | semeion | 1593 | 25 | 2 |
| Dermatology | 358 | 34 | 7 | wdbc | 569 | 30 | 2 |
| IonosphereEW | 351 | 34 | 2 | DLBCL | 267 | 22 | 2 |

**Table 19. Characteristics of UCI high dimensional datasets**

| Datasets | Samples | Features | Classes | Datasets | Samples | Features | Classes |
|---|---|---|---|---|---|---|---|
| Colon | 62 | 2000 | 2 | Prostate_Tumor | 102 | 10509 | 2 |
| Leukemia2 | 72 | 11225 | 3 | Leukemia | 72 | 7130 | 2 |
| Lung_Cancer | 203 | 12600 | 3 | Brain_Tumor1 | 90 | 5920 | 5 |
| Tumors_9 | 60 | 5726 | 9 | Brain_Tumor2 | 50 | 10367 | 4 |
| Tumors_11 | 174 | 12533 | 11 | CNS | 60 | 7129 | 2 |
| Leukemia1 | 72 | 5327 | 3 | DLBCL | 77 | 5469 | 4 |

**Table 20. Characteristics of SBCB high dimensional datasets**

| Datasets | Samples | Features | Classes | Datasets | Samples | Features | Classes |
|---|---|---|---|---|---|---|---|
| Brain_GSE50161 | 130 | 54675 | 5 | Leukemia_GSE9476 | 64 | 22283 | 5 |
| Brain_GSE15824 | 37 | 54675 | 4 | Leukemia_GSE28497 | 281 | 22283 | 7 |
| Breast_GSE10797 | 66 | 22277 | 3 | Liver_GSE14520_U133_2 | 41 | 22277 | 2 |
| Breast_GSE22820 | 139 | 33579 | 2 | Liver_GSE14520_U133A | 357 | 22277 | 2 |
| Colorectal_GSE44861 | 105 | 22277 | 2 | Lung_GSE7670 | 51 | 22283 | 2 |
| Colorectal_GSE77953 | 55 | 22283 | 4 | Lung_GSE63459 | 65 | 24526 | 2 |


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**Table 21. Parameters of competing algorithms**

| Algorithm | |
|---|---|
| bMFO | $a = 2$; $b = 1$ |
| bGWO | $a = [0,2]$ |
| bSMA | $z = 0.03$ |
| bALO | – |
| BBA | $a = 0.5$; $r = 0.5$ |
| BSSA | – |
| bWOA | $a = [0,2]$; $b = 1$ |
| bHHO | $c = [0,2]$ |

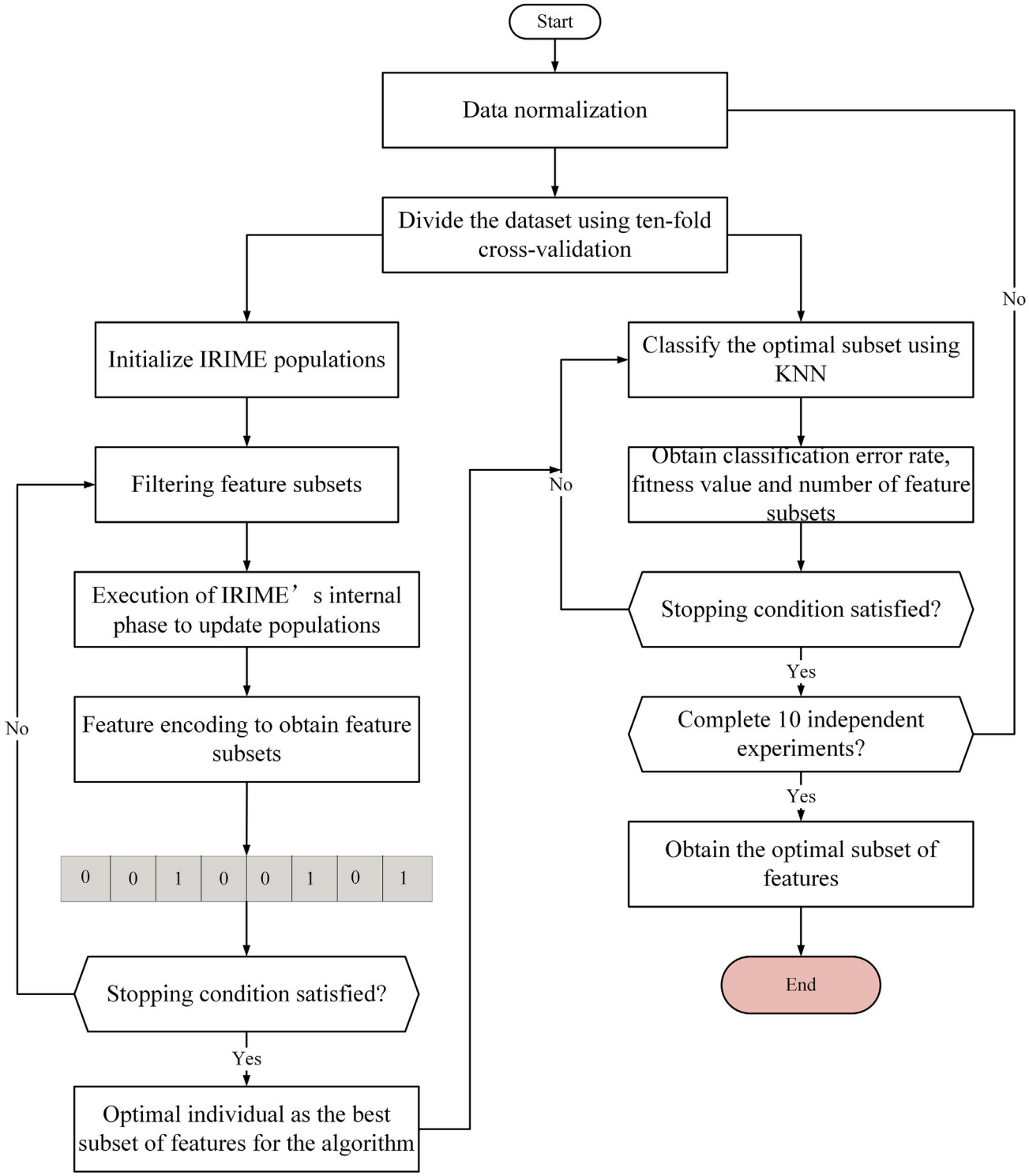



**Figure 15. IRIME for feature selection block diagram**

https://doi.org/10.1016/j.eswa.2021.114864.

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**Table 22. Average fitness value of IRIME and other algorithms**

| Datasets | Metric | bRLRUN | bMFO | bGWO | bSMA | bALO | BBA | BSSA | bWOA | bHHO |
|---|---|---|---|---|---|---|---|---|---|---|
| BreastEW | STD | 7.9815E-03 | 2.5737E-02 | 1.3499E-02 | 1.2733E-02 | 8.1784E-03 | 1.8250E-02 | 8.9557E-03 | 1.7751E-02 | 1.2013E-02 |
| | AVG | **1.4000E-02** | 4.6885E-02 | 1.7667E-02 | 3.5473E-02 | 2.5089E-02 | 6.3953E-02 | 4.0835E-02 | 3.3167E-02 | 3.3696E-02 |
| clean1 | STD | 8.0113E-03 | 3.0677E-02 | 1.3548E-02 | 2.7793E-02 | 1.4896E-02 | 4.3625E-02 | 4.2877E-02 | 2.6929E-02 | 2.3570E-02 |
| | AVG | **9.1950E-03** | 8.7581E-02 | 1.1962E-02 | 6.8613E-02 | 3.8623E-02 | 1.0669E-01 | 6.9863E-02 | 6.7017E-02 | 5.6441E-02 |
| clean2 | STD | 2.0256E-03 | 6.5742E-03 | 5.8958E-03 | 5.9095E-03 | 7.1673E-03 | 8.0765E-03 | 5.6477E-03 | 5.4483E-03 | 2.1300E-02 |
| | AVG | **9.6195E-03** | 5.9140E-02 | 1.4324E-02 | 5.1541E-02 | 3.4053E-02 | 5.4659E-02 | 4.5573E-02 | 4.8422E-02 | 4.3298E-02 |
| Dermatology | STD | 1.5579E-03 | 3.2107E-03 | 7.8873E-03 | 3.9944E-03 | 3.0847E-03 | 2.3074E-02 | 2.7730E-03 | 5.4674E-03 | 9.6025E-03 |
| | AVG | **9.8529E-03** | 2.5147E-02 | 1.2492E-02 | 1.9706E-02 | 1.2059E-02 | 3.6433E-02 | 2.3529E-02 | 2.1471E-02 | 2.0874E-02 |
| IonosphereEW | STD | 1.6695E-03 | 2.5263E-02 | 8.3861E-03 | 1.8966E-02 | 2.5900E-02 | 2.1877E-02 | 2.2691E-02 | 1.0195E-02 | 1.7295E-02 |
| | AVG | **7.6471E-03** | 6.7648E-02 | 1.1244E-02 | 3.5408E-02 | 2.4673E-02 | 7.9065E-02 | 5.1711E-02 | 4.0904E-02 | 2.7185E-02 |
| JPNdata | STD | 4.1973E-02 | 3.9584E-02 | 4.1074E-02 | 3.2128E-02 | 3.2662E-02 | 3.3562E-02 | 4.4824E-02 | 4.7941E-02 | 3.5860E-02 |
| | AVG | **2.7208E-02** | 6.1307E-02 | 5.4042E-02 | 3.4104E-02 | 4.6771E-02 | 1.0361E-01 | 4.6979E-02 | 4.8625E-02 | 4.1342E-02 |
| penglungEW | STD | 3.6117E-04 | 9.0751E-02 | 4.9428E-04 | 6.6442E-02 | 6.6761E-02 | 8.5537E-02 | 6.0959E-02 | 5.6188E-02 | 5.8670E-02 |
| | AVG | **9.5385E-04** | 7.2404E-02 | 1.7077E-03 | 7.1400E-02 | 6.6839E-02 | 7.9025E-02 | 3.8278E-02 | 4.2958E-02 | 4.3981E-02 |
| WineEW | STD | 2.5960E-03 | 3.7157E-03 | 2.5641E-03 | 4.0744E-03 | 3.1404E-03 | 1.6826E-02 | 4.9488E-03 | 4.2327E-03 | 3.3677E-03 |
| | AVG | **1.0385E-02** | 1.6923E-02 | 1.1538E-02 | 1.4231E-02 | 1.1538E-02 | 2.6432E-02 | 1.5000E-02 | 1.5769E-02 | 1.1923E-02 |
| segment | STD | 4.0264E-03 | 7.6491E-03 | 6.8076E-03 | 5.9976E-03 | 5.5373E-03 | 6.3285E-03 | 5.6081E-03 | 6.4091E-03 | 7.8559E-03 |
| | AVG | **2.3472E-02** | 3.7996E-02 | 2.3999E-02 | 3.2470E-02 | 2.6318E-02 | 4.5317E-02 | 3.7174E-02 | 3.2552E-02 | 3.2667E-02 |
| semeion | STD | 1.1349E-03 | 9.1866E-03 | 2.7014E-03 | 5.0273E-03 | 3.2573E-03 | 8.9377E-03 | 6.9273E-03 | 3.7990E-03 | 6.0018E-03 |
| | AVG | **7.5849E-03** | 3.6688E-02 | 8.8553E-03 | 2.9001E-02 | 2.4024E-02 | 3.4877E-02 | 3.2268E-02 | 2.8356E-02 | 2.5773E-02 |
| SpectEW | STD | 2.5801E-02 | 5.0296E-02 | 2.3642E-02 | 3.7956E-02 | 3.1472E-02 | 5.7054E-02 | 3.8109E-02 | 4.2316E-02 | 3.4303E-02 |
| | AVG | **6.4373E-02** | 9.8474E-02 | 6.8878E-02 | 9.3054E-02 | 7.0309E-02 | 1.3453E-01 | 9.1260E-02 | 9.0159E-02 | 9.0684E-02 |
| wdbc | STD | 8.4911E-03 | 1.1968E-02 | 7.7347E-03 | 1.3064E-02 | 1.1282E-02 | 1.7392E-02 | 1.2655E-02 | 1.0739E-02 | 1.0621E-02 |
| | AVG | **8.6667E-03** | 3.1167E-02 | 1.2971E-02 | 2.3393E-02 | 1.4000E-02 | 3.7089E-02 | 2.6863E-02 | 2.4530E-02 | 2.1060E-02 |
| Brain_ GSE15824 | STD | 7.5101E-02 | 2.2845E-01 | 1.6307E-01 | 2.2375E-01 | 2.2070E-01 | 2.3333E-01 | 2.2150E-01 | 2.3873E-01 | 1.7940E-01 |
| | AVG | **2.3760E-02** | 2.7655E-01 | 2.5446E-01 | 3.0016E-01 | 2.2860E-01 | 3.0551E-01 | 2.6169E-01 | 2.3637E-01 | 2.3515E-01 |
| Brain_ GSE50161 | STD | 7.9444E-04 | 1.0133E-01 | 1.3569E-01 | 1.3112E-01 | 1.2586E-01 | 1.5356E-01 | 1.3274E-01 | 1.3839E-01 | 1.4238E-01 |
| | AVG | **6.0062E-04** | 2.9848E-01 | 2.2520E-01 | 2.8648E-01 | 2.9762E-01 | 2.9273E-01 | 2.7187E-01 | 2.6971E-01 | 2.0587E-01 |
| Brain_ Tumor1 | STD | 5.1502E-04 | 5.2538E-02 | 4.7623E-02 | 8.0800E-02 | 5.4370E-02 | 6.8528E-02 | 6.6492E-02 | 6.6683E-02 | 6.6470E-02 |
| | AVG | **7.5000E-04** | 8.7679E-02 | 3.4915E-02 | 8.2239E-02 | 7.4680E-02 | 9.1747E-02 | 6.8237E-02 | 7.5196E-02 | 6.0576E-02 |
| Brain_ Tumor2 | STD | 6.7161E-04 | 9.9264E-02 | 9.0702E-02 | 9.7213E-02 | 1.2296E-01 | 1.2498E-01 | 6.9413E-02 | 1.0473E-01 | 9.0844E-02 |
| | AVG | **4.5722E-04** | 1.3717E-01 | 4.8412E-02 | 1.1624E-01 | 1.1373E-01 | 1.1387E-01 | 4.1818E-02 | 7.6558E-02 | 6.4574E-02 |
| Breast_ GSE10797 | STD | 6.2274E-02 | 1.4917E-01 | 1.2966E-01 | 8.6308E-02 | 1.4945E-01 | 1.4455E-01 | 1.7530E-01 | 2.0822E-01 | 1.6484E-01 |
| | AVG | **2.9855E-02** | 2.7758E-01 | 1.7319E-01 | 2.7517E-01 | 2.9292E-01 | 2.9566E-01 | 2.5801E-01 | 2.6357E-01 | 2.1799E-01 |

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**Table 22.** *Continued*

| Datasets | Metric | bRLRUN | bMFO | bGWO | bSMA | bALO | BBA | BSSA | bWOA | bHHO |
|---|---|---|---|---|---|---|---|---|---|---|
| Breast_ GSE22820 | STD | 4.2233E-06 | 9.6618E-05 | 2.1338E-04 | 8.5628E-05 | 6.9469E-05 | 2.8593E-03 | 9.0516E-03 | 2.5629E-03 | 8.0285E-04 |
| | AVG | **6.8495E-06** | 2.4786E-02 | 5.7423E-03 | 2.4585E-02 | 2.4083E-02 | 1.8268E-02 | 8.8250E-03 | 1.1224E-02 | 5.5063E-03 |
| CNS | STD | 3.0828E-04 | 1.1663E-01 | 2.5095E-04 | 1.1950E-01 | 7.6410E-02 | 1.0827E-01 | 1.1331E-01 | 5.1080E-02 | 1.0908E-01 |
| | AVG | **4.4887E-04** | 2.0079E-01 | 5.9616E-03 | 1.4978E-01 | 7.1066E-02 | 2.0477E-01 | 1.1291E-01 | 1.6154E-01 | 8.6624E-02 |
| Colon | STD | 1.3375E-04 | 2.3220E-01 | 7.9046E-02 | 1.6408E-01 | 1.2419E-01 | 1.6361E-01 | 1.1769E-01 | 1.5715E-01 | 1.0511E-01 |
| | AVG | **2.2000E-04** | 2.1891E-01 | 6.4971E-02 | 2.1131E-01 | 1.7304E-01 | 2.0421E-01 | 1.4447E-01 | 1.5245E-01 | 1.1528E-01 |
| Colorectal_ GSE44861 | STD | 4.7573E-02 | 1.3523E-01 | 8.4068E-02 | 1.0603E-01 | 8.8658E-02 | 1.1494E-01 | 8.8834E-02 | 1.1544E-01 | 1.4981E-01 |
| | AVG | **5.5397E-02** | 2.1172E-01 | 1.6615E-01 | 2.0507E-01 | 2.0234E-01 | 2.2692E-01 | 1.9172E-01 | 1.9228E-01 | 1.8551E-01 |
| Colorectal_ GSE77953 | STD | 9.8112E-05 | 1.4855E-01 | 6.9790E-02 | 1.7012E-01 | 1.1539E-01 | 1.3445E-01 | 1.1875E-01 | 1.1285E-01 | 1.3783E-01 |
| | AVG | **8.8857E-05** | 1.6819E-01 | 3.8405E-02 | 1.1959E-01 | 1.2529E-01 | 1.8073E-01 | 9.1832E-02 | 1.3292E-01 | 1.0747E-01 |
| DLBCL | STD | 6.7466E-05 | 4.2949E-02 | 3.7536E-02 | 3.7527E-02 | 3.7584E-02 | 5.5531E-02 | 4.8047E-03 | 5.4671E-02 | 3.6925E-02 |
| | AVG | **1.0697E-04** | 3.8334E-02 | 1.7209E-02 | 3.5939E-02 | 3.4550E-02 | 4.4220E-02 | 5.0622E-03 | 3.8731E-02 | 1.9087E-02 |
| Leukemia_ GSE9476 | STD | 1.7988E-05 | 7.1088E-02 | 6.2127E-02 | 9.4898E-02 | 7.3056E-02 | 7.3603E-02 | 8.0684E-02 | 7.1131E-02 | 5.7598E-02 |
| | AVG | **3.7248E-05** | 6.8578E-02 | 3.5054E-02 | 7.8771E-02 | 6.9155E-02 | 7.4000E-02 | 6.7760E-02 | 5.5499E-02 | 3.3527E-02 |
| Leukemia_ GSE28497 | STD | 4.0929E-02 | 7.0408E-02 | 4.7071E-02 | 3.3890E-02 | 6.7803E-02 | 4.6150E-02 | 7.7667E-02 | 5.0513E-02 | 6.8309E-02 |
| | AVG | **5.4689E-02** | 1.8486E-01 | 9.5053E-02 | 1.7731E-01 | 1.6724E-01 | 1.8078E-01 | 1.5189E-01 | 1.5928E-01 | 1.3127E-01 |
| Leukemia1 | STD | 8.7998E-04 | 3.6799E-03 | 3.4717E-04 | 4.2325E-04 | 2.0229E-04 | 5.2337E-02 | 5.2211E-03 | 3.7791E-03 | 2.9042E-03 |
| | AVG | **5.6786E-04** | 2.6291E-02 | 5.2431E-03 | 2.4129E-02 | 2.2738E-02 | 4.2726E-02 | 5.6796E-03 | 1.5124E-02 | 7.3925E-03 |
| Leukemia2 | STD | 1.2603E-04 | 5.7290E-02 | 2.1923E-04 | 4.2902E-02 | 2.5406E-04 | 6.8639E-02 | 4.8773E-02 | 4.3369E-02 | 4.4738E-03 |
| | AVG | **1.4477E-04** | 5.1794E-02 | 5.4552E-03 | 3.7868E-02 | 2.3536E-02 | 6.1354E-02 | 2.2450E-02 | 2.5908E-02 | 7.7639E-03 |
| Liver_ GSE14520_ U133_2 | STD | 1.5149E-06 | 7.5132E-02 | 7.5081E-02 | 7.5093E-02 | 7.5094E-02 | 1.2344E-01 | 1.0897E-01 | 7.4785E-02 | 9.0972E-02 |
| | AVG | **3.8156E-06** | 4.8469E-02 | 2.9365E-02 | 4.8297E-02 | 4.7669E-02 | 9.0935E-02 | 5.5549E-02 | 3.5035E-02 | 4.7669E-02 |
| Liver_ GSE14520_ U133A | STD | 1.9078E-02 | 2.1906E-02 | 1.5255E-02 | 4.3383E-02 | 2.1969E-02 | 3.9167E-02 | 2.5007E-02 | 3.3678E-02 | 3.3828E-02 |
| | AVG | **1.1138E-02** | 6.2681E-02 | 3.5134E-02 | 6.1369E-02 | 6.0968E-02 | 5.6843E-02 | 5.3275E-02 | 5.1179E-02 | 4.0948E-02 |
| Lung_Cancer | STD | 1.4101E-03 | 3.0019E-02 | 2.4070E-02 | 3.3055E-02 | 3.2334E-02 | 3.6533E-02 | 3.0129E-02 | 2.7963E-02 | 3.2113E-02 |
| | AVG | **1.6659E-03** | 5.8019E-02 | 2.4535E-02 | 5.2618E-02 | 4.2477E-02 | 5.4925E-02 | 4.2002E-02 | 3.6842E-02 | 2.2758E-02 |
| Lung_ GSE7670 | STD | 6.3289E-06 | 1.2490E-01 | 1.0692E-01 | 8.7083E-02 | 8.5529E-02 | 8.1515E-02 | 9.2001E-02 | 8.7004E-02 | 1.3341E-01 |
| | AVG | **8.5267E-06** | 7.8513E-02 | 5.3048E-02 | 7.8280E-02 | 6.3476E-02 | 6.9025E-02 | 6.9140E-02 | 6.4658E-02 | 8.1411E-02 |
| Lung_ GSE63459 | STD | 4.2794E-02 | 1.4580E-01 | 1.5776E-01 | 1.2649E-01 | 1.9248E-01 | 1.2000E-01 | 1.4982E-01 | 1.9898E-01 | 1.2405E-01 |
| | AVG | **1.3940E-02** | 3.4092E-01 | 1.8239E-01 | 3.0298E-01 | 2.7123E-01 | 3.5521E-01 | 2.3901E-01 | 2.4614E-01 | 2.5828E-01 |
| Prostate_ Tumor | STD | 5.1841E-04 | 7.2804E-02 | 3.0095E-02 | 6.5441E-02 | 8.4864E-02 | 6.9476E-02 | 4.0830E-02 | 6.8358E-02 | 4.9091E-02 |
| | AVG | **5.9663E-04** | 1.0209E-01 | 1.5606E-02 | 7.9993E-02 | 6.8506E-02 | 1.2328E-01 | 3.6777E-02 | 6.5066E-02 | 4.8100E-02 |

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**Table 22.** *Continued*

| Datasets | Metric | bRLRUN | bMFO | bGWO | bSMA | bALO | BBA | BSSA | bWOA | bHHO |
|---|---|---|---|---|---|---|---|---|---|---|
| Tumors_9 | STD | 5.4431E-02 | 8.4402E-02 | 8.8788E-02 | 1.2171E-01 | 1.3037E-01 | 1.5507E-01 | 1.4641E-01 | 9.0359E-02 | 1.2999E-01 |
| | AVG | **3.0789E-02** | 6.6583E-02 | 5.9178E-02 | 1.8182E-01 | 1.5710E-01 | 2.4602E-01 | 1.6895E-01 | 1.6403E-01 | 1.3934E-01 |
| Tumors_11 | STD | 2.2165E-02 | 5.4079E-02 | 2.3720E-02 | 5.8568E-02 | 4.5486E-02 | 5.2819E-02 | 6.3385E-02 | 4.9406E-02 | 5.8841E-02 |
| | AVG | **1.5782E-02** | 9.2876E-02 | 1.7671E-02 | 8.4907E-02 | 7.0923E-02 | 1.1335E-01 | 8.6000E-02 | 7.7339E-02 | 6.4372E-02 |
| Leukemia | STD | 2.0834E-04 | 2.7844E-03 | 3.3545E-04 | 1.7732E-04 | 2.7192E-04 | 4.1094E-03 | 2.8667E-03 | 3.1979E-03 | 1.7701E-03 |
| | AVG | **1.3324E-04** | 2.5518E-02 | 5.4369E-03 | 2.4106E-02 | 2.2933E-02 | 1.7079E-02 | 3.6950E-03 | 1.3627E-02 | 6.6115E-03 |
| | AVG | **1** | 7.97 | 2.46 | 6.51 | 4.86 | 8.31 | 4.94 | 5.09 | 3.83 |
| | Rank | **1** | 8 | 2 | 7 | 4 | 9 | 5 | 6 | 3 |

**Table 23. Average number of features obtained by IRIME and other algorithms**

| Datasets | Metric | bIRIMR | bMFO | bGWO | bSMA | bALO | BBA | BSSA | bWOA | bHHO |
|---|---|---|---|---|---|---|---|---|---|---|
| BreastEW | STD | 1.5776 | 2.5734 | 2.0656 | 2.4967 | 3.1972 | 2.8983 | 2.6352 | 2.9981 | 3.4254 |
| | AVG | **6.4** | 17.2 | 6.6 | 13.3 | 9 | 13.2 | 16.5 | 14.9 | 13.2 |
| clean1 | STD | 5.1088 | 6.8508 | 3.5277 | 6.9769 | 8.4781 | 5.4985 | 23.3771 | 14.1221 | 21.9396 |
| | AVG | **17.1** | 111.4 | 20 | 81.7 | 62.1 | 69.7 | 93.4 | 90.1 | 61.7 |
| clean2 | STD | 2.8597 | 15.8328 | 4.2426 | 6.6072 | 6.168 | 5.9861 | 26.1621 | 20.0544 | 17.9679 |
| | AVG | **26.2** | 90.7 | 27 | 77.9 | 68.6 | 63.5 | 44.7 | 64.2 | 44.8 |
| Dermatology | STD | 1.0593 | 2.1833 | 1.8288 | 2.7162 | 2.0976 | 2.9814 | 1.8856 | 3.7178 | 2.8363 |
| | AVG | **6.7** | 17.1 | 6.7 | 13.4 | 8.2 | 14 | 16 | 14.6 | 12.4 |
| IonosphereEW | STD | 1.1353 | 1.8288 | 1.3166 | 2.9059 | 1.3499 | 2.9515 | 5.2239 | 3.9285 | 4.383 |
| | AVG | **5.2** | 18.3 | 5.8 | 13 | 7.6 | 13.6 | 16.8 | 13.1 | 11.1 |
| JPNdata | STD | 0.78881 | 1.1595 | 0.73786 | 0.73786 | 0.73786 | 1.2293 | 1.1595 | 1.075 | 1.1972 |
| | AVG | **1.8** | 3.7 | 2.1 | 3.1 | 3.1 | 3.2 | 3.3 | 2.6 | 3.1 |
| penglungEW | STD | 2.3476 | 5.9479 | 3.2128 | 7.3907 | 6.0452 | 20.0832 | 51.8739 | 32.2897 | 21.5726 |
| | AVG | **6.2** | 148.4 | 11.1 | 138.2 | 106.1 | 123 | 68.7 | 102.8 | 52.6 |
| WineEW | STD | 0.67495 | 0.96609 | 0.66667 | 1.0593 | 0.8165 | 1.9692 | 1.2867 | 1.1005 | 0.8756 |
| | AVG | **2.7** | 4.4 | 3 | 3.7 | 3 | 5.9 | 3.9 | 4.1 | 3.1 |
| segment | STD | 0.67495 | 1.5811 | 1.1005 | 1.1738 | 0.94281 | 2.8363 | 1.2693 | 1.5239 | 1.792 |
| | AVG | **4.7** | 8.5 | 4.9 | 6.4 | 5 | 7.4 | 8.5 | 6.9 | 7.1 |
| semeion | STD | 6.0148 | 22.1883 | 3.7476 | 7.2793 | 9.7502 | 6.4842 | 40.8799 | 19.3509 | 15.3148 |
| | AVG | **40.2** | 143.9 | 40.6 | 122.1 | 105.2 | 110.4 | 123.5 | 118.7 | 82.9 |
| SpectEW | STD | 1.1353 | 1.8856 | 2.1187 | 1.5811 | 1.5492 | 2.1731 | 3.8586 | 2.0656 | 3.5917 |
| | AVG | **4.8** | 12 | 5.4 | 9.5 | 5.8 | 7.5 | 9 | 9.6 | 8.7 |
| wdbc | STD | 2.2998 | 1.6364 | 1.9322 | 1.8257 | 1.7764 | 2.3664 | 4.4083 | 3.199 | 3.0623 |
| | AVG | **4.2** | 13.7 | 4.8 | 11 | 6.4 | 12.4 | 13.1 | 9.7 | 9.6 |
| Brain_GSE15824 | STD | 11.6065 | 65.3031 | 153.118 | 117.1173 | 112.1333 | 1007.2227 | 10851.9124 | 3637.7805 | 3019.6378 |
| | AVG | **10.4** | 27118.6 | 6424.7 | 26962.3 | 26629.1 | 21879.3 | 12600.6 | 12613.1 | 6085.6 |
| Brain_GSE50161 | STD | 353.9549 | 70.7041 | 85.5706 | 78.8799 | 52.6081 | 484.9491 | 3973.6415 | 1191.9019 | 858.0189 |
| | AVG | **267.6** | 11042.8 | 2581.1 | 10937.4 | 10661.5 | 9041.4 | 6445.4 | 5282.2 | 3038.7 |
| Brain_Tumor1 | STD | 60.9787 | 448.2748 | 57.8624 | 39.2458 | 53.3912 | 158.3239 | 742.5944 | 579.1508 | 936.8074 |
| | AVG | **88.8** | 3132.5 | 634.5 | 2863.3 | 2718.2 | 2343.3 | 955.5 | 1779.5 | 1298.2 |
| Brain_Tumor2 | STD | 139.251 | 85.9613 | 53.1882 | 55.7838 | 59.9941 | 205.6697 | 1105.0287 | 766.008 | 626.9682 |
| | AVG | **94.8** | 5131.7 | 1173.9 | 5060.5 | 4868.8 | 4238.6 | 2104.8 | 2742 | 1570.4 |
| Breast_GSE10797 | STD | 121.6189 | 932.2376 | 117.3364 | 105.8715 | 38.6586 | 319.2611 | 2425.964 | 936.5835 | 569.7187 |
| | AVG | **147.5** | 8388.1 | 1903.5 | 8043.9 | 7783.6 | 6852.1 | 2274.7 | 4538.3 | 2057.1 |
| Breast_GSE22820 | STD | 2.8363 | 64.8867 | 143.2994 | 57.506 | 46.6543 | 1912.4869 | 6078.8883 | 1721.1711 | 539.1781 |
| | AVG | **4.6** | 16645.5 | 3856.4 | 16510.5 | 16173.8 | 13633.8 | 5926.7 | 7538.1 | 3697.9 |
| CNS | STD | 43.9545 | 536.9503 | 35.7802 | 46.8686 | 38.8829 | 208.6891 | 1449.3973 | 523.8594 | 579.0644 |
| | AVG | **64** | 3795.5 | 850 | 3489 | 3360.1 | 2945.4 | 2424.8 | 2586.4 | 1385.7 |
| Colon | STD | 5.35 | 33.8633 | 9.6609 | 17.9938 | 18.2589 | 45.6172 | 301.1333 | 213.3695 | 84.6493 |
| | AVG | **8.8** | 975.5 | 156 | 943 | 859.5 | 802.6 | 259.8 | 579.1 | 268.2 |
| Colorectal_ GSE44861 | STD | 100.6106 | 1663.5714 | 103.2098 | 88.8525 | 70.9926 | 586.2611 | 1695.3644 | 1609.8696 | 1227.7201 |
| | AVG | **55.4** | 11795.4 | 2585.4 | 10947.9 | 10692.2 | 9420.6 | 3714.3 | 5634.7 | 3001.1 |
| Colorectal_ GSE77953 | STD | 43.7244 | 76.2848 | 82.6169 | 93.6872 | 67.4105 | 600.6082 | 3477.4347 | 2159.3564 | 1118.6136 |
| | AVG | **39.6** | 11045.4 | 2600 | 10957.8 | 10677.2 | 8947 | 3830 | 5913.6 | 2734 |

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**Table 23. *Continued***

| Datasets | Metric | bIRIMR | bMFO | bGWO | bSMA | bALO | BBA | BSSA | bWOA | bHHO |
|---|---|---|---|---|---|---|---|---|---|---|
| DLBCL | STD | 7.3794 | 43.2904 | 32.3495 | 27.2334 | 15.943 | 361.8856 | 525.5336 | 358.0402 | 274.3796 |
| | AVG | **11.7** | 2708.5 | 583.4 | 2632.1 | 2480.2 | 2195.1 | 553.7 | 1453.1 | 788.8 |
| Leukemia_ GSE9476 | STD | 8.0166 | 1230.3573 | 76.3036 | 65.7115 | 68.4297 | 346.9749 | 4050.7056 | 1775.9559 | 1037.8521 |
| | AVG | **16.6** | 11409.7 | 2517.7 | 10912 | 10658.8 | 9040.7 | 3988.7 | 5580.7 | 2845 |
| Leukemia_ GSE28497 | STD | 453.0615 | 2093.6994 | 100.9667 | 152.4565 | 86.8179 | 412.9731 | 4868.3445 | 2632.1426 | 1536.9839 |
| | AVG | **1385.7** | 12364.7 | 2884.6 | 11031.1 | 10761.7 | 9353.9 | 7078.2 | 7723.1 | 4167.8 |
| Leukemia1 | STD | 93.7529 | 392.0516 | 36.9871 | 45.0926 | 21.5523 | 302.6517 | 556.252 | 402.6275 | 309.4131 |
| | AVG | **60.5** | 2801 | 558.6 | 2570.7 | 2422.5 | 2072.6 | 605.1 | 1611.3 | 787.6 |
| Leukemia2 | STD | 28.2931 | 32.7102 | 49.2162 | 41.1183 | 57.0369 | 206.3349 | 1892.2037 | 583.1634 | 1004.3603 |
| | AVG | **32.5** | 5534.2 | 1224.7 | 5454.6 | 5283.9 | 4651.1 | 1993.3 | 2769.6 | 1743 |
| Liver_GSE14520_ U133_2 | STD | 0.67495 | 35.2844 | 78.3579 | 58.6402 | 23.9372 | 534.9238 | 4079.327 | 1012.2107 | 302.7592 |
| | AVG | **1.7** | 11013.1 | 2501.8 | 10936.7 | 10656.9 | 8792.2 | 3586.1 | 5027.9 | 2191.7 |
| Liver_GSE14520_ U133A | STD | 84.0806 | 1238.4951 | 198.1242 | 55.69 | 56.9928 | 1228.6671 | 3153.1031 | 1583.0377 | 1396.8465 |
| | AVG | **125** | 11396 | 2617.9 | 10945.6 | 10666.2 | 8725.3 | 3614.3 | 6274.7 | 2892.4 |
| Lung_Cancer | STD | 355.3536 | 727.6364 | 94.772 | 74.8031 | 47.5904 | 213.324 | 3748.5934 | 1691.9311 | 484.3701 |
| | AVG | **419.8** | 6463.9 | 1508.8 | 6185.8 | 5967.2 | 5040 | 3497.6 | 4478.9 | 2126.8 |
| Lung_GSE7670 | STD | 2.8206 | 51.8112 | 57.2539 | 20.6669 | 33.1678 | 578.7427 | 4161.6603 | 1142.9429 | 628.1154 |
| | AVG | **3.8** | 10998.8 | 2472.7 | 10894.7 | 10648.1 | 9097.7 | 4704.6 | 4824.2 | 2411.3 |
| Lung_GSE63459 | STD | 225.7063 | 1871.412 | 105.4702 | 283.9564 | 62.5372 | 527.5762 | 4625.5503 | 1503.2684 | 1285.2096 |
| | AVG | **180.7** | 13004.6 | 2923.2 | 12149.1 | 11831.7 | 10319 | 4344.3 | 6457.6 | 3537.9 |
| Prostate_Tumor | STD | 108.9589 | 1067.7715 | 48.8653 | 91.198 | 61.3753 | 173.2097 | 2300.4042 | 1161.1786 | 585.1257 |
| | AVG | **125.4** | 6029.1 | 1283.4 | 5195.8 | 4959.6 | 4401.4 | 1921.1 | 3873.5 | 2122.9 |
| Tumors_9 | STD | 288.3897 | 516.3061 | 83.0793 | 81.4777 | 48.7785 | 335.6235 | 1163.4018 | 724.2532 | 406.3999 |
| | AVG | **611.8** | 3092 | 689.8 | 2818.8 | 2669.7 | 2230.5 | 2174.7 | 2492 | 1399.7 |
| Tumors_11 | STD | 512.3098 | 1115.9752 | 89.4884 | 81.3853 | 77.025 | 317.9088 | 2373.6164 | 821.3965 | 1469.4393 |
| | AVG | **1310** | 7279.1 | 1618.2 | 6214.7 | 5987.2 | 5091 | 5200.9 | 5493.2 | 3587.2 |
| Leukemia | STD | 29.7097 | 397.0611 | 47.8355 | 25.2861 | 38.7751 | 222.0476 | 408.7861 | 456.0148 | 252.4211 |
| | AVG | **19** | 3638.8 | 775.3 | 3437.5 | 3270.2 | 2832.7 | 526.9 | 1943.2 | 942.8 |
| | AVG | **1** | 8.94 | 2.14 | 7.31 | 5.92 | 5.94 | 4.81 | 5.36 | 3.39 |
| | Rank | **1** | 9 | 2 | 8 | 6 | 7 | 4 | 5 | 3 |

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**Table 24. Classification error rate of IRIME and other algorithms**

| Datasets | Metric | bRLRUN | bMFO | bGWO | bSMA | bALO | BBA | BSSA | bWOA | bHHO |
|---|---|---|---|---|---|---|---|---|---|---|
| BreastEW | STD | 7.3971E-03 | 2.7868E-02 | 1.2267E-02 | 1.3837E-02 | 9.1414E-03 | 3.7780E-02 | 7.4011E-03 | 1.7050E-02 | 1.4456E-02 |
| | AVG | **3.5088E-03** | 1.9177E-02 | 7.0175E-03 | 1.4007E-02 | 1.0620E-02 | 7.0243E-02 | 1.4037E-02 | 8.7719E-03 | 1.2312E-02 |
| clean1 | STD | 8.9776E-03 | 3.3100E-02 | 1.4061E-02 | 2.9450E-02 | 1.3891E-02 | 4.4568E-02 | 4.5523E-02 | 2.7819E-02 | 2.4970E-02 |
| | AVG | **4.2572E-03** | 5.6871E-02 | 6.2500E-03 | 4.6321E-02 | 2.0966E-02 | 1.4685E-01 | 4.3927E-02 | 4.1977E-02 | 3.9849E-02 |
| clean2 | STD | 1.9951E-03 | 8.2843E-03 | 5.6298E-03 | 6.6392E-03 | 5.6738E-03 | 9.0642E-03 | 1.0893E-02 | 5.4336E-03 | 6.6965E-03 |
| | AVG | **1.8189E-03** | 3.3495E-02 | 6.5172E-03 | 2.9555E-02 | 1.4095E-02 | 4.5766E-02 | 3.3800E-02 | 3.0616E-02 | 3.1372E-02 |
| Dermatology | STD | 0.0000E+00 | 0.0000E+00 | 8.7841E-03 | 0.0000E+00 | 0.0000E+00 | 6.9606E-02 | 0.0000E+00 | 0.0000E+00 | 8.7841E-03 |
| | AVG | **0.0000E+00** | 0.0000E+00 | 2.7778E-03 | 0.0000E+00 | 0.0000E+00 | 7.2036E-02 | 0.0000E+00 | 0.0000E+00 | 2.7778E-03 |
| IonosphereEW | STD | 0.0000E+00 | 2.7619E-02 | 9.0351E-03 | 1.9984E-02 | 2.7722E-02 | 3.7861E-02 | 2.6593E-02 | 1.2007E-02 | 1.4765E-02 |
| | AVG | **0.0000E+00** | 4.2880E-02 | 2.8571E-03 | 1.7148E-02 | 1.4206E-02 | 8.5420E-02 | 2.8427E-02 | 2.2778E-02 | 1.1433E-02 |
| JPNdata | STD | 4.2682E-02 | 4.2930E-02 | 4.4618E-02 | 3.1553E-02 | 3.4719E-02 | 1.1073E-01 | 4.4879E-02 | 5.2705E-02 | 3.5206E-02 |
| | AVG | **1.9167E-02** | 4.5060E-02 | 4.5833E-02 | 1.9583E-02 | 3.2917E-02 | 2.2673E-01 | 3.2083E-02 | 3.7500E-02 | 2.7202E-02 |
| penglungEW | STD | 0.0000E+00 | 9.2854E-02 | 0.0000E+00 | 6.9537E-02 | 7.0397E-02 | 1.0462E-01 | 6.2268E-02 | 7.0722E-02 | 6.3115E-02 |
| | AVG | **0.0000E+00** | 9.1468E-02 | 0.0000E+00 | 5.2778E-02 | 5.3175E-02 | 9.1071E-02 | 2.9167E-02 | 6.6468E-02 | 3.7778E-02 |
| WineEW | STD | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 5.7457E-02 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 |
| | AVG | **0.0000E+00** | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 4.4771E-02 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 |
| segment | STD | 4.1069E-03 | 7.8508E-03 | 8.1756E-03 | 5.6994E-03 | 5.6994E-03 | 5.8501E-02 | 3.0269E-03 | 5.8615E-03 | 5.8437E-03 |
| | AVG | **1.1688E-02** | 1.6450E-02 | 1.1688E-02 | 1.6450E-02 | 1.3853E-02 | 6.5801E-02 | 1.5584E-02 | 1.5152E-02 | 1.4719E-02 |
| semeion | STD | 0.0000E+00 | 7.3437E-03 | 2.6518E-03 | 5.9204E-03 | 3.0327E-03 | 1.1005E-02 | 9.5200E-03 | 5.1139E-03 | 5.9432E-03 |
| | AVG | **0.0000E+00** | 1.0039E-02 | 1.2579E-03 | 6.2775E-03 | 4.3947E-03 | 2.5095E-02 | 9.4381E-03 | 6.2736E-03 | 1.0664E-02 |
| SpectEW | STD | 2.6996E-02 | 5.3540E-02 | 9.1270E-02 | 4.0268E-02 | 3.2490E-02 | 6.1361E-02 | 3.8675E-02 | 4.2238E-02 | 3.8752E-02 |
| | AVG | **5.6278E-02** | 7.4949E-02 | 5.9585E-02 | 7.5224E-02 | 6.0134E-02 | 2.2498E-01 | 7.4532E-02 | 7.1937E-02 | 7.4644E-02 |
| wdbc | STD | 5.5479E-03 | 1.2405E-02 | 8.4262E-03 | 1.2017E-02 | 1.1096E-02 | 3.6333E-02 | 1.1878E-02 | 1.2430E-02 | 8.5758E-03 |
| | AVG | **1.7544E-03** | 8.7719E-03 | 5.2329E-03 | 5.3258E-03 | 3.5088E-03 | 4.9373E-02 | 5.2945E-03 | 8.8033E-03 | 5.3258E-03 |
| Brain_GSE15824 | STD | 7.9057E-02 | 2.3100E-01 | 1.7160E-01 | 2.3560E-01 | 2.3233E-01 | 2.4417E-01 | 2.3528E-01 | 2.4930E-01 | 1.9023E-01 |
| | AVG | **2.5000E-02** | 3.1500E-01 | 2.6167E-01 | 2.9000E-01 | 2.1500E-01 | 3.0333E-01 | 2.6333E-01 | 3.3667E-01 | 2.4167E-01 |
| Brain_GSE50161 | STD | 0.0000E+00 | 1.0675E-01 | 1.4288E-01 | 1.3798E-01 | 1.3254E-01 | 1.8059E-01 | 1.3659E-01 | 1.4686E-01 | 1.5008E-01 |
| | AVG | **0.0000E+00** | 2.8810E-01 | 2.3095E-01 | 2.7571E-01 | 2.8810E-01 | 3.3524E-01 | 2.7095E-01 | 2.7143E-01 | 2.0952E-01 |
| Brain_Tumor1 | STD | 0.0000E+00 | 5.2326E-02 | 5.0185E-02 | 8.4984E-02 | 5.7485E-02 | 1.2373E-01 | 7.2208E-02 | 6.6882E-02 | 7.2587E-02 |
| | AVG | **0.0000E+00** | 7.5556E-02 | 3.1111E-02 | 6.1111E-02 | 5.4444E-02 | 1.4111E-01 | 6.3333E-02 | 8.8333E-02 | 5.2222E-02 |
| Brain_Tumor2 | STD | 0.0000E+00 | 1.0436E-01 | 9.5598E-02 | 1.0238E-01 | 1.2959E-01 | 1.9487E-01 | 7.0273E-02 | 1.1434E-01 | 9.6609E-02 |
| | AVG | **0.0000E+00** | 1.1833E-01 | 4.5000E-02 | 9.6667E-02 | 9.5000E-02 | 2.9333E-01 | 3.3333E-02 | 8.6667E-02 | 6.0000E-02 |
| Breast_GSE10797 | STD | 6.5494E-02 | 1.5685E-01 | 1.3664E-01 | 9.0690E-02 | 1.5732E-01 | 1.8436E-01 | 1.8492E-01 | 2.1991E-01 | 1.7343E-01 |
| | AVG | **3.0952E-02** | 2.6524E-01 | 1.7619E-01 | 2.6381E-01 | 2.8333E-01 | 3.5238E-01 | 2.6429E-01 | 2.6286E-01 | 2.2286E-01 |

(*Continued on next page*)

**Table 24. *Continued***

| Datasets | Metric | bRLRUN | bMFO | bGWO | bSMA | bALO | BBA | BSSA | bWOA | bHHO |
|---|---|---|---|---|---|---|---|---|---|---|
| Breast_GSE22820 | STD | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 2.2588E-02 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 |
| | AVG | **0.0000E+00** | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 7.1429E-03 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 |
| CNS | STD | 0.0000E+00 | 1.2298E-01 | 0.0000E+00 | 1.2565E-01 | 8.0508E-02 | 1.8829E-01 | 1.1793E-01 | 5.4758E-02 | 1.1621E-01 |
| | AVG | **0.0000E+00** | 1.8333E-01 | 0.0000E+00 | 1.3190E-01 | 5.0000E-02 | 3.3286E-01 | 1.0095E-01 | 1.5095E-01 | 8.0952E-02 |
| Colon | STD | 0.0000E+00 | 2.4467E-01 | 8.3277E-02 | 1.7281E-01 | 1.3043E-01 | 1.7368E-01 | 1.2267E-01 | 1.6551E-01 | 1.1156E-01 |
| | AVG | **0.0000E+00** | 2.0476E-01 | 6.4286E-02 | 1.9762E-01 | 1.5952E-01 | 2.1190E-01 | 1.4524E-01 | 1.4524E-01 | 1.1429E-01 |
| Colorectal_GSE44861 | STD | 5.0197E-02 | 1.4438E-01 | 8.8567E-02 | 1.1162E-01 | 9.3352E-02 | 1.3306E-01 | 9.5036E-02 | 1.2315E-01 | 1.5787E-01 |
| | AVG | **5.8182E-02** | 1.9500E-01 | 1.6879E-01 | 1.9000E-01 | 1.8773E-01 | 2.3909E-01 | 1.9303E-01 | 1.8909E-01 | 1.8818E-01 |
| Colorectal_GSE77953 | STD | 0.0000E+00 | 1.5646E-01 | 7.3525E-02 | 1.7916E-01 | 1.2151E-01 | 1.3572E-01 | 1.2236E-01 | 1.2062E-01 | 1.4470E-01 |
| | AVG | **0.0000E+00** | 1.5095E-01 | 3.4286E-02 | 1.0000E-01 | 1.0667E-01 | 2.1357E-01 | 8.7619E-02 | 1.2595E-01 | 1.0667E-01 |
| DLBCL | STD | 0.0000E+00 | 4.5175E-02 | 3.9528E-02 | 3.9528E-02 | 3.9528E-02 | 9.6806E-02 | 0.0000E+00 | 5.6626E-02 | 3.9528E-02 |
| | AVG | **0.0000E+00** | 1.4286E-02 | 1.2500E-02 | 1.2500E-02 | 1.2500E-02 | 8.0357E-02 | 0.0000E+00 | 2.6786E-02 | 1.2500E-02 |
| Leukemia_GSE9476 | STD | 0.0000E+00 | 7.3128E-02 | 6.5494E-02 | 9.9887E-02 | 7.6947E-02 | 8.0781E-02 | 8.0312E-02 | 7.3128E-02 | 6.0234E-02 |
| | AVG | **0.0000E+00** | 4.5238E-02 | 3.0952E-02 | 5.7143E-02 | 4.7619E-02 | 7.6190E-02 | 6.1905E-02 | 4.5238E-02 | 2.8571E-02 |
| Leukemia_GSE28497 | STD | 4.3607E-02 | 7.3975E-02 | 4.9652E-02 | 3.5586E-02 | 7.1267E-02 | 7.5138E-02 | 7.8860E-02 | 5.2083E-02 | 7.1357E-02 |
| | AVG | **5.4295E-02** | 1.6539E-01 | 9.3242E-02 | 1.6058E-01 | 1.5062E-01 | 2.0426E-01 | 1.4316E-01 | 1.4942E-01 | 1.2834E-01 |
| Leukemia1 | STD | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 1.0623E-01 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 |
| | AVG | **0.0000E+00** | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 6.2500E-02 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 |
| Leukemia2 | STD | 0.0000E+00 | 6.0234E-02 | 0.0000E+00 | 4.5175E-02 | 0.0000E+00 | 1.0773E-01 | 4.5175E-02 | 4.5175E-02 | 0.0000E+00 |
| | AVG | **0.0000E+00** | 2.8571E-02 | 0.0000E+00 | 1.4286E-02 | 0.0000E+00 | 1.1131E-01 | 1.4286E-02 | 1.4286E-02 | 0.0000E+00 |
| Liver_GSE14520_U133_2 | STD | 0.0000E+00 | 7.9057E-02 | 7.9057E-02 | 7.9057E-02 | 7.9057E-02 | 1.3006E-01 | 1.0541E-01 | 7.9057E-02 | 9.5598E-02 |
| | AVG | **0.0000E+00** | 2.5000E-02 | 2.5000E-02 | 2.5000E-02 | 2.5000E-02 | 7.8333E-02 | 5.0000E-02 | 2.5000E-02 | 4.5000E-02 |
| Liver_GSE14520_U133A | STD | 1.9977E-02 | 2.3361E-02 | 1.6042E-02 | 4.5679E-02 | 2.3189E-02 | 4.3890E-02 | 2.3007E-02 | 3.5082E-02 | 3.4871E-02 |
| | AVG | **1.1429E-02** | 3.9056E-02 | 3.0798E-02 | 3.8739E-02 | 3.8977E-02 | 4.4700E-02 | 4.7540E-02 | 3.9048E-02 | 3.6270E-02 |
| Lung_Cancer | STD | 0.0000E+00 | 3.2547E-02 | 2.5218E-02 | 3.4707E-02 | 3.4035E-02 | 4.5504E-02 | 2.5589E-02 | 2.5986E-02 | 3.4761E-02 |
| | AVG | **0.0000E+00** | 3.4073E-02 | 1.9524E-02 | 2.9549E-02 | 1.9787E-02 | 8.3358E-02 | 2.9603E-02 | 2.0072E-02 | 1.5072E-02 |
| Lung_GSE7670 | STD | 0.0000E+00 | 1.3152E-01 | 1.1249E-01 | 9.1692E-02 | 9.0010E-02 | 8.6353E-02 | 1.0124E-01 | 9.1692E-02 | 1.3984E-01 |
| | AVG | **0.0000E+00** | 5.6667E-02 | 5.0000E-02 | 5.6667E-02 | 4.1667E-02 | 5.3333E-02 | 6.1667E-02 | 5.6667E-02 | 8.0000E-02 |
| Lung_GSE63459 | STD | 4.5175E-02 | 1.5520E-01 | 1.6602E-01 | 1.3330E-01 | 2.0265E-01 | 1.0966E-01 | 1.5199E-01 | 2.0970E-01 | 1.3159E-01 |
| | AVG | **1.4286E-02** | 3.3095E-01 | 1.8571E-01 | 2.9286E-01 | 2.6012E-01 | 4.0238E-01 | 2.4226E-01 | 2.4524E-01 | 2.6429E-01 |
| Prostate_Tumor | STD | 0.0000E+00 | 7.5727E-02 | 3.1623E-02 | 6.8862E-02 | 8.9463E-02 | 1.3870E-01 | 4.6906E-02 | 7.0052E-02 | 5.1640E-02 |
| | AVG | **0.0000E+00** | 7.7273E-02 | 1.0000E-02 | 5.8182E-02 | 4.7273E-02 | 2.0818E-01 | 2.9091E-02 | 4.9091E-02 | 4.0000E-02 |
| Tumors_9 | STD | 5.6626E-02 | 2.7035E-01 | 9.3914E-02 | 1.2790E-01 | 1.3735E-01 | 2.0722E-01 | 1.5429E-01 | 9.2272E-02 | 1.3791E-01 |
| | AVG | **2.6786E-02** | 3.2750E-01 | 5.5952E-02 | 1.6548E-01 | 1.4083E-01 | 5.0810E-01 | 1.5786E-01 | 1.4976E-01 | 1.3381E-01 |

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**Table 24.** *Continued*

| Datasets | Metric | bRLRUN | bMFO | bGWO | bSMA | bALO | BBA | BSSA | bWOA | bHHO |
|---|---|---|---|---|---|---|---|---|---|---|
| Tumors_11 | STD | 2.3424E-02 | 8.7234E-02 | 2.4942E-02 | 6.1668E-02 | 4.7832E-02 | 4.7091E-02 | 6.2645E-02 | 5.3297E-02 | 6.3592E-02 |
| | AVG | **1.1111E-02** | 9.3989E-02 | 1.1806E-02 | 6.3278E-02 | 4.9513E-02 | 1.3793E-01 | 6.8685E-02 | 5.8342E-02 | 5.2696E-02 |
| Leukemia | STD | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 7.5190E-02 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 |
| | AVG | **0.0000E+00** | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 | 5.7738E-02 | 0.0000E+00 | 0.0000E+00 | 0.0000E+00 |
| | AVG | **1** | 6.39 | 2.33 | 4.92 | 3.64 | 8.75 | 4.78 | 4.78 | 3.81 |
| | Rank | **1** | 8 | 2 | 7 | 3 | 9 | 5 | 5 | 4 |

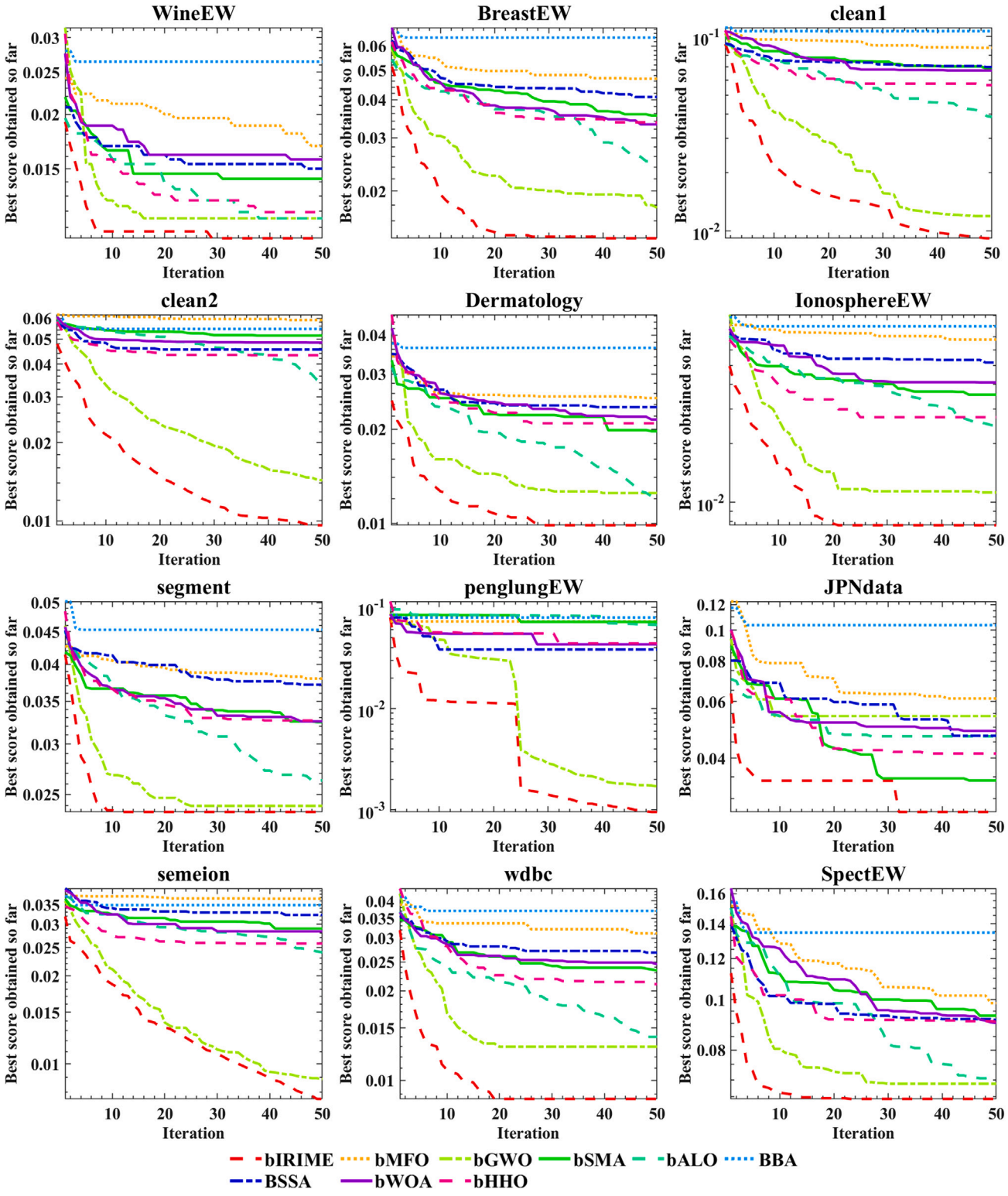



Figure 16. Convergence curves of bIRIME and other algorithms for UCI low dimensional datasets

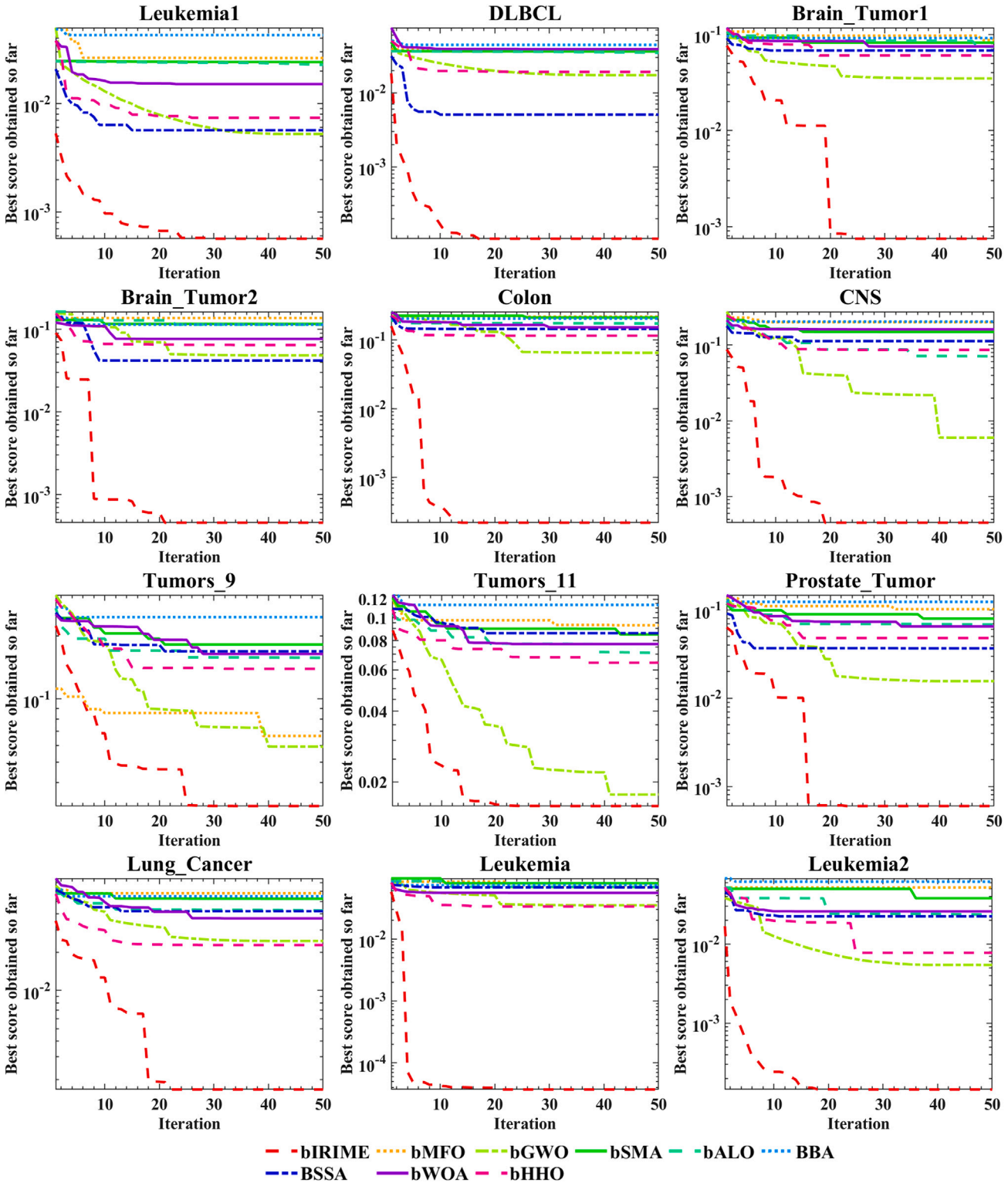



Figure 17. Convergence curves of bIRIME and other algorithms for UCI high dimensional datasets

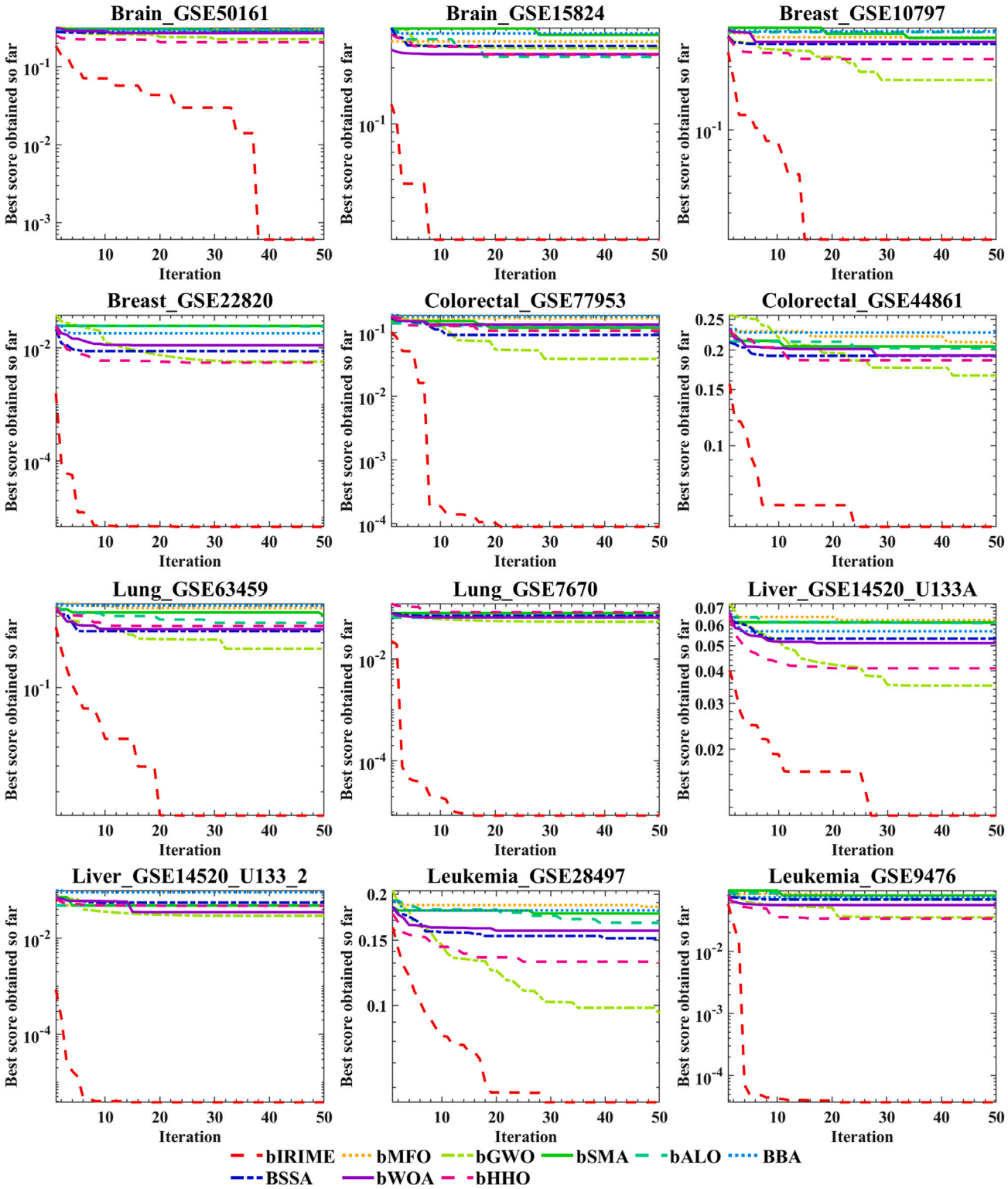



Figure 18. Convergence curves of bIRIME and other algorithms for SBCB high dimensional datasets

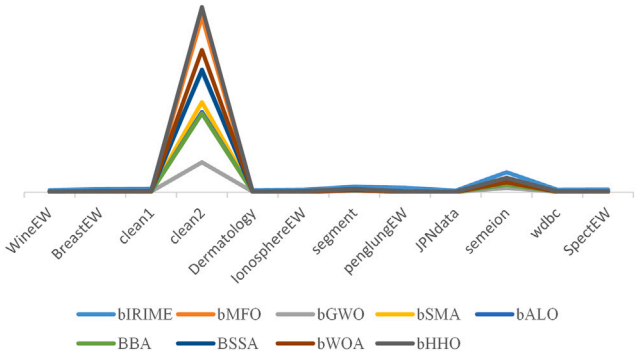



**Figure 19. The time cost of bIRIME and other algorithms on low dimensional datasets**

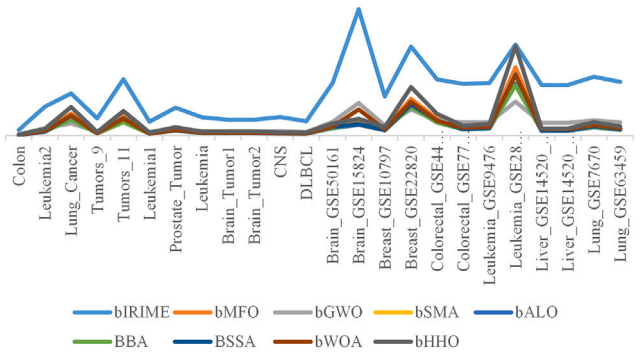



**Figure 20. The time cost of bIRIME and other algorithms on high dimensional datasets**

https://doi.org/10.1007/s11277-023-10578-y.

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**Algorithm 1. Pseudo-code for RIME**

1. **Initialization**: population $X$, $X^b$, $T$, $F_i$, $E$
2. While $t<T$
3. if $r_1 < E$
4. soft-rime: update $X_i^{new}$ by Equations 1, 2, 3, and 4
5. end if
6. if $r_2 < N(F_i)$
7. hard-rime: update $X_i^{new}$ by Equation 5
8. end if
9. if *fitness* of $(X_i^{new}) < F_i$
10. greedy selection: $X_i = X_i^{new}$
11. end if
12. update $t$, $F_i$, $X^b$
13. end while
14. return $X^b$

**Algorithm 2. Pseudo-code for IRIME**

Initialization: population $X$, $X^b$, $T$, $F_i$, $E$, $count(i)$
  While $t<T$
    if $|E_0| > 1$
      SB: update $X_i^{new}$ by Equation 6
    else
      if $r_1 < E$
        soft-rime: update $X_i^{new}$ by Equations 1, 2, 3, and 4
      end if
      if $r_2 < N(F_i)$
        hard-rime: update $X_i^{new}$ by Equation 5
      end if
    end if
    if *fitness* of $(X_i^{new}) < F_i$
      greedy selection: $X_i = X_i^{new}$
    else
      $count(i)=count(i)+1$
    end if
    CMS: Choose best position in *X1 X2* and *X3* as $X_i^{new}$
    if *fitness* of $(X_i^{new}) < F_i$
      $X_i = X_i^{new}$
    else
      $count(i)=count(i)+1$
    end if
    if $count(i)>50$
      RS: Choose best position in *Y1* and *Y2* as $X_i$
      $count(i)=0$
    end if
    update $t$, $F_i$, $X^b$
  end while
  return $X^b$

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## STAR★METHODS

### KEY RESOURCES TABLE

| REAGENT or RESOURCE | SOURCE | IDENTIFIER |
|---|---|---|
| Software and algorithms | | |
| RIME | Ali Asgher Heidari | https://aliasgharheidari.com/RIME.html |
| IRIME | This paper | https://github.com/TingJin0/IRIME.git |
| Deposited data | | |
| UCI dataset | UCI Repository | https://archive.ics.uci.edu/ |
| SBCB dataset | SBCB Lab | https://sbcb.inf.ufrgs.br/ |

### RESOURCE AVAILABILITY

#### Lead contact

For further inquiries for information, please direct them to the lead contact, Huiling Chen, who will handle them accordingly. You can reach him via email at chenhuiling.jlu@gmail.com.

#### Materials availability

This study did not generate new materials.

#### Data and code availability

- The dataset for this study is publicly accessible online and can be shared by the primary contact upon request. Links to the code and DOIs are included in the resources table.
- The data presented in this paper will be made available through the primary contact upon request.
- The paper does not include the original code directly, but it can be accessed from the designated contact upon request for reanalysis purposes.

### METHOD DETAILS

The proposed method primarily comprises RIME, SB, and CMS-RS, along with the relevant theories and content regarding feature selection.

#### RIME

RIME is a new physics-based metaheuristic algorithm whose main idea is to simulate the growth process of rice-ice. Its modeling process is divided into three parts: soft-time, hard-rime, and greedy selection.

##### *Soft-rime*

Due to relatively low wind forces, rime-ice primarily grows outward from the center of the frost during the initial stages of rime-ice formation. This growth process does not continue indefinitely but gradually reaches a stable state. Expressed mathematically, it can be represented as follows:

$$X_{i,j}^{new} = X_j^b + 2 \times (h_1 - 0.5) \times \cos\theta \times \beta \times \left(h_2 \times \left(ub_j - lb_j\right) + lb_j\right), r_1 < E \quad \text{(Equation 1)}$$

$$\theta = t\pi/10T \quad \text{(Equation 2)}$$

$$\beta = 1 - [tw / T]/w \quad \text{(Equation 3)}$$

$$E = \sqrt{t/T} \quad \text{(Equation 4)}$$

where, $X_{i,j}^{new}$ represents the generated new candidate solution, $i$ and $j$ denote the $i$-th particle and the $j$-th dimension, respectively. $X_j^b$ signifies the value of the optimal solution's $j$-th dimension discovered thus far. $h_1$, $h_2$ and $r_1$ are uniformly distributed random numbers between [0, 1]. $ub_j$ and $lb_j$ represent the upper and lower bounds in $j$-th dimension. $[\cdot]$ denotes rounding, while $t$ and $T$ depict the present iteration number

and the total iteration count, respectively. In the original paper, $w$ is set to 5, and this paper follows the original setting. A visual representation of soft-rime is depicted in Figure S1.

### *Hard-rime*

In high wind conditions, rime-ice tends to grow in the same direction and exhibits crossing phenomena. According to the inspiration, the authors proposed the rime puncture mechanism, represented mathematically as follows:

$$X_{i,j}^{new} = X_j^b, r_2 < N(F_i) \quad \text{(Equation 5)}$$

where $X_{i,j}^{new}$ represents the generated new candidate solution, $r_2$ is a random number uniformly distributed from 0 to 1. $F_i$ represents the fitness value of the $i$-th individual, and $N(F_i)$ is the Z-score standardized value of $F_i$. A schematic diagram of hard-rime puncture is shown in Figure S2.

### *Greedy selection*

After generating a new candidate solution, its fitness value is computed to ensure its reliability. If the fitness value is less than (all problems considered in this paper are minimization problems) the fitness value of the current solution, the algorithm replaces the original individual with the generated candidate solution. The pseudocode of RIME is shown in Algorithm 1.

After reviewing the entire algorithm during the soft-rime process, although it can move towards the newly generated rime particles, increasing the population's diversity as described in Equation 1, it only searches around the optimal point. Its influence on population diversity is limited, and RIME lacks effective communication among individuals, affecting its ability to discover potential optimal solutions among them. Additionally, in the hard-rime phase, higher fitness values tend to assign better values to certain dimensions after Z-score normalization. Coupled with greedy selection, although it somewhat enhances RIME's exploitation capability, it severely diminishes its exploration ability. This tendency makes RIME prone to falling into local optima when dealing with complex problems. Therefore, suitable methods are needed to solve the problems that exist in RIME.

## The proposed IRIME

RIME lacks interaction among individuals, quickly falling into local optima and experiencing low convergence accuracy, mainly when dealing with complex problems like feature selection. The main reason for choosing SB is its potential to enhance RIME's global capabilities. SB is inspired by HHO, which has strong global search abilities. Integrating SB aims to enhance RIME's global capabilities and exploration abilities. The main reason for selecting CMS-RS is because it strengthens population communication abilities and can also break out of local optima. CMS-RS is inspired by CoDE, and enhancing population information exchange through individual differential operations enables new spaces to be explored through restart mechanisms. Including SB and CMS-RS enables effective individual communication, expands RIME's search space, and enhances its convergence accuracy. When RIME gets trapped in local optima, a restart strategy can be initiated to explore the problem further. Let us delve into detailed explanations of SB and CMS-RS.

### *SB*

The concept of SB is primarily derived from HHO.[28] Hawks, while hunting, hover in the air to observe the positions of other individuals and prey, aiming to find the optimal attack position. The specific mathematical expression for this concept is given in Equation 6:

$$X_{i,j}^{new} = \begin{cases} X_A - k_1|X_B - 2k_2X_i|, p \geq 0.5 \\ X^b - X^m - k_3\left(k_4\left(ub_j - lb_j\right) + lb_j\right), p < 0.5 \end{cases} \quad |E_0| > 1 \quad \text{(Equation 6)}$$

where $X_i^{new}$ represents the newly generated candidate solution, $X_A$ and $X_B$ denote two distinct individuals within the population. $k_1$, $k_2$, $k_3$, $k_4$, and $p$ are uniformly distributed random numbers within the interval [0, 1]. $X^m$ signifies the mean of all individuals, and $X_i$ denotes the $i$-th individual. $ub_j$ and $lb_j$ represent the upper and lower bounds in $j$-th dimension, respectively. $|\cdot|$ indicates the absolute value. It can be observed that SB updates the newly generated candidate solution with equal probability. This method involves multiple individuals, making full use of information within the population, and its randomness contributes to increasing population diversity. $E_0$ as Equation 7:

$$E_0 = (a - 1) \times 2(1 - t / T) \quad \text{(Equation 7)}$$

where $t$ represents the current iteration number, $T$ denotes the total iteration count, and $a$ stands for a uniformly distributed random number within the interval [0, 2].

### *CMS-RS*

The inclusion of CMS-RS is primarily derived from the CoDE,[139] which has been further explored by other researchers as well.[140] CMS-RS can significantly enhance the precision of RIME. When RIME gets stuck in a local optimum, CMS-RS can assist in restarting RIME to explore other

dimensions within the solution space. Firstly, CMS enhances information exchange between individuals through three different differentiation methods. The specific mathematical model is as follows:

$$X1_j = \begin{cases} X_{r1,j} + F_1\left(X_{r2,j} - X_{r3,j}\right) if \quad rand < C_{r1} or j = j_{rand} \\ X_{i,j} \quad else \end{cases} \quad \text{(Equation 8)}$$

where $X1_j$ represents the newly generated solution's value in dimension $j$, where $r1$, $r2$, and $r3$ are three different integers ranging from 1 to $N$, with $N$ being the population size. $X_{r1,j}$, $X_{r2,j}$, and $X_{r3,j}$ denote the values in the $j$-th dimension for three distinct individuals within the population. $X_{i,j}$ signifies the value of the $i$-th individual in the $j$-th dimension. In this section, the subsequent $X_{i,j}$ represents the same concept. $rand$ refers to a uniformly distributed random number between 0 and 1. The value for $C_{r1}$ is 0.1, and $F_1$ is 1. $j$ represents the $j$-th dimension, while $j_{rand}$ is an integer ranging from 1 to $D$. $D$ signifies the problem's dimension.

$$X2_j = \begin{cases} X_{r4,j} + F_2\left(X_{r5,j} - X_{r6,j}\right) + F_2\left(X_{r7,j} - X_{r8,j}\right) if \quad rand < C_{r2} or j = j_{rand} \\ X_{i,j} \quad else \end{cases} \quad \text{(Equation 9)}$$

where $X2_j$ represents the newly generated solution's value in dimension $j$. $r4$-$r8$ are four distinct integers ranging between 1 and $N$, with $N$ being the population size. $X_{r4,j}$-$X_{r8,j}$ respectively denote the values in the $j$-th dimension for four distinct individuals within the population. $rand$ refers to a uniformly distributed random number within the range of 0 to 1. Here, the value for $C_{r2}$ is 0.2, which is an increased probability compared to $C_{r1}$, providing a greater chance for differentiation. Similarly, the number of individuals selected for differentiation has increased from three to four, fully utilizing the information within the population. Here, $F_2$ is valued at 0.8. $j$ represents the $j$-th dimension, while $j_{rand}$ is an integer ranging from 1 to $D$. $D$ signifies the problem's dimension.

$$X3_j = \begin{cases} X_{i,j} + rand\left(X_{r9,j} - X_{i,j}\right) + F_3\left(X_{r10,j} - X_{r11,j}\right) if \quad rand < C_{r3} or j = j_{rand} \\ X_{i,j} \quad else \end{cases} \quad \text{(Equation 10)}$$

where, $X3_j$ represents the newly generated solution's value in dimension $j$. $r9$-$r11$ are three distinct integers ranging between 1 and $N$, with $N$ being the population size. $X_{r9,j}$, $X_{r10,j}$, and $X_{r11,j}$ respectively denote the values in the $j$-th dimension for three distinct individuals within the population. $rand$ refers to a uniformly distributed random number within the range of 0 to 1. Here, the value for $C_{r3}$ is 0.9, indicating an increased probability for differentiation compared to the previous two differential mechanisms, aiming to enhance population diversity. $F_3$ is valued at 1. $j$ represents the $j$-th dimension, while $j_{rand}$ is a random integer ranging from 1 to $D$. $D$ signifies the problem's dimension. After the completion of CMS, the optimal individual selected is denoted as $X_i^{new}$. The subsequent section pertains to the restart strategy, outlined as follows.

During the entire operation of RIME, we employed a variable $count(i)$ to track how long each individual had gone without further updating its fitness value and the duration since the CMS result had not outperformed the current fitness value. Here, $i$ represents an integer within the range of [1, $N$], denoting the $i$-th individual. Upon exceeding a certain threshold, set at 50 in this study, we initiate RS. There are two specific RS strategies, with the first one being random restart, as depicted in Equation 11:

$$Y1_j = rand\left(ub_j - lb_j\right) + lb_j \quad \text{(Equation 11)}$$

where $Y1_j$ represents the newly generated solution's value in dimension $j$, $rand$ is a uniformly distributed random number within [0, 1], and $ub_j$ and $lb_j$ represent the upper and lower bounds in $j$-th dimension. The second strategy involves opposition restart, as illustrated in Equation 12:

$$Y2_j = rand\left(ub_j + lb_j\right) - X_{i,j} \quad \text{(Equation 12)}$$

where $Y2_j$ represents the newly generated solution's value in dimension $j$, $rand$ is a uniformly distributed random number within [0,1], and $ub_j$ and $lb_j$ represent the upper and lower bounds in $j$-th dimension. Upon the completion of the restart strategy, the best-performing individual is selected as $X_i^{new}$.

## Framework of proposed IRIME

During the execution of IRIME, the population's update is determined by $E_0$. $E_0$ determines whether the population is updated through SB or according to the original RIME method. Based on the nature of $E_0$, the algorithm is more likely to update the population using the SB method in the initial stages. Indeed, this form is not found in algorithms like PSO and DE. DE works on a simple concept of differential and mutation, and PSO operates on the singular idea of swarm intelligence. Both of these methods have a straightforward optimization process. As iterations progress, the probability of executing SB decreases gradually until it reaches zero. This approach facilitates extensive information exchange within the population, allowing RIME's convergence capability to prevail in the later stages. Following each population update, CMS is employed to leverage the information within the population further and explore potential solutions among individuals. The best-performing individual is greedily selected based on fitness; if it surpasses the current individual, a replacement strategy is performed. When the algorithm becomes trapped in local optima, an RS restarts and replaces individuals within the population. RS is not present in simple DE and PSO, which is why they are particularly prone to falling into local optima in the later stages. However, the RS in this paper can constantly

monitor the running state of IRIME. When an individual falls into a local optimum, it performs a restart. In summary, the combination of these mechanisms is not straightforward and does indeed differ from classical algorithms such as DE and PSO. The specific pseudocode is provided in Algorithm 2, and the process flow is depicted in Figure S3.

### Computational complexity analysis

Computational complexity is a critical metric for assessing the efficiency of an algorithm.[141,142] If the dimension of the problem is $D$, the population size is $N$, and the maximum number of iterations is $T$, the main processes of IRIME include population initialization, fitness value calculation, and position updating. The complexity of initialization is $O(D\times N)$, and the complexity of fitness value calculation is $O(T\times N\times F)$ + $O(\log(T)\times N\times 3F+T\times N/50\times 2F)$ where $F$ is the complexity of calculating fitness values. The complexity of position updating is $O(T\times N\times D)$. Therefore, the overall complexity of the algorithm is $O(T\times N\times D + T\times N\times F + \log(T)\times N\times 3F+T\times N/50\times 2F + D\times N)$. It is worth noting that compared to RIME, IRIME primarily increases the complexity of the CMS-RS calculation, which is $O(\log(T)\times N\times 3F+T\times N/50\times 2F)$.

### K-nearest neighbor classifier

The K-nearest neighbor classifier (KNN)[143] is a commonly used classification method. Its main concept involves finding the K nearest individuals in proximity to a given point, which then defines its class as the one most frequently present among these neighbors. For proximity measurement, this study adopts the Euclidean distance, expressed by the specific formula:

$$Dis(X,Y) = \sqrt{\sum_{k=1}^{n}(X_k - Y_k)^2} \quad \text{(Equation 13)}$$

where $X$ represents the training samples, $Y$ represents the test samples, and $n$ denotes the number of samples.

### Binary IRIME

The feature selection problem can be regarded as a discrete combinatorial optimization problem, where we use "0" or "1" to represent not selecting or selecting a particular feature. In this manner, algorithms need to be adjusted to handle binary problems. The following equation provides the specific transformation function:

$$T(x) = \frac{1}{1+e^{-10(x-0.5)}} \quad \text{(Equation 14)}$$

where $x$ represents the value of an individual on a certain dimension. It is noteworthy that Equation 14 was proposed in bGWO.[130] For the sake of experimental rigor, a 10-fold cross-validation was employed during the experimentation process. The specific transformation process is detailed in Equation 15:

$$X_{i,j}(t+1) = \begin{cases} 1 & rand < T(x) \\ 0 & rand > T(x) \end{cases} \quad \text{(Equation 15)}$$

where $t$ represents the current iteration count, $i$ and $j$ denote the $i$-th individual within the population, and the $j$-th dimension of an individual respectively. *rand* stands for a uniformly distributed random number within the interval [0,1]. This paper treats the feature selection problem as a single-objective problem,[144] where the objective function is defined as follows:

$$Fitness = \alpha * err + (1-\alpha) * \frac{N}{M} \quad \text{(Equation 16)}$$

where, $\alpha$ takes a value of 0.95, err represents the error rate, $M$ stands for the dimensionality of the dataset, and $N$ denotes the dimensionality after feature selection. It is evident from the formula that the error rate of *Fitness* is related to the size of the selected feature subset. Additionally, the smaller the *Fitness*, the better the algorithm performs in feature selection applications.